STUDY TEXTBOOK

Physics for Engineers

Mechanics, fields, and circuits

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CHAPTER 1

Kinematics: describing motion

Mechanics begins with description, not explanation: before we ask *why* things move (chapter 2), we need language precise enough to say *how* they move. That language is calculus applied to a position function $x(t)$, and the payoff for learning it carefully is that the hardest parts of physics — momentum, orbits, wave motion — reuse exactly these ideas. This chapter derives the constant-acceleration formulas from the definitions (never quotes them), treats projectile motion as two independent one-dimensional problems, and builds the centripetal-acceleration formula from a picture of velocity differences. Engineering practice shows up at the end: inertial navigation is this chapter run continuously.

1.1 Position, displacement, and velocity

Fix a line and an origin. A point particle's *position* $x(t)$ is a signed distance from the origin; the sign encodes direction. The *displacement* over $[t_1, t_2]$ is $\Delta x = x(t_2) - x(t_1)$ — which is emphatically not the distance travelled: run around a track once and the displacement is zero while the distance is a lap. The *average velocity* is the slope of the secant,

$$\bar{v} = \frac{\Delta x}{\Delta t},$$

and the *instantaneous velocity* is what the average converges to as the interval shrinks:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{dx}{dt}.$$

Velocity is a slope on the $x(t)$ graph. Speed is its magnitude — "60 mph east" is a velocity, "60 mph" is a speed. Sign matters: in one dimension $v > 0$ means moving toward $+x$.

READING AN $x(t)$ GRAPH

Slope of the curve at t = instantaneous velocity. Slope of the secant over $[t_1, t_2]$ = average velocity. A horizontal tangent = momentarily at rest — and that is all it means; what happens next depends on the acceleration.

Exercise 1.1 (easy)**.** A particle sits at $x(t) = 3t^2 - 12t$. Find its average velocity over $[0, 4]$ and its instantaneous velocity at $t = 2$. (AP-C)

1.2 Acceleration and the calculus loop

Acceleration is velocity's velocity:

$$a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2},$$

and figure 1.1 shows all three functions for a thrown object. Read them vertically: when v crosses zero the object is at the top of its climb, and a — a constant $-9.8\,\mathrm{m/s^2}$ — does not care.

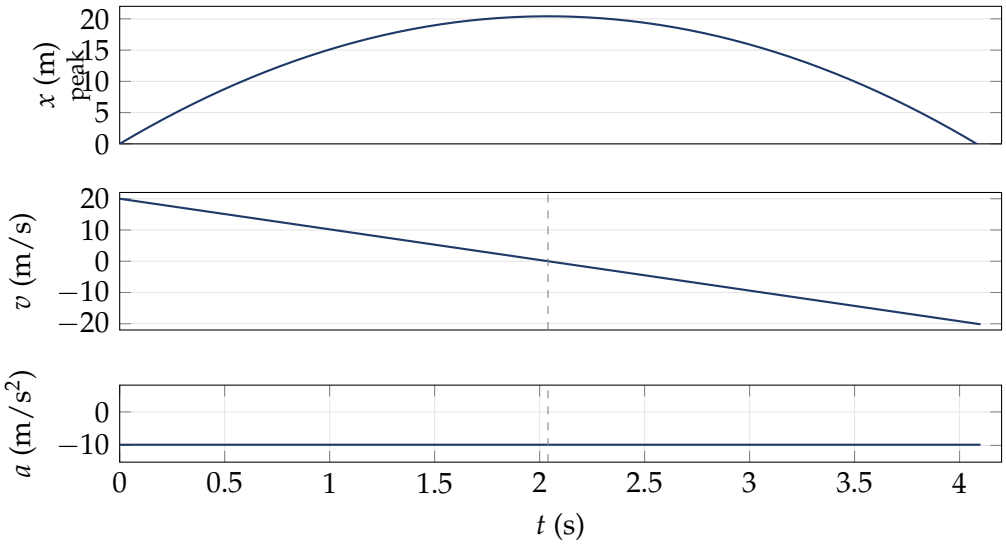


Figure 1.1: $x(t) = 20t - 4.9t^2$ with its first two derivatives, drawn on one time axis. Notice the three-way alignment: the peak of x sits exactly above the zero of v, while a stays fixed at $-9.8\,\mathrm{m/s^2}$ the whole flight. A horizontal tangent in x means "stopped *then*," never "stopped forever."

The loop of figure 13.3 is the chapter's engine: differentiate to go from x to v to a; integrate to come back:

$$v(t) = v(t_0) + \int_{t_0}^{t} a(t')\,dt', \qquad x(t) = x(t_0) + \int_{t_0}^{t} v(t')\,dt'.$$

Kinematics is nothing more — and nothing less — than choosing a direction around that loop.

Example 1.1 (Slopes off a measured $x(t)$)**.** A radar logs a dragster's position every half second:

t (s)	0	0.5	1.0	1.5	2.0	2.5	3.0
x (m)	0	3.1	9.8	22.0	39.5	62.4	90.7

Estimate $v(1.0)$ and $a(1.0)$ without a formula for $x(t)$.

Strategy. With data, derivatives are differences: centre the secant for v (symmetric difference quotients are second-order accurate), then difference the velocities for a.

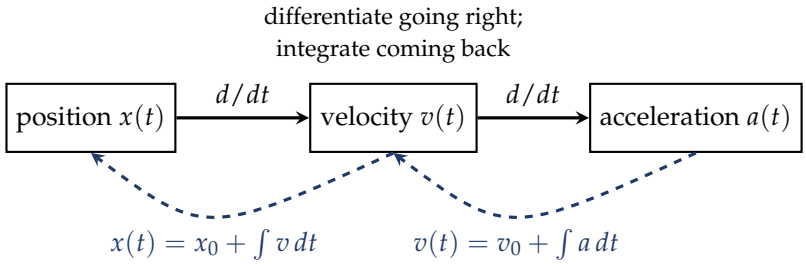


Figure 1.2: The calculus loop. Notice that going right loses information (the constants x_0, v_0 are invisible in $a(t)$), so coming back needs initial conditions: two numbers must be supplied before any $x(t)$ is unique.

Solution. $v(1.0) \approx \frac{x(1.5)-x(0.5)}{1.0} = \frac{22.0-3.1}{1.0} = 18.9\,\text{m/s}$. For a: estimate $v(0.5) \approx 9.8/1.0 = 9.8$ (from $x(1.0) - x(0)$ over 1 s $\Rightarrow$ centre it at 0.5) and $v(1.5) \approx (39.5 - 9.8)/1.0 = 29.7$; then $a(1.0) \approx (29.7 - 9.8)/1.0 = 19.9\,\text{m/s}^2 \approx 2g$. ✓ Same calculus, no closed form needed — this is exactly how IMU and encoder data get turned into velocities.

1.3 Constant acceleration, derived

When a is constant the loop collapses to three formulas — derived here so their assumptions are visible. Integrating a once, $v(t) = v_0 + at$; integrating again,

$$x(t) = x_0 + v_0 t + \tfrac{1}{2}at^2. \tag{1.1}$$

Eliminating t between them (solve $v = v_0 + at$ for t, substitute into (1.1)) gives the time-free relation

$$v^2 = v_0^2 + 2a\,(x - x_0). \tag{1.2}$$

That is the whole subject of "constant-acceleration kinematics." Everything else — free fall, braking distance, the meeting-time problem — is one of these two equations with different letters.

Example 1.2 (Braking distance and the v^2 trap)**.** A car at 100 km/h brakes at a steady $a = -8\,\text{m/s}^2$ (good tires, dry road). How far does it travel before stopping — and how far if the driver needs 0.5 s to react first?

Strategy. No time is asked or given: equation (1.2) answers it directly. Convert units before plugging; add the reaction-time distance afterwards (during it, $a = 0$).

Solution. $100\,\text{km/h} = 27.8\,\text{m/s}$. From (1.2), $0 = v_0^2 + 2a\,\Delta x$, so $\Delta x = \frac{v_0^2}{2|a|} = \frac{27.8^2}{16} = 48.3\,\text{m}$. Reaction adds $v_0 \Delta t = 27.8 \times 0.5 = 13.9\,\text{m}$ — total 62.2 m, a third of it before the brakes touch. ✓ Notice the v^2: at 140 km/h the braking part alone grows to 94 m, not 67 — doubling speed quadruples the stop.

Exercise 1.2 (medium)**.** Two trains 400 m apart approach at 20 m/s each. When does a bee flying between them at 30 m/s get squashed — and how far does the bee travel? Solve without summing the series. (F=ma)

1.4 Projectile motion: two problems at once

With gravity the only acceleration, the vector equation $\mathbf{a} = (0, -g)$ splits: horizontal motion is constant-velocity, vertical is constant-acceleration, and they share only the clock:

$$x(t) = v_{0x}t, \qquad y(t) = v_{0y}t - \tfrac{1}{2}gt^2.$$

Independence is the deep point — a dropped ball and a fired one land together — and figure 1.3 shows the trajectory with its velocity components at three instants.

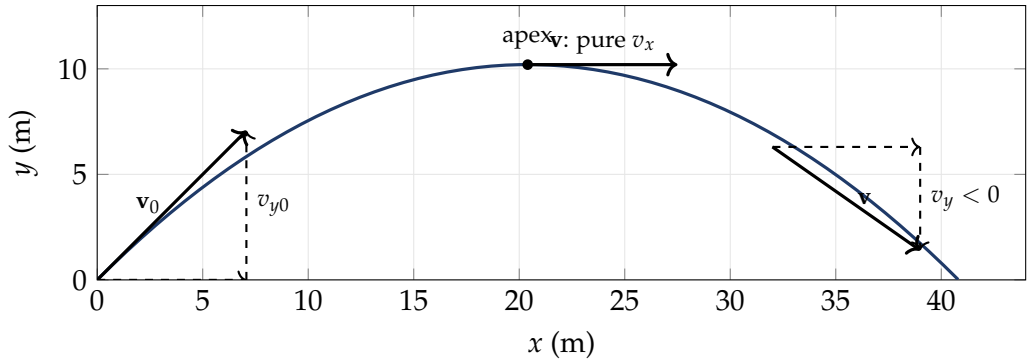


Figure 1.3: Projectile with $v_0 = 20\,\mathrm{m/s}$ at 45°. Notice what changes along the path: v_x never does; v_y drains to zero at the apex and goes negative after. The parabola is a parametric curve — $y(x)$ has no time in it; recover t from $x = v_{0x}t$ whenever it is needed.

The flat-ground range $R = \frac{v_0^2 \sin 2\theta}{g}$ follows by setting $y = 0$ and eliminating t. The generalised version — landing at height h — is worth more:

Example 1.3 (Landing on a ledge)**.** The launch of figure 1.3 ($v_0 = 20\,\mathrm{m/s}$, 45°) lands on a platform $h = 5\,\mathrm{m}$ above the launch level. Find the range.

Strategy. Do not memorise a second range formula: solve $y(t) = h$ for the *later* crossing time, then $R = v_{0x}t$.

Solution. $v_{0x} = v_{0y} = 20/\sqrt{2} = 14.14\,\mathrm{m/s}$. Solve $14.14t - 4.9t^2 = 5$, i.e. $4.9t^2 - 14.14t + 5 = 0$: $t = \frac{14.14 \pm \sqrt{199.9 - 98}}{9.8} = \frac{14.14 \pm 10.10}{9.8}$; the later root $t \approx 2.47\,\mathrm{s}$ (the earlier one, 0.41 s, is the upward crossing). Then $R = 14.14 \times 2.47 \approx 34.9\,\mathrm{m}$ — *shorter* than the flat-ground 40.8 m, because the ledge intercepts the descending arc early. Landing 5 m *below* gives $R \approx 45.3\,\mathrm{m}$; the simulator prints both signs of h in section 1.7. ✓

1.5 Circular motion: acceleration without speeding up

Uniform circular motion — constant speed v on a circle of radius r — is accelerated: the velocity vector turns even though its length never changes. Figure 1.4 draws the velocity at two nearby instants: $\mathbf{v}(t)$ and $\mathbf{v}(t + \Delta t)$ are equal-length vectors differing by a small rotation, so $\Delta\mathbf{v}$ points *inward*, and in the limit $\Delta t \to 0$ it points exactly at the centre. The magnitude comes from arc geometry: the velocity vector turns through the same angle the radius does, $\Delta\theta = v\Delta t/r$, so $|\Delta\mathbf{v}| \approx v\,\Delta\theta = v^2\Delta t/r$, giving

$$a_c = \frac{v^2}{r} \quad \text{toward the centre.} \tag{1.3}$$

This is *centripetal acceleration* — required by geometry, supplied by whatever real force (tension, friction, gravity) is handy; chapter 2 matches it to its source.

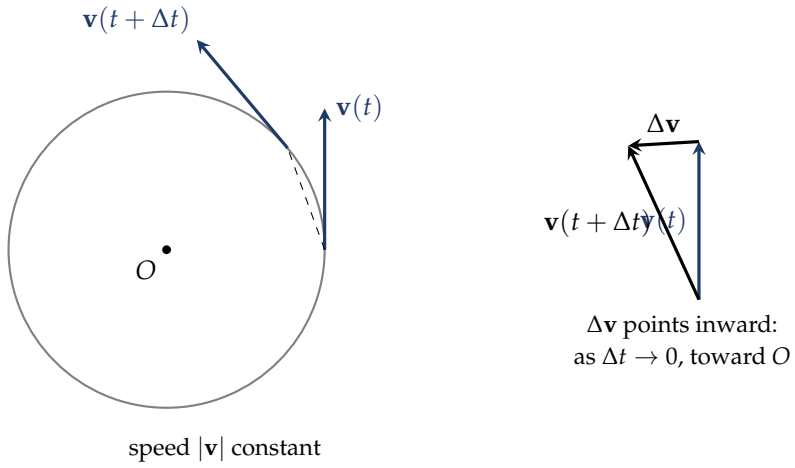


Figure 1.4: Where $a_c = v^2/r$ comes from. Notice the argument needs only that $|\mathbf{v}|$ is constant: the velocity arrow rotates with the radius, so $\Delta\mathbf{v}$ is perpendicular to $\mathbf{v}$ and aimed at the centre.

Example 1.4 (A car on a curve)**.** A car takes a radius-50 m curve at 20 m/s (72 km/h). How large is the sideways acceleration, in units of g?

Strategy. Equation (1.3) directly; compare with g for intuition (your body feels acceleration in g's).

Solution. $a_c = 20^2/50 = 8\,\mathrm{m/s^2} \approx 0.8g$. The tires must supply every newton of it — on ice, 20 m/s around this curve is impossible, which is why ramps are banked (chapter 2). ✓

Exercise 1.3 (medium)**.** A CD spins at 500 rpm. Find a_c at its 6 cm rim in SI and in units of g. (AP-C)

1.6 Relative velocity and Galilean frames

All velocities are relative to a frame; switching frames adds a constant vector:

$$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C},$$

the *Galilean velocity addition* — read it as "A as seen from C equals A as seen from B plus B as seen from C". The subscripts chain like a telescope. Figure 1.5 is the one picture to carry: two perpendicular legs and a resultant.

Example 1.5 (River crossing)**.** A boat that makes 4 m/s in still water aims straight across a 200 m river flowing at 3 m/s. When does it land, and where? Then: what heading would land it straight across?

Strategy. Components decouple, exactly like projectile motion: crossing time depends on the across-component only. For the straight crossing, aim upstream so the upstream component of boat velocity cancels the current.

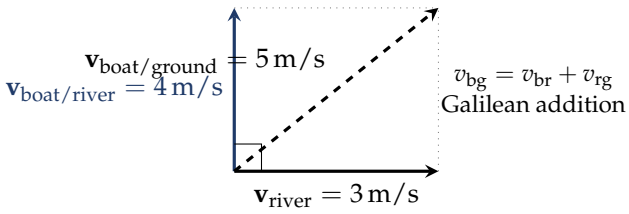


Figure 1.5: Relative velocity as vector addition. Notice the frame B = river does all the work: the boat's thrust is measured relative to the water it pushes against, and the bank sees the sum.

Solution. Across-component 4 m/s: crossing time $t = 200/4 = 50$ s. Drift $= 3 \times 50 = 150$ m downstream — a $3{-}4{-}5$ triangle, ground speed 5 m/s. To land straight across, aim at angle θ upstream with $4\sin\theta = 3$, i.e. $\theta = \arcsin 0.75 \approx 48.6°$; the effective across-speed drops to $4\cos\theta = \sqrt{7} \approx 2.65$ m/s, so $t \approx 75.7$ s — straighter costs slower. ✓

1.7 The simulator: integration is not optional

Every claim above is checkable by *numerically* running the calculus loop — and in real engineering (drag, control forces, planetary n-body) numerical integration is the only option. The listing below integrates projectile motion by forward Euler, measures the step-size error, then adds quadratic drag — the first preview of chapter 2's terminal velocity.

Checkpoint 1.1 *Predict before you run.* Euler overshoots: v is updated after x, so each step uses a stale velocity. Guess the sign of the range error before looking.

```
def euler_range(v0, th, dt, k=0.0):
    x = y = 0.0
    vx, vy = v0*math.cos(th), v0*math.sin(th)
    while y >= -1e-12:
        s = math.hypot(vx, vy)
        ax, ay = -k*s*vx, -G - k*s*vy # gravity + quadratic drag
        x += vx*dt; y += vy*dt # update position first
        vx += ax*dt; vy += ay*dt # then velocity: forward Euler
    return x
```

Listing 1.1: `code/ch01_projectile.py` — Euler integration vs closed form, then with quadratic drag.

Output of a real run:

```
analytic range: 40.8163 m
euler dt=0.1 range= 42.4264 m error=+1.6101 m
euler dt=0.01 range= 41.0122 m error=+0.1959 m
euler dt=0.001 range= 40.8425 m error=+0.0262 m
with quadratic drag k=0.02: range=25.8593 m (36.6% shorter)
landing at h= +5.0 m: range=34.9825 m (vs 40.8163 on flat ground)
landing at h= -5.0 m: range=45.3195 m (vs 40.8163 on flat ground)
```

Notice two things. First, the error shrinks roughly with h — Euler is first-order — and overshoots positive, exactly as predicted: the stale velocity keeps climbing after it should

turn. Second, drag at $k = 0.02\,\mathrm{m}^{-1}$ cuts the range by a third: real ballistics is not the parabola, and chapter 2's drag force is why.

1.8 In engineering practice: dead reckoning

An inertial measurement unit outputs acceleration, never position — so every drone, submarine, and phone navigation chip lives inside the right half of figure 13.3: $v(t) = v_0 + \int a$, $x(t) = x_0 + \int v$, run thousands of times a second on a tiny processor. The cost is *drift*: each integration accumulates the sensor's bias error, and a 1 mg accelerometer bias becomes a $\sim 0.5\,\mathrm{m}$ position error in ten seconds. GPS exists to reset that integral: satellite fixes pin $x(t)$ periodically so the loop restarts from truth. Dead reckoning — this chapter, continuously — is the engineering version of "kinematics is the calculus loop."

1.9 Common mistakes

SPEED $\neq$ VELOCITY

Distance travelled and displacement differ the moment the path bends or reverses. "How fast" questions can mean speed or velocity — check the sign convention of the problem before answering.

$a_c = v^2/r$ IS NOT A NEW FORCE

It is a *required* acceleration. The forces are tension/friction/ gravity; their sum must equal mv^2/r inward, not the other way around.

INCONSISTENT SIGN OF g

Pick an axis once. With $+y$ up, $a = -9.8$; with $+y$ down, $a = +9.8$. Mixing both inside one solution is the most common source of wrong answers in §1.3–1.4.

CONSTANT-a FORMULAS ON VARYING a

Equations (1.1)–(1.2) need $a =$ const. With drag, springs, or powered flight they are wrong; integrate the loop instead (section 1.7).

ADDING VELOCITIES AS SCALARS

$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$ is vector addition. A 4-m/s boat in a 3-m/s river does *not* move at 7 m/s — it moves at 5 m/s diagonally (figure 1.5).

1.10 Summary

Idea	One-line version
calculus loop	$x \xrightarrow{d/dt} v \xrightarrow{d/dt} a$; integrate back, paying two constants
constant a	$v = v_0 + at$, $x = x_0 + v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a\Delta x$
projectile	x: const. velocity; y: const. accel $-g$; shared only by t
range on height h	solve $y(t) = h$, take the later root, $R = v_{0x}t$
circular motion	$a_c = v^2/r$ toward centre — required, not a force
relative velocity	$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$ (vectors)
speed rule	v, a same sign ⇒ speeding up; opposite ⇒ slowing (fig. 1.6)

Table 1.1: Kinematics cheat sheet.

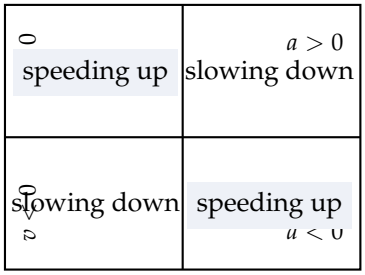


Figure 1.6: The (v, a) sign map. Notice it never asks about position: only the *relative* signs of velocity and acceleration set whether the object speeds up or slows down.

DEFINITION OF DONE

You can now: read v and a off an $x(t)$ graph and off a data table; derive the three constant-acceleration formulas; split projectile motion into components and land it on a ledge; explain where $a_c = v^2/r$ comes from; add velocities through a frame chain; and run the calculus loop numerically with a first-order integrator.

Notes

The calculus-first ordering and the derivations follow MIT 8.01SC weeks 1–3 [2]; AP Physics C unit 1 covers the same ground in the MC/FR split [4]. OpenStax College Physics chapters 3–4 [3] supply the projectile and relative-velocity treatment; the centripetal derivation by velocity difference is Feynman I §8-5 [5]. Halliday–Resnick [6] has the largest drill inventory; Morin [7] is the harder recommendation for anyone who found this chapter easy. The dead-reckoning paragraph sketches the strapdown inertial navigation problem; the bias-drift scaling is standard.

WATCH INSTEAD

1. *1D Kinematics — Speed, Velocity, Acceleration* — Walter Lewin, MIT 8.01x Lecture 2, 50:42, https://youtu.be/q9IWoQ199_o — §1.1–§1.3 at full length, same notation (accessed 2026-10-11).
2. *Why distance is area under the velocity–time line* — Khan Academy, 9:25, https://youtu.be/d-_eqgj5-K8 — the integration half of §1.2 in eight minutes (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 1–3 — the source course; recitation videos solve the same problems. Free at https://ocw.mit.edu, course 8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 3 "Two-Dimensional Kinematics" — free text, more worked projectile examples. At https://openstax.org, search "College Physics 2e" (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 1 — the exam blueprint this chapter aligns to; FR verb practice. At https://apstudents.collegeboard.org, course page "AP Physics C: Mechanics" (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics*, vol. I, ch. 8 "Motion" — the prose version, still the best motivation for why derivatives run physics. Free online at Caltech.
- D. Morin, *Introduction to Classical Mechanics*, ch. 2 — kinematics for people who want the hard version; ISBN 978-0521876223.
- Khan Academy, *Physics — one-dimensional motion* — drill exercises matching §1.1–§1.3, at https://www.khanacademy.org/science/physics. Verified 2026-10-11.
- The Physics Classroom, *1-D and 2-D Kinematics* tutorials — readable remediation with check-yourself quizzes, at https://www.physicsclassroom.com (verified 2026-10-11).

Additional exercises

Exercise 1.4 (easy)**.** From the $v(t)$ graph of figure 1.1: read off the launch velocity, the time of the peak, and the landing time — then confirm all three from $x(t) = 20t - 4.9t^2$. (AP-C)

Exercise 1.5 (easy)**.** Convert these accelerations to SI: $32\,\mathrm{km/h/s}$, $0.05\,\mathrm{miles/h^2}$ (in $\mathrm{m/s^2}$, approximately), and $9.8\,\mathrm{m/s^2}$ in $\mathrm{km/h^2}$. (AP-C)

Exercise 1.6 (easy)**.** A ball is dropped from rest at height h. Find its speed after t seconds and after falling d metres — which of the three constant-a formulas did each answer need? (F=ma)

Exercise 1.7 (easy)**.** A train accelerates uniformly from rest to $30\,\mathrm{m/s}$ over $600\,\mathrm{m}$. Find a and the elapsed time. (AP-C)

Exercise 1.8 (easy)**.** Displacement vs. distance: a runner does one lap of a 400 m track in 80 s. Compute average velocity and average speed. (AP-C)

Exercise 1.9 (medium)**.** Show that for a launch on level ground the maximum range at fixed v_0 comes at $\theta = 45^\circ$, then show the max range equals $\frac{v_0^2}{g}$ — by maximising $R(\theta)$, not by quoting it. (AP-C FR-style)

Exercise 1.10 (medium)**.** A second crossing: the boat of example 1.5 wants minimum *time* rather than zero drift. Show the fastest crossing points straight across ($t = 50$ s), and compute where it lands. Then find the heading for minimum *total distance travelled over ground*. (F=ma)

Exercise 1.11 (medium)**.** From the dragster table of example 1.1, estimate $v(2.0)$ and $a(2.0)$ by the same symmetric-difference method, then fit $x(t) \approx \frac{1}{2}at^2$ and compare. (F=ma, data analysis)

Exercise 1.12 (medium)**.** A ball is thrown horizontally at 15 m/s off a 30 m cliff. Find the time to land, the range, and the velocity vector (magnitude and direction) at impact. (AP-C)

Exercise 1.13 (medium)**.** On the sign map (figure 1.6): an object has $v < 0$ and $a < 0$. Is it speeding up? Give the everyday example of each quadrant. (AP-C)

Exercise 1.14 (medium)**.** An airplane flies at 200 m/s north relative to air, in a 40 m/s east wind. Find the ground track (heading and ground speed). (F=ma)

Exercise 1.15 (medium)**.** Two cars start 500 m apart and drive toward each other at 25 m/s and 20 m/s. When and where do they meet — solve in the ground frame, then in the frame of the slower car. (F=ma)

Exercise 1.16 (hard)**.** Generalise example 1.3: for a launch at (v_0, θ) onto a ledge at height h, find the angle θ^* that maximises range. Show $\theta^* \to 45^\circ$ as $h \to 0$ and find the leading correction for small h/v_0^2. (USAPhO-style)

Exercise 1.17 (hard)**.** A pursuit with a cap: a cheetah accelerates at 5 m/s^2 but is speed-limited to 30 m/s; a gazelle, 20 m ahead, accelerates at 3 m/s^2 capped at 25 m/s. When does the cheetah catch it? (Track the phase where both are at cap.) (APhO-style)

Exercise 1.18 (hard)**.** From first principles: derive $a_c = v^2/r$ without the velocity-difference argument — write $\mathbf{r}(t) = (r\cos\omega t,\ r\sin\omega t)$ and differentiate twice. Confirm both methods agree on direction and magnitude. (USAPhO-style)

CHAPTER 2

Newton's laws and applications

Kinematics (chapter 1) described motion; *dynamics* explains it. Newton's three laws are exactly three statements — a definition of inertial frames, a relation $\mathbf{F} = m\mathbf{a}$, and a pairing rule — but the engineering content is in the method that wraps them: the free-body diagram, the constraint bookkeeping of strings and surfaces, and the differential equations that friction and drag leave behind.

2.1 The three laws, and the frame assumption

First law. A body with no net force moves at constant velocity — *in an inertial frame*. The law looks circular (how do you know the frame is inertial? check whether first-law behaviour holds) and it is: Newton's first law is the *definition* of the arena his other two laws live in. Accelerating cars and rotating merry-go-rounds are non-inertial; inside them you feel "forces" (centrifugal, elevator heaviness) that have no third-law partner.

Second law.

$$\sum \mathbf{F} = m\mathbf{a}. \tag{2.1}$$

This is a *vector* equation — three scalar equations when you pick axes — and the most compressed sentence in physics: acceleration is what the *sum* of forces buys. Units: force in newtons, $1\,\mathrm{N} = 1\,\mathrm{kg}\cdot\mathrm{m/s^2}$.

Third law. Forces come in pairs on *different* bodies: the Earth pulls you down; you pull the Earth up. The pair never cancels on one body's diagram — a point section 2.3 returns to.

2.2 The force inventory

Only a handful of forces appear in this book:

Force	Rule	Notes
gravity	$m\mathbf{g}$ down	uniform near Earth's surface
normal	whatever it must be	contact; perpendicular to surface
tension	along the rope	pulls both ends; massless rope $\Rightarrow$ same T
spring	$-kx$	Hooke's law, linear
friction	$\le \mu_s N$ static, $= \mu_k N$ kinetic	opposes *impending/actual* sliding
drag	cv or cv^2	opposes velocity; §2.7

The discipline: every entry except gravity is a *contact* force — if nothing touches the body, only $m\mathbf{g}$ acts. "The force of the throw" left the ball when your hand did.

2.3 Free-body diagrams: the discipline

A free-body diagram (FBD) draws *one* body and every force *on* it — nothing else. Not the force it exerts, not its velocity, not its acceleration (draw $m\mathbf{a}$ beside the diagram, never in it). Figure 2.1 is the canonical gallery.

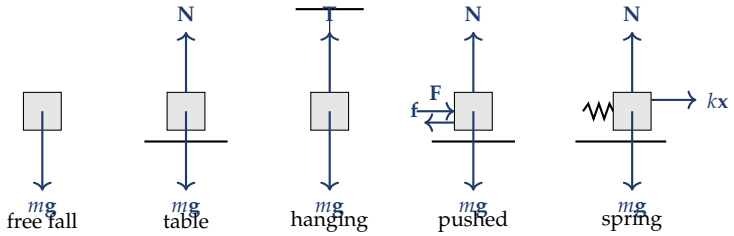


Figure 2.1: Five canonical FBDs. Notice only the body's own forces appear — the Earth's pull counts, the body's pull on the Earth does not. The pushed box's friction arrow points *back* because friction opposes sliding.

The method: (1) isolate one body; (2) enumerate contacts + gravity; (3) pick axes that simplify components; (4) write equation (2.1) per axis; (5) add constraint equations (rope length, contact, $f \le \mu_s N$); (6) solve. Skipping step 2's enumeration is where forces get invented or dropped.

Example 2.1 (Atwood machine)**.** Masses $m_1 = 3\,\mathrm{kg}$ and $m_2 = 5\,\mathrm{kg}$ hang from a light frictionless pulley (figure 2.2). Find the acceleration and tension.

Strategy. Two FBDs, one constraint: the rope's fixed length means the accelerations are equal and opposite. Take each mass's "down is positive" axis along its own motion.

Solution. For m_1 (accelerating up): $T - m_1 g = m_1 a$. For m_2 (accelerating down): $m_2 g - T = m_2 a$. Add to eliminate T:

$$a = \frac{m_2 - m_1}{m_1 + m_2} g = \frac{2}{8} \times 9.8 = 2.45\,\mathrm{m/s^2}, \qquad T = m_1(g + a) = 3 \times 12.25 = 36.75\,\mathrm{N}.$$

Check limits: $m_1 = m_2 \Rightarrow a = 0, T = mg$ (hanging balance); $m_2 \gg m_1 \Rightarrow a \to g$ (small mass is a brake). $T = 36.75\,\mathrm{N}$ lies between $m_1 g = 29.4$ and $m_2 g = 49$ — the moving rope always splits the difference. ✓

Exercise 2.1 (easy)**.** In example 2.1, what is the force the pulley's axle must exert on the ceiling? Why isn't it $m_1 g + m_2 g$? (AP-C)

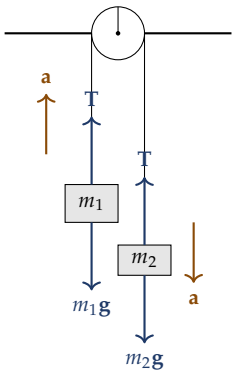


Figure 2.2: Atwood machine: two FBDs, one rope constraint. Notice the tension is the *same* T on both sides — the massless-frictionless ideal pulley transmits, it does not eat.

2.4 Constraints: inclines, pulleys, contact

On an incline the useful axes are *down-slope* and *outward* (figure 2.3): gravity decomposes as $mg\sin\theta$ down-slope, $mg\cos\theta$ into the surface, and when the surface doesn't accelerate, $\mathbf{N} = mg\cos\theta$ *only then*. That "only then" is the section's lesson — N is determined, not given.

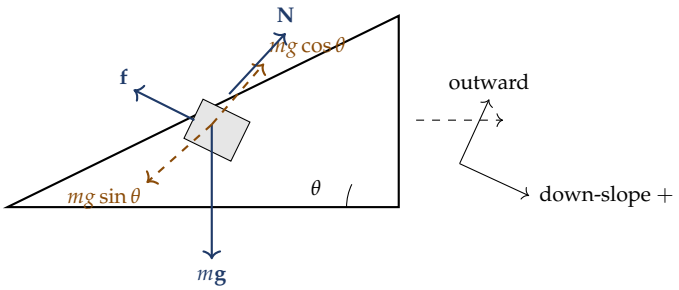


Figure 2.3: Incline decomposition. Notice the axes are rotated, not the force: mg splits into a down-slope driver $mg\sin\theta$ and a pressed-in part $mg\cos\theta$ that the normal force answers.

Example 2.2 (Apparent weight on an accelerating surface)**.** A 60 kg student stands on a scale in an elevator accelerating up at $2\,\mathrm{m/s^2}$. What does the scale read? What if the cable snaps?

Strategy. The scale reads the *normal* force, not gravity. One FBD, one axis (up $+$).

Solution. $N - mg = ma \Rightarrow N = m(g+a) = 60 \times 11.8 = 708\,\mathrm{N}$ — the scale reads 708 N "of weight", i.e. a 72.3 kg equivalent. Free fall $a = -g$: $N = 0$, weightlessness — not "no gravity", just nothing to push against. The ISS floats exactly this way, inside $0.9g$ of gravity. ✓

Exercise 2.2 (medium)**.** A block slides down a frictionless incline at $\theta = 30^\circ$ while the *incline itself* is free to slide on a frictionless floor. Qualitatively: does the block's acceleration exceed or fall short of $g\sin 30^\circ$? Why? (F=ma)

2.5 Circular dynamics: the net force bends

Section 1.5 showed circular motion *needs* $a_c = v^2/r$ inward; Newton's second law says the *net* force must supply it. The mantra: $\sum F_{\text{inward}} = mv^2/r$ — centripetal is what the net force *does*, never an entry on the FBD.

Example 2.3 (Banked curve, two boundary cases)**.** A curve of radius $r = 80\,\text{m}$ is banked at $\theta = 15°$. (a) What speed rounds it with *no* friction? (b) With $\mu_s = 0.4$, what are the max and min speeds?

Strategy. (a) FBD has only mg and N; resolve into vertical (no accel) and horizontal (centripetal). (b) Friction acts along the surface — down-slope at high speed (fighting the tendency to slide *up* the bank), up-slope at low speed.

Solution. (a) Vertical: $N\cos\theta = mg$. Horizontal: $N\sin\theta = mv^2/r$. Divide:

$$v = \sqrt{rg\tan\theta} = \sqrt{80 \times 9.8 \times 0.268} = 14.5\,\text{m/s} \approx 52\,\text{km/h}.$$

(b) At high speed, friction points down-slope and adds an inward component; the max-speed condition becomes (Morin's derivation) $v_{\max}^2 = rg\,\frac{\sin\theta+\mu_s\cos\theta}{\cos\theta-\mu_s\sin\theta} = 80 \times 9.8 \times \frac{0.259+0.4\times0.966}{0.966-0.4\times0.259}$, giving $v_{\max} = \sqrt{784 \times 0.747} = 24.2\,\text{m/s} \approx 87\,\text{km/h}$. The design window 14.5–24.2 m/s is what the bank buys. ✓

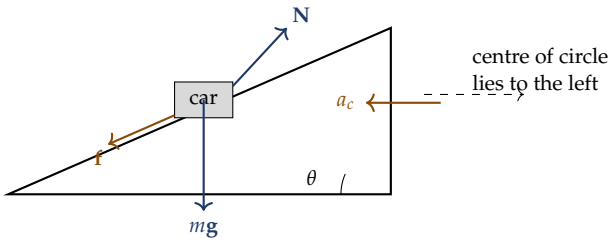


Figure 2.4: Banked curve. Notice the normal force leans inward — the bank's job is to lend a horizontal component to N so friction has less to do.

2.6 Friction: the force that adapts

Static friction *matches* applied force up to a ceiling: $f_s \le \mu_s N$; kinetic friction is a fixed drag $f_k = \mu_k N$ opposing sliding, with $\mu_k < \mu_s$. Figure 2.5 shows the characteristic: a 45-degree ramp that saturates, then drops to the kinetic plateau. The *angle of repose* — the steepest angle before slipping — sets $\tan\theta_{\max} = \mu_s$, a fast way to measure μ_s with a protractor and patience.

Checkpoint 2.1 *Friction direction is derived, not guessed.* Two stacked blocks, the bottom one pushed: which way does friction act on the *top* block? If you said "opposes motion", you've guessed — friction on the top block points *forward*: it's the only horizontal force on it, and it's what accelerates it.

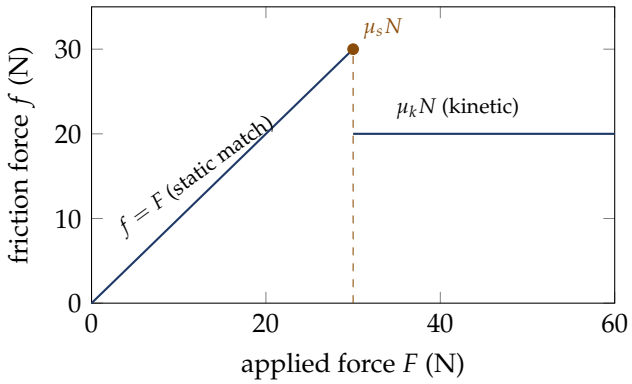


Figure 2.5: Static friction matches, saturates at $\mu_s N$, then drops to the kinetic level $\mu_k N$. Notice the discontinuity: "friction is weaker once you're moving" is why antilock brakes exist.

Example 2.4 (Conical pendulum)**.** A 0.5 kg mass on a 1.2 m string swings in a horizontal circle at 30° from vertical. Find its speed and period.

Strategy. FBD has mg and T only; T's horizontal component is the centripetal supply. Radius is $L\sin\theta$, not L.

Solution. Vertical: $T\cos\theta = mg \Rightarrow T = 5.66\,\text{N}$. Horizontal: $T\sin\theta = mv^2/r$ with $r = 1.2\sin 30^\circ = 0.6\,\text{m}$: $v = \sqrt{Tr\sin\theta/m} = \sqrt{5.66\times 0.6\times 0.5/0.5} = 1.84\,\text{m/s}$. Period $\tau = 2\pi r/v = 2.05\,\text{s}$ — independent of the mass, a Galileo-clock property. ✓

2.7 Drag: terminal velocity as a fixed point

Drag opposes velocity with a strength that grows with speed: cv (linear, small spheres at low Reynolds number) or cv^2 (quadratic, everything macroscopic in air or water). With quadratic drag,

$$m\dot{v} = mg - c_d v^2$$

is an ODE with a *fixed point*: at $v_t = \sqrt{mg/c_d}$ the right side vanishes and the fall settles. The closed form

$$v(t) = v_t \tanh\left(\frac{gt}{v_t}\right)$$

reaches 76% of v_t at $t = v_t/g$ — for an 80-kg skydiver with $c_d = 0.30$, $v_t = 51.1\,\text{m/s}$, the answer a real dive converges to.

Example 2.5 (Deriving $v(t)$ for quadratic drag)**.** Separate variables in $m\dot{v} = mg - c_d v^2$ and integrate.

Strategy. Write it as $\frac{dv}{g - cv^2/m} = dt$ and substitute $v_t^2 = mg/c_d$ so the denominator is $g(1 - v^2/v_t^2)$ — a standard form giving artanh.

Solution.

$$\int \frac{dv}{1 - v^2/v_t^2} = g\int dt \;\Rightarrow\; v_t\,\text{artanh}\,\frac{v}{v_t} = gt \;\Rightarrow\; v(t) = v_t\tanh\frac{gt}{v_t}. \quad \checkmark$$

The listing below confirms the derivation numerically — Euler-Cromer converges to the analytic curve.

```
def eulercromer(cd_over_m, dt, t_end):
    v, x, t = 0.0, 0.0, 0.0
    while t < t_end:
        v += (G - cd_over_m * v * v) * dt # update velocity first
        x += v * dt # then position with NEW v
        t += dt
    return v, x
```

Listing 2.1: `code/ch02_drag_fall.py` — Euler-Cromer on quadratic drag vs the tanh solution.

Output of a real run ($m = 80\,\text{kg}$, $c_d = 0.30\,\text{kg/m}$):

```
terminal velocity analytic: 51.1208 m/s
euler-cromer dt=0.5 v(20s)=51.0967 m/s err=+0.0237 fall=858.0 m
euler-cromer dt=0.1 v(20s)=51.0785 m/s err=+0.0055 fall=841.8 m
euler-cromer dt=0.01 v(20s)=51.0736 m/s err=+0.0006 fall=838.1 m
at t=vt/g=5.22s: v=38.9333 m/s = 76.2% of vt
v= 5 m/s: linear drag a=1.000 m/s^2, quadratic a= 0.094 m/s^2
v= 20 m/s: linear drag a=4.000 m/s^2, quadratic a= 1.500 m/s^2
v= 50 m/s: linear drag a=10.000 m/s^2, quadratic a= 9.375 m/s^2
```

Notice the linear/quadratic crossover in the last lines: linear drag dominates slow objects, quadratic dominates fast ones — a skydiver's v_t sits in the quadratic regime (figure 2.6).

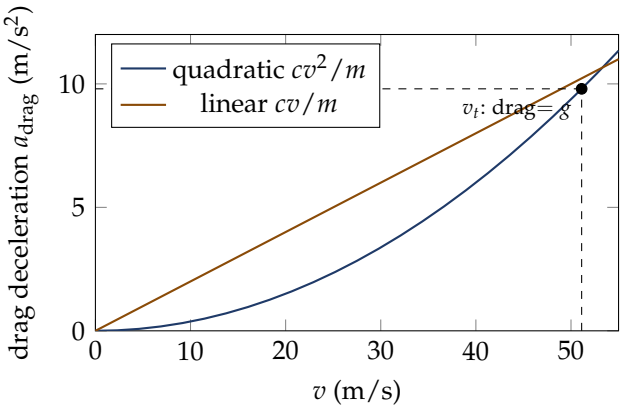


Figure 2.6: Drag deceleration vs speed for the two models. Terminal velocity is the fixed point where drag equals g (marked, $v_t = 51.1\,\text{m/s}$).

2.8 In engineering practice: load paths

Every load-bearing design — bridge truss, building frame, airframe — is a walk through this chapter's FBD chain: gravity enters at the top, travels through members as tension and compression, and exits into the ground. Structural engineers literally draw the discipline of section 2.3 as *free-body cuts*: isolate a joint, draw the member forces, solve $\sum F = 0$, repeat. The banked-curve analysis is the same trick highway designers run for every off-ramp, and the friction ceiling of figure 2.5 is why ABS brakes pump: holding the tire near μ_s, not sliding on the weaker μ_k plateau.

2.9 Common mistakes

mv^2/r DRAWN AS A FORCE

It is the *required net inward force*. If your FBD shows mg, N, *and* a centripetal arrow, double-count.

$N = mg$ ON ACCELERATING SURFACES

N is determined by the constraint, not by gravity. Elevator (example 2.2) and inclines (figure 2.3) both violate $N = mg$.

GUESSING THE FRICTION DIRECTION

Friction opposes *impending sliding at the contact* — which can be forward, as the checkpoint's stacked blocks show. Derive it from the constraint, never from intuition about "motion".

THIRD-LAW PAIRS ON ONE BODY

The pair acts on *different* bodies. "Normal cancels gravity" is not third-law bookkeeping — those two are both on the block.

COMPONENTS BEFORE THE FBD

Picking axes and decomposing first is backwards: enumerate the real forces on the real body, *then* choose axes that simplify the algebra.

2.10 Summary

Idea	One-line version
three laws	inertial frames; $\sum \mathbf{F} = m\mathbf{a}$; pairs on different bodies
FBD method	isolate, enumerate contacts, pick axes, write $\sum F$ per axis
inclines	$mg\sin\theta$ drives, $N = mg\cos\theta$ only when surface is still
constraints	rope length $\Rightarrow$ equal-a; contact $\Rightarrow N \geq 0$
circular dynamics	$\sum F_{\text{inward}} = mv^2/r$; centripetal is a role, not a force
friction	f_s matches up to $\mu_s N$; $f_k = \mu_k N$; $\tan\theta_{\text{repose}} = \mu_s$
drag	cv or cv^2; fixed point $v_t = \sqrt{mg/c_d}$, $v = v_t \tanh(gt/v_t)$

Table 2.1: Newton's laws cheat sheet.

DEFINITION OF DONE

You can now: draw a correct FBD for any of the five canonical cases; solve Atwood-style constraint problems; get the apparent weight in accelerating frames; apply $\sum F_{\text{inward}} = mv^2/r$ to curves and pendulums; predict friction direction from the constraint; and derive + numerically verify terminal velocity under quadratic drag.

Notes

The three-laws presentation and the FBD discipline follow MIT 8.01SC weeks 2–4 [2] and AP Physics C unit 2 [4]. OpenStax College Physics chapters 4–6 [3] cover the same material with a large worked set. The banked-curve and conical-pendulum examples follow Morin chapters 1–3 [7]; the friction characteristic and repose angle are standard [6]. Drag models and the tanh solution follow 8.01SC's resistive-media module; the skydiver numbers are representative, not measured.

WATCH INSTEAD

1. *Newton's Laws* — Walter Lewin, MIT 8.01x Lecture 6, 49:15, `https://youtu.be/oduZsA0Tk58` — §2.1–§2.5 at full length, same notation (accessed 2026-10-11).
2. *Types of forces and free body diagrams* — Khan Academy, 8:03, `https://youtu.be/MaabUHLIIXA` — the §2.3 discipline in eight minutes (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 2–4 — Newton's laws, FBDs, friction and drag with recitation solutions. Free at https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 4 "Dynamics" — free text, a second pass at the same laws with more worked FBDs. At https://openstax.org, search "College Physics 2e" (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 2 — the exam blueprint for this chapter's FR style. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 1–3 — FBDs, statics and accelerated frames at the level this book is aiming for; ISBN 978-0521876223.
- The Physics Classroom, *Newton's Laws* tutorial — readable remediation for §2.1–§2.6, at https://www.physicsclassroom.com/class/newtlaws (verified 2026-10-11).
- Khan Academy, *Forces and Newton's laws of motion* — drills matching every section here, at https://www.khanacademy.org/science/physics (verified 2026-10-11).

Additional exercises

Exercise 2.3 (easy). Which of these belongs on a block-on-table FBD: (a) gravity on the block, (b) the block's pull on the table, (c) friction on the block, (d) the block's weight on the table. Justify each keep/drop. (AP-C)

Exercise 2.4 (easy). When does $N = mg\cos\theta$ hold on an incline? Give a case where it fails. (AP-C)

Exercise 2.5 (easy). A 70 kg engineer weighs "70 kg" on Earth. Convert to SI force; then compute what a scale reads on the Moon ($g = 1.6\,\mathrm{m/s^2}$) and inside an elevator accelerating down at $3\,\mathrm{m/s^2}$. (AP-C)

Exercise 2.6 (easy). A rope holds a 10 kg mass static. Find the tension; find it again while the mass accelerates up at $2\,\mathrm{m/s^2}$. (F=ma)

Exercise 2.7 (easy). A car's brakes can produce deceleration $0.8g$ on dry pavement ($\mu_k = 0.8$). Estimate μ_k on ice if the same car needs $5\times$ the dry distance to stop. (AP-C)

Exercise 2.8 (medium). Two stacked blocks: $m = 2\,\mathrm{kg}$ on $M = 5\,\mathrm{kg}$, frictionless floor, $\mu_s = 0.4$ between them. What is the largest horizontal force on M for which m does not slip? (F=ma)

Exercise 2.9 (medium). Conical pendulum redux: keep L and θ of example 2.4 but double the mass. Which of v, T, τ change, and how? (AP-C)

Exercise 2.10 (medium). Atwood on a table: $m_1 = 2\,\mathrm{kg}$ on a frictionless table linked over a pulley to hanging $m_2 = 3\,\mathrm{kg}$. Find a and T; repeat with $\mu_k = 0.2$ on the table. (AP-C)

Exercise 2.11 (medium). The elevator of example 2.2 accelerates at $+2\,\mathrm{m/s^2}$ for 3 s, cruises, then brakes at $-3\,\mathrm{m/s^2}$ to rest. Sketch the scale reading vs. time, and compute the peak reading. (F=ma)

Exercise 2.12 (medium)**.** Show that on the banked curve of example 2.3, the minimum speed with friction is $v_{\min}^2 = rg\,\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta}$, and evaluate it. What happens when $\mu_s \geq \tan\theta$? (F=ma)

Exercise 2.13 (medium)**.** A skydiver with $c_d/m = 0.004\,\mathrm{m}^{-1}$ reaches v_t; estimate v_t, the time to 90% v_t, and the distance fallen by then. (APhO-style)

Exercise 2.14 (hard)**.** A rope of mass m and length L lies on a frictionless table with a length x hanging over the edge. Show its acceleration is $a = gx/L$ and find $x(t)$ — the classic variable-length element problem. (USAPhO-style)

Exercise 2.15 (hard)**.** A wedge of mass M and angle θ is free to slide on a frictionless floor; a block m slides on its frictionless face. Set up the four equations (two bodies, two axes + constraint) and solve for both accelerations. (APhO-style)

Exercise 2.16 (hard)**.** Estimate the maximum range of a projectile under quadratic drag: linearise for small $c_d v_0^2/(mg)$ and show the leading correction to the no-drag range is $-\frac{4}{3}c_d v_0^3/(m^2 g^2)\cdot(\text{shape factor})$. Compare with the 36.6% loss the ch01 code printed at $k = 0.02$. (USAPhO-style)

Exercise 2.17 (medium)**.** Rederive the angle-of-repose rule: show that a block on an incline starts sliding exactly when $\tan\theta = \mu_s$, and use it to compute μ_s if sliding starts at 22°. (AP-C)

CHAPTER 3

Momentum, impulse and collisions

Force is what changes momentum: $\mathbf{F} = d\mathbf{p}/dt$ is Newton's second law in its original form, and the reason $\mathbf{p} = m\mathbf{v}$ is the right bookkeeping unit for a chapter about things that hit each other. Momentum is the first *conserved* quantity in the book — the thing that survives collisions unchanged — and the discipline of *when* it survives (the system-boundary rule) is the real lesson.

3.1 Momentum and impulse

$\mathbf{p} = m\mathbf{v}$ is a vector. Impulse is momentum's denomination of payment:

$$\mathbf{J} = \int \mathbf{F}\, dt = \Delta\mathbf{p}. \tag{3.1}$$

A cricket ball stopped in a glove and a foam wall deliver the same Δp — impulse spreads the payment over different times (figure 3.1), which is why crumple zones, airbags, and bent knees all buy *time*.

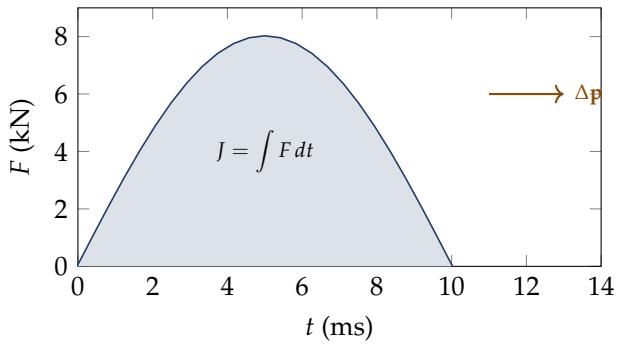


Figure 3.1: Impulse is area under the $F(t)$ curve. Notice the same shaded area can be a tall narrow spike or a long gentle hill — same Δp, very different peak force.

Example 3.1 (Impulse from a real $F(t)$ curve)**.** A 0.15 kg baseball hits a bat with the half-sine force profile $F(t) = 8000 \sin(\pi t/10\,\text{ms})$ N. Find the impulse and the exit speed if the ball arrived at 40 m/s.

Strategy. Area under the curve: the half-sine's average height is $2/\pi$ of peak over its 10 ms width.

Solution. $J = \int_0^{10} F\, dt = \frac{2}{\pi} \times 8000 \times 0.010 = 50.9\,\text{N s}$. Reversing the ball needs $\Delta p = 2mv = 2 \times 0.15 \times 40 = 12\,\text{N s}$ — the bat over-supplies: exit speed satisfies $m(v_{\text{out}} + 40) = 50.9$,

giving $v_{\text{out}} = 299\,\text{m/s}$. (Real bats don't: much of J deforms the ball, so effective impulse is lower — but the method is the point.) ✓

3.2 Systems: the boundary is a choice

Internal forces come in third-law pairs and cancel inside the sum; only *external* forces move the total momentum. Drawing the dashed line of figure 3.2 is a *decision*: put both blocks inside and the contact force vanishes from the bookkeeping; put it outside and the same force is external and changes $\mathbf{p}$.

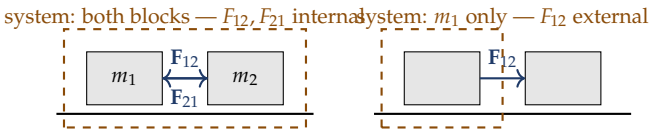


Figure 3.2: The same collision under two system choices. Internal forces cancel by the third law; external ones change the system's $\mathbf{p}$.

3.3 Conservation of momentum: the applicability rule

MOMENTUM CONSERVATION CONDITION

$\sum \mathbf{p}$ is conserved for a system exactly when the *net external impulse* during the event is zero. Gravity during a 10 ms collision contributes $\sim 10^{-3}\,\text{N s}$ — negligible. Gravity during a 10 s flight is not.

This is why "momentum is conserved in collisions" has the silent rider "while the collision lasts". The ballistic pendulum's swing, the projectile's arc — neither conserves $\mathbf{p}$; the *impact* does.

3.4 Centre of mass: the system's average representative

$$\mathbf{r}_{\text{cm}} = \frac{1}{M}\sum_i m_i \mathbf{r}_i, \qquad M\mathbf{A}_{\text{cm}} = \mathbf{F}_{\text{ext}}.$$

The CM obeys single-particle mechanics under the external force alone — internal explosions, splits, and collisions can't move it (figure 3.3). A wrench thrown across the room tumbles in a mess; its CM traces the clean parabola of chapter 1.

Example 3.2 (Person on a movable cart)**.** A 60 kg person stands on a 20 kg cart (wheels frictionless) and walks 3 m forward along the cart. How far does the cart move under them?

Strategy. No external horizontal force $\Rightarrow$ the CM of person+cart doesn't move. The person moves 3 m relative to the *cart*, not the ground — the classic mistake.

Solution. Let the cart move $-x$ (backward) while the person moves $3 - x$ forward on the ground. CM fixed: $60(3-x) + 20(-x) = 0 \Rightarrow 180 = 80x \Rightarrow x = 2.25\,\text{m}$. The cart slides back

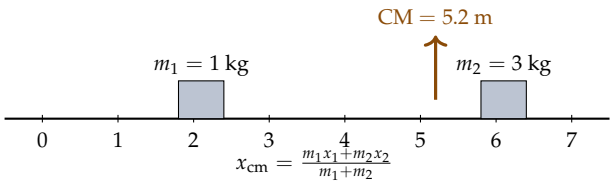


Figure 3.3: CM of two masses: x_{cm} weights each position by mass. Notice it sits nearer the heavy block — CM is the mass-weighted average, not the midpoint.

2.25 m; the person gains only 0.75 m of ground. This is why stepping off a light boat onto a dock is hazardous. ✓

Exercise 3.1 (easy)**.** Redo example 3.2 with a 90 kg person and a 10 kg cart — does the cart move more or less? Compute it. (AP-C)

3.5 1-D collisions: the two canonical cases

Perfectly inelastic (they stick): momentum alone fixes the answer — the shared speed is total momentum over total mass, $v' = (m_1 v_1 + m_2 v_2)/(m_1 + m_2)$ — and K drops by the fraction that went into deformation. **Elastic**: momentum plus energy conservation (or equivalently, relative velocity reverses). The full elastic solution needs K, so §4.7 of the next chapter closes it; this chapter keeps the momentum half. Figure 3.4 shows the three phases — and that the force between the blocks is internal either way.

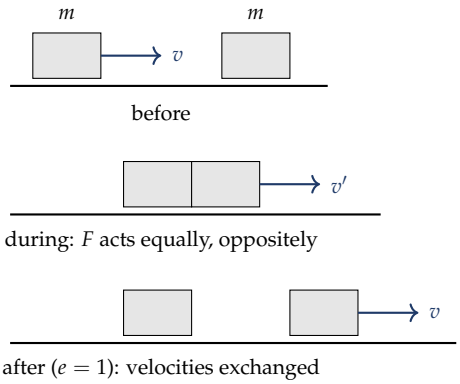


Figure 3.4: Collision sequence. The interaction force is internal to the system whatever the elasticity — p survives, K doesn't always.

Checkpoint 3.1 *Quick K audit.* Equal masses, one moving: perfectly inelastic collision loses exactly half the kinetic energy. Run the numbers yourself before the code below does.

3.6 2-D collisions and explosions

In two dimensions, momentum conservation is two scalar equations — x and y separately (figure 3.5). Explosions are collisions run backward: a stationary object splitting into fragments has $\sum \mathbf{p} = 0$ after as before, so the pieces fly apart with their momenta summing to zero — heavier fragments go slower.

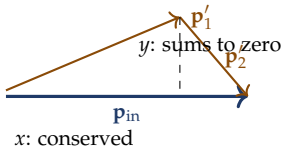


Figure 3.5: A 2-D collision as a vector triangle: $\mathbf{p}_{\text{in}} = \mathbf{p}_1' + \mathbf{p}_2'$. Notice the y-components must cancel — perpendicular momentum can't appear from nothing.

Example 3.3 (Glancing collision)**.** A puck at 6 m/s strikes an identical stationary puck and leaves at 4 m/s, 30° above the original direction. Find the second puck's velocity.

Strategy. Equal masses: velocity conservation is vector conservation. Components: x must return 6, y must net 0.

Solution. x: $4\cos 30^\circ + v_{2x} = 6 \Rightarrow v_{2x} = 6 - 3.464 = 2.54\,\text{m/s}$. y: $4\sin 30^\circ + v_{2y} = 0 \Rightarrow v_{2y} = -2\,\text{m/s}$. $|\mathbf{v}_2| = \sqrt{2.54^2 + 4} = 3.2\,\text{m/s}$ at -38.2°. The angle sum $30 + 38.2 \approx 68^\circ$ — not the elastic equal-mass 90°, so this collision lost energy. ✓

Exercise 3.2 (medium)**.** A 4 kg firework at rest explodes into three pieces: 1 kg north at 12 m/s, 1.5 kg east at 8 m/s. Find the third piece's velocity. (F=ma)

3.7 Continuous mass transfer: the rocket equation

When mass leaves continuously, momentum bookkeeping over one ejection (figure 3.6) gives

$$0 = (m - \Delta m)\,\Delta v - u\,\Delta m \xrightarrow{\Delta m \to 0} m\,dv = -u\,dm \;\Rightarrow\; \Delta v = u \ln \frac{m_0}{m_f},$$

the *Tsiolkovsky rocket equation* — u the exhaust speed relative to the rocket. The logarithm is why rockets are mostly fuel tank: each equal Δv costs a multiplicative mass ratio.

Example 3.4 (Staging estimate)**.** A rocket exhausts at $u = 3\,\text{km/s}$. (a) Single stage to orbit ($\Delta v \approx 9\,\text{km/s}$): mass ratio? (b) Two equal stages?

Strategy. $m_0/m_f = e^{\Delta v/u}$ each stage; stages multiply.

Solution. (a) $e^{9/3} = e^3 \approx 20$: the rocket is 95% propellant. (b) Each stage needs $\Delta v = 4.5\,\text{km/s}$: $e^{1.5} \approx 4.48$ per stage — payload structure is dramatically better. This arithmetic, not optimism, is why real rockets stage. ✓

Example 3.5 (Water-jet thrust)**.** A fire hose delivers 20 L/s of water ($\rho = 1\,\text{kg/L}$) at 15 m/s. What force does the jet exert on a wall it splashes against?

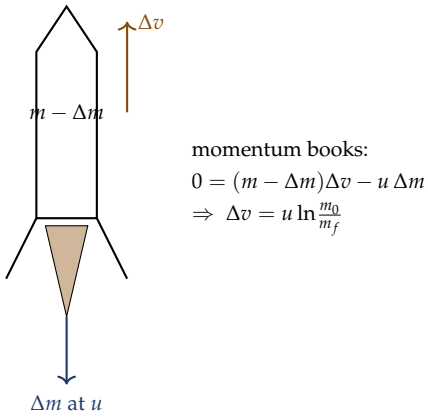


Figure 3.6: One ejection event: throw Δm backward at speed u and conservation of **p** pushes the rest forward.

Strategy. Momentum flux: force equals mass flow rate times the speed destroyed, $F = \dot{m}v$.

Solution. $F = (20\,\mathrm{kg/s})(15\,\mathrm{m/s}) = 300\,\mathrm{N}$ — a refrigerator's weight, and the reason hoses need two operators. The same $\dot{m}v$ formula gives a rocket's thrust before the logarithm does the integration. ✓

```
def collide(m1, v1, m2, v2, e):
    v1p = (m1*v1 + m2*v2 - m2*e*(v1-v2)) / (m1+m2)
    v2p = (m1*v1 + m2*v2 + m1*e*(v1-v2)) / (m1+m2)
    return v1p, v2p
```

Listing 3.1: `code/ch03_collision_chain.py` — event-stepped collisions: p survives, K audits the loss.

Output of a real run:

```
initial: p= 4.000 kg m/s K= 8.000 J
after e=0 (1kg+2kg): p= 4.000 kg m/s K= 2.667 J (K lost 5.333 J =
    66.7%)
after e=1 (3kg+3kg): p= 4.000 kg m/s K= 2.667 J v1'=+0.000 v2'=+1.333
e=0.0: va'=+2.000 vb'=+2.000 K=8.000 J (lost 50.0%)
e=0.5: va'=+1.000 vb'=+3.000 K=10.000 J (lost 37.5%)
e=1.0: va'=+0.000 vb'=+4.000 K=16.000 J (lost 0.0%)
```

Read it in order: momentum is conserved to machine precision in every line; kinetic energy is not (the 66.7% loss of the perfectly inelastic hit is the chapter's promise made numeric). The elastic equal-mass exchange ($e = 1$: velocities swap) and the intermediate loss at $e = 0.5$ complete the restitution family.

3.8 In engineering practice: the crash budget

Vehicle crash engineering is impulse management: the occupant's Δp is fixed by the speed, so design maximises collision *time* — crumple zones, airbags, and collapsible steering columns all stretch Δt to shrink F. The same bookkeeping runs in reverse for spacecraft: a Falcon 9's landing burn spends $\dot{m}v$ of thrust exactly as example 3.5 computes, and mass-ratio arithmetic of example 3.4 is why propellant dominates every launch manifest.

3.9 Common mistakes

CONSERVING p WHEN AN EXTERNAL IMPULSE EXISTS

Momentum conservation needs *zero external impulse for the duration* — gravity during a long swing, friction during a long skid, and the dock during a boat step-off all break it.

WRONG SYSTEM BOUNDARY

If the fragment you need sits outside your dashed line, conservation "fails" — draw the boundary to include every piece of the event.

SIGN ERRORS IN 1-D

$\mathbf{p}$ is signed. A rebound at $-v$ contributes $\Delta p = 2mv$, not zero.

BELIEVING K SURVIVES INELASTIC COLLISIONS

Sticking is maximally lossy — the equal-mass case loses half. K audits, $\mathbf{p}$ conserves.

THINKING THE CM PAUSES FOR EXPLOSIONS

The CM keeps its velocity through the breakup — the fragments' weighted sum never moved.

3.10 Summary

DEFINITION OF DONE

You can now: compute impulse from a force–time curve; choose a system boundary and state when $\mathbf{p}$ is conserved; move a cart under a walking person without moving the CM; solve inelastic and 2-D collisions; derive the rocket equation; and read a restitution sweep's K losses.

Idea	One-line version
impulse	$\mathbf{J} = \int F\,dt = \Delta\mathbf{p}$ — area under $F(t)$
boundary rule	internal forces cancel; $\mathbf{p}$ conserved iff external impulse $= 0$
centre of mass	$M\mathbf{A}_{\text{cm}} = \mathbf{F}_{\text{ext}}$; internal events can't move it
collisions 1-D	inelastic: p only; elastic: $p + K$ (§4.7)
collisions 2-D	conserve x and y separately
rocket	$\Delta v = u \ln(m_0/m_f)$ — staging is arithmetic
thrust	$F = \dot{m}v$ for continuous flux

Table 3.1: Momentum cheat sheet.

Notes

The momentum-first presentation follows MIT 8.01SC weeks 3 and 5–6 [2] and AP Physics C unit 4 [4]. OpenStax College Physics ch. 8–9 [3] covers collisions and CM at this level; Morin ch. 5 [7] is the source of the rocket staging and variable-mass exercises. The restitution-sweep numbers are the listing's real output; the baseball numbers are representative.

WATCH INSTEAD

1. *Momentum, Conservation of Momentum, Center of Mass* — Walter Lewin, MIT 8.01x Lecture 15, 52:17, https://youtu.be/aIhScO3_I50 — §3.1–§3.4 at full length (accessed 2026-10-11).
2. *Elastic and inelastic collisions* — Khan Academy Physics, 14:24, https://youtu.be/8OB8eIPgEkQ — §3.5–§3.6 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 3, 5–6 — momentum and collisions with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 8 "Linear Momentum and Collisions" — free text, the same laws with more worked explosions. At https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 4 — the exam blueprint this chapter aligns to. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, ch. 5 — momentum and rockets at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Momentum and Its Conservation* — readable remediation for the whole chapter, at https://www.physicsclassroom.com/class/momentum (verified 2026-10-11).

- Khan Academy, *Impacts and linear momentum* — drills matching §3.1–§3.6, at `https://www.khanacademy.org/science/physics` (verified 2026-10-11).

Additional exercises

Exercise 3.3 (easy)**.** A constant 500 N force acts for 0.2 s on a 2 kg mass initially at rest. Find the impulse and final velocity. (AP-C)

Exercise 3.4 (easy)**.** A 1200 kg car moving at 15 m/s is stopped by an identical parked car they stick to. Find the shared velocity. (F=ma)

Exercise 3.5 (easy)**.** Find the CM of three masses: 1 kg at $x = 0$, 2 kg at $x = 3$, 3 kg at $x = 6$. (AP-C)

Exercise 3.6 (easy)**.** Explain why momentum is *not* conserved while a thrown ball is in the air, in one sentence using the boundary rule. (AP-C)

Exercise 3.7 (easy)**.** A cannon fires a 5 kg shell at 400 m/s; the cannon recoils at 2 m/s. Find the cannon's mass. (F=ma)

Exercise 3.8 (medium)**.** Ballistic pendulum, momentum stage only: a 10 g bullet at v stops inside a 2 kg block, which then holds together. Find the block's post-impact speed if $v = 300$ m/s. (The swing is ch4's problem.) (AP-C)

Exercise 3.9 (medium)**.** From the impulse graph of figure 3.1: if the same J were delivered as a *constant* force over 20 ms, what would F be? How does the peak change? (F=ma)

Exercise 3.10 (medium)**.** Water jet on a bucket: the hose of example 3.5 hits a bucket moving away at 5 m/s. Redo the thrust using relative flux. (USAPhO-lite)

Exercise 3.11 (medium)**.** A 70 kg astronaut throws a 2 kg wrench at 6 m/s while floating. Find her recoil, then her drift over the 30 s EVA repair. (F=ma)

Exercise 3.12 (medium)**.** In the CM frame: transform the equal-mass elastic collision of the code listing into the CM frame, show the velocities merely *reverse sign*, and transform back. (F=ma)

Exercise 3.13 (medium)**.** Two pucks collide on air hockey as in example 3.3, but now the second puck emerges at 45° below the original direction with speed 3 m/s. Recompute puck 1's exit. (AP-C)

Exercise 3.14 (medium)**.** A rocket has $u = 2.5$ km/s and needs $\Delta v = 6$ km/s in one stage. Find m_0/m_f, then the propellant fraction. (F=ma)

Exercise 3.15 (hard)**.** A uniform chain of length L and mass m hangs with its lower end just touching a table and is released. Show that when a length x has landed, the table's reaction is $R = \frac{3mx}{L}g$ — the flux term triples the weight. (APhO-style)

Exercise 3.16 (hard)**.** Two-stage optimisation: with total propellant fraction fixed at 90% and $u = 3$ km/s, sketch how splitting it 40/50 vs 60/30 between stages changes total Δv — set up, don't solve, the constrained optimum. (USAPhO-style)

Exercise 3.17 (medium)**.** Superball bounce: a ball hits the floor at v and rebounds with restitution $e = 0.9$. Find the impulse delivered and interpret the loss. (F=ma)

CHAPTER 4

Work, energy and power

Force is local; energy is the accounting trick that lets you skip the details. $W = \int \mathbf{F} \cdot d\mathbf{r}$ accumulates what forces deposit, $K = \frac{1}{2}mv^2$ is the currency it arrives in, and potential energy is the bank where conservative forces store it. The chapter's payoff is the right to answer "how fast" without ever solving for the trajectory.

4.1 Work: force collected along a path

$$W = \int \mathbf{F} \cdot d\mathbf{r}. \tag{4.1}$$

Work is a *path integral*: only the force component along the motion counts. The centripetal force of §1.5 does *no* work — always perpendicular, always at right angles to $d\mathbf{r}$, which is why uniform circular motion keeps its speed. Figure 4.1 is the one-dimensional version: area under $F(x)$.

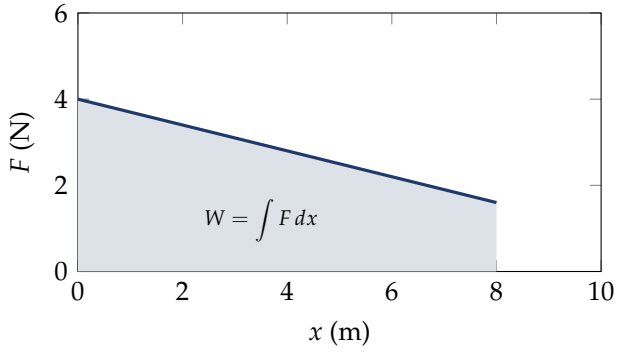


Figure 4.1: Work as area under the $F(x)$ curve. Notice variable forces need the integral; the constant-force $W = Fd\cos\theta$ is only its rectangle.

4.2 The work–kinetic-energy theorem

Start from $\mathbf{F} = m\,d\mathbf{v}/dt$ and dot with $d\mathbf{r}$:

$$\mathbf{F} \cdot d\mathbf{r} = m\frac{d\mathbf{v}}{dt} \cdot d\mathbf{r} = m\,\mathbf{v} \cdot d\mathbf{v} = d\left(\tfrac{1}{2}mv^2\right),$$

so

$$W_{\text{net}} = \Delta K. \tag{4.2}$$

Every joule of net work reappears exactly as kinetic energy — the theorem is a tautology dressed as a tool, and it works for any path, including ones too ugly to integrate force over directly.

4.3 Potential energy: where conservative forces bank work

A force is *conservative* when the work it does depends only on the endpoints. Then you can declare a stored quantity: $\Delta U = -W_{\text{by force}}$. The two workhorses:

$$U_g = mgy, \qquad U_s = \tfrac{1}{2}kx^2.$$

Friction fails the test — the longer the path, the more it takes — so there is no "potential energy of friction"; its work leaves the mechanical books entirely (section 4.8).

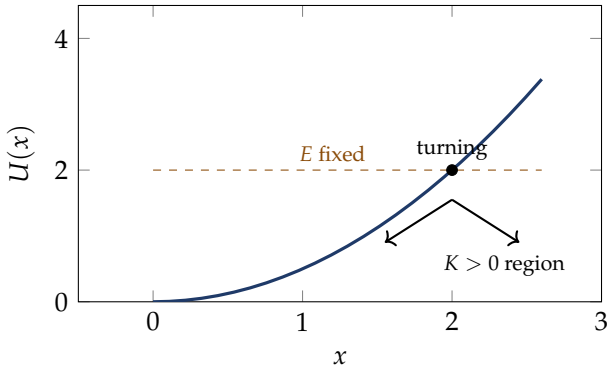


Figure 4.2: Reading a $U(x)$ diagram. With E fixed, motion lives where $K = E - U > 0$; the turning point is where the parabola meets the line. The restoring force is $-dU/dx$, the slope reversed.

4.4 Conservation of mechanical energy — and its failure modes

When only conservative forces work,

$$K_i + U_i = K_f + U_f. \tag{4.3}$$

Figure 4.3 is the account statement: the bars trade height, never total. The failure modes are equally instructive: friction converts, deformation absorbs, and anything with a motor adds from outside — all of them break equation (4.3), all handled by section 4.8's audit.

Example 4.1 (Minimum release height for the loop-the-loop)**.** A cart starts from rest at height h on a frictionless track containing a loop of radius r (figure 13.3). Minimum h to make it around?

Strategy. Two chapters meet: energy sets speeds (equation (4.3)), circular dynamics (ch2) sets the requirement — at the top, $N \geq 0$ means $v^2 \geq gr$.

Solution. Top of loop is at height $2r$; borderline case has $v_{\text{top}}^2 = gr$. Energy: $mgh = mg(2r) + \tfrac{1}{2}m\,gr$, so

$$h = \frac{5}{2}r.$$

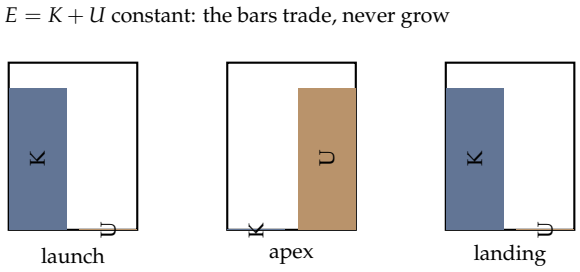


Figure 4.3: Energy bars along a trajectory: K and U trade, the sum doesn't move.

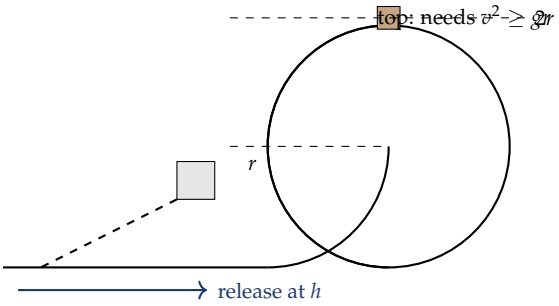


Figure 4.4: The loop-the-loop energy budget. The $5r/2$ answer is the classic meeting point of ch1's $a_c = v^2/r$ and this chapter's bookkeeping.

Real tracks need more — friction and the cart's own rotation (ch5) both take a share. ✓

Example 4.2 (Spring-launched block up a rough incline)**.** A spring ($k = 200\,\text{N/m}$) compressed 0.3 m launches a 0.5 kg block up a $\theta = 30^\circ$ incline with $\mu_k = 0.2$. How far along the slope does it go?

Strategy. Energy audit with friction present: spring's deposit minus friction's withdrawal equals the height gained.

Solution. $U_s = \frac{1}{2}(200)(0.09) = 9\,\text{J}$ available. Along distance d: gravity takes $mgd\sin 30^\circ = 2.45d\,\text{J}$, friction takes $\mu_k mg\cos 30^\circ \cdot d = 0.849d\,\text{J}$. So $d = 9/(2.45 + 0.849) = 2.73\,\text{m}$ — vs 3.67 m frictionless; friction's share is a quarter of the budget. ✓

Example 4.3 (Pendulum speed at the bottom)**.** A 0.5 kg mass on a 1 m string is released from horizontal. Speed at the bottom?

Strategy. Pure energy — the tension does no work (always perpendicular to the arc), so mechanical energy conserves.

Solution. $mgL = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gL} = \sqrt{19.6} = 4.43\,\text{m/s}$ — independent of mass and of the path taken to get there. The same $\sqrt{2gh}$ a frictionless slide gives: energy doesn't care about trajectories. ✓

Exercise 4.1 (easy)**.** Find the pendulum's speed when the string passes 45° from vertical. (AP-C)

4.5 Energy diagrams: motion read off a picture

Given $U(x)$ (figure 4.2): total energy E draws a horizontal line; kinetic energy is the gap below it; turning points are where the line crosses the curve; equilibria sit at the curve's extrema (stable at minima, unstable at maxima). A particle in a quartic double well can *never* leave one side if its line dips below the saddle — that is the whole qualitative solution, no ODE required.

4.6 Power: the rate of doing work

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}. \tag{4.4}$$

A 1000 kg car climbing a 6% grade at 20 m/s burns $P = mg \sin\theta \cdot v \approx 11.8$ kW on gravity alone — before drag's share, which figure 4.5 adds.

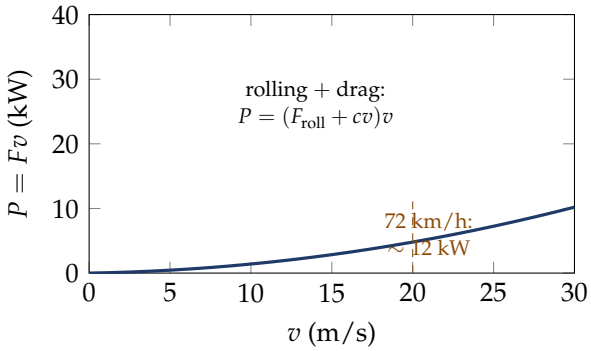


Figure 4.5: $P = \mathbf{F} \cdot \mathbf{v}$: climbing power is linear in v, drag power quadratic-ish. Notice how quickly the sum leaves the city-driving budget.

Example 4.4 (Stair-climb power measurement)**.** You (65 kg) run up a 4 m stair flight in 5 s. Estimate your mechanical power output.

Strategy. $P = \Delta U / t$ — the simplest honest measurement in the book.

Solution. $\Delta U = mgh = 65 \times 9.8 \times 4 = 2548$ J, so $P \approx 510$ W ≈ 0.7 hp. A cyclist sustains ~ 250 W; elite sprinters peak near 2 kW for seconds. ✓

Exercise 4.2 (medium)**.** A pump lifts 200 L/min of water 10 m. Minimum motor power? (F=ma)

4.7 Elastic collisions: the deferred full treatment

Chapter 3 owned momentum; energy completes the elastic case. The v_{rel}-flip shortcut: in an elastic collision the relative velocity *reverses* — $v_{2f} - v_{1f} = -(v_{2i} - v_{1i})$ — which with momentum alone solves everything.

Example 4.5 (Elastic collision, three ways)**.** $m_1 = 2\,\text{kg}$ at $6\,\text{m/s}$ hits stationary $m_2 = 2\,\text{kg}$. Solve by (a) lab-frame algebra, (b) the CM frame, (c) the flip rule.

Strategy. Three routes to the same answer — each teaches a different skill.

Solution. (a) $v_1' + v_2' = 6$ (momentum) and $v_1'^2 + v_2'^2 = 36$ (energy): substituting gives $v_1' v_2' = 0$, so one stops, one carries $6\,\text{m/s}$ — velocities swap. (b) CM moves at $3\,\text{m/s}$; in its frame the blocks approach at ± 3 and must leave at ∓ 3 (reversal is the only K-preserving option). Transforming back: 0 and 6. (c) Flip: $v_2' - v_1' = 6$ plus $v_1' + v_2' = 6$ gives $v_1' = 0$, $v_2' = 6$. All three agree; the equal-mass swap is figure 4.6. ✓

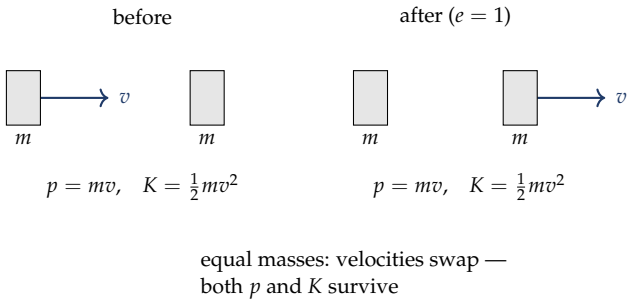


Figure 4.6: Bookkeeping for the equal-mass elastic hit: both p and K survive — a rare event, reserved for $e = 1$.

Example 4.6 (General elastic formula)**.** Apply the flip rule to unequal masses: $m_1 = 5\,\text{kg}$ at $8\,\text{m/s}$ hits $m_2 = 1\,\text{kg}$ at rest.

Strategy. Two equations: momentum and the reversed relative velocity.

Solution. $5v_1' + v_2' = 40$ and $v_2' - v_1' = 8$: subtracting, $6v_1' = 32 \Rightarrow v_1' = 5.33\,\text{m/s}$, $v_2' = 13.33\,\text{m/s}$. Check K: $\frac{1}{2}(5)(28.4) + \frac{1}{2}(1)(177.7) = 71 + 88.9 = 160\,\text{J} = \frac{1}{2}(5)(64)$. ✓

4.8 Non-conservative work: the energy audit

With friction present, equation (4.3) becomes the audit

$$K_i + U_i + W_{\text{nc}} = K_f + U_f, \qquad W_{\text{nc}} = -f_k d < 0 \text{ (usually)}.$$

The missing energy isn't gone — it's heat, sound, and deformation. The conveyor belt is the classic trap: dragging a block up to belt speed heats the belt-block interface by exactly the kinetic energy the block gains ($Q = K$) — half the motor's work becomes motion, half becomes warmth.

```
if scheme == "euler":
    x_new = x + v * dt
    v = v - (k/m) * x * dt # BOTH updates use OLD values
    x = x_new
else: # symplectic
    v += -(k/m) * x * dt # update v first
    x += v * dt # then x with the NEW v
```

Listing 4.1: `code/ch04_energy_check.py` — plain Euler vs symplectic Euler on a block-spring.

Output of a real run ($k = 4$, $m = 1$, released from $x = 1$):

```
E0 = 2.0000 J
scheme dt periods E_end drift
euler 0.1 10 446168.12 22308306.0%
euler 0.05 10 1034.79 51639.6%
symplectic 0.1 10 1.97 -1.5%
symplectic 0.05 10 2.00 0.1%
euler 100 periods, dt=0.1: E=446168.120 J (x 223084.1 of original --
    secular growth)
sympl 100 periods, dt=0.1: E=1.971 J (bounded oscillation, no secular
    drift)
```

The checkpoint's prediction is now a number: plain Euler's error compounds multiplicatively (the orbit spirals *out*), while symplectic Euler — update v before x — conserves a nearby shadow Hamiltonian and merely oscillates around the true energy. Every orbit simulation after this chapter uses the symplectic variant.

4.9 In engineering practice: the energy audit

Regenerative braking is this chapter's audit run as accounting: an electric car descending a hill deposits into the battery at 70–85% efficiency — not 100%, because the audit of section 4.8 still applies (aerodynamic drag, inverter losses, battery chemistry each take a line). Roller-coaster designers run example 4.1 backward: given a required top speed, the lift hill is $h = 5r/2$ plus an empirically tuned friction allowance — which is why first hills tower over every loop.

4.10 Common mistakes

SIGN OF WORK VS ΔU

$W_{\text{by}} = +\Delta K$; $\Delta U = -W_{\text{by conservative}}$. Mixing "work done on" with "work done by" flips every sign in the audit.

MIXING PE ZERO POINTS

$U = mgh$ measures against *your* chosen floor; two different floors in one problem corrupt the difference.

CONSERVING MECHANICAL ENERGY THROUGH FRICTION

Friction withdraws before gravity deposits — example 4.2 lost a quarter of its budget to it.

$P = Fv$ WITH THE WRONG v

The velocity in $P = \mathbf{F} \cdot \mathbf{v}$ is the *point of application's* — for a moving belt problem it's the belt, not the block.

"CONTACT FORCES DO NO WORK" ON DEFORMABLES

True for rigid bodies; false for cars, balls, and people — deformation absorbs real joules.

4.11 Summary

Idea	One-line version
work	$W = \int \mathbf{F} \cdot d\mathbf{r}$; only the along-motion part counts
WKE theorem	$W_{\text{net}} = \Delta K$ — derived, not assumed
potential energy	$\Delta U = -W_{\text{cons}}$; mgy and $\frac{1}{2}kx^2$
conservation	$K + U$ constant iff only conservative forces work
energy diagram	turning points where E meets U; equilibria at U' extrema
power	$P = \mathbf{F} \cdot \mathbf{v}$
elastic collisions	relative velocity flips; with p it solves everything
energy audit	W_{nc} bridges the books; friction $\Rightarrow -f_k d$

Table 4.1: Energy cheat sheet.

DEFINITION OF DONE

You can now: integrate variable forces; derive the work–energy theorem; read motion off $U(x)$; size a loop-the-loop hill; audit friction's share; solve elastic collisions three ways; and explain why symplectic integrators run the book's orbits.

Notes

The work–energy ordering follows MIT 8.01SC weeks 7–9 [2] and AP Physics C unit 3 [4]. OpenStax College Physics ch. 6–7 [3] has parallel treatment; Morin ch. 5 [7] is the source of the energy-diagram exercises. The integrator comparison is the standard motivation for symplectic schemes; loop-the-loop numbers are the classic result.

WATCH INSTEAD

1. *Work, Kinetic & Potential Energy, Gravitation, Conservative Forces* — Walter Lewin, MIT 8.01x Lecture 11, 49:06, https://youtu.be/9gUdDM6LZGo — §4.1–§4.4 at full length (accessed 2026-10-11).
2. *Work-energy theorem* — Khan Academy India, 12:34, https://youtu.be/nnMirzeTepQ — the §4.2 derivation worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 7–9 — work, energy and collisions with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 6–7 "Work, Energy, and Energy Resources" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 3 — the exam blueprint for this chapter. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, ch. 5 — energy methods at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Work, Energy, and Power* — readable remediation with roller-coaster walkthroughs, at https://www.physicsclassroom.com (verified 2026-10-11).
- Khan Academy, *Work and energy* — drills for this whole chapter, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 4.3 (easy)**.** Sign reasoning: classify the work of gravity, normal, and friction as $+$, $-$, or 0 for (a) a block sliding down an incline, (b) carried up the same incline at constant speed, (c) a satellite in circular orbit. (AP-C)

Exercise 4.4 (easy)**.** A crane lifts a 500 kg load at 0.5 m/s steadily. Find the motor's power. (AP-C)

Exercise 4.5 (easy)**.** A spring ($k = 400\,\mathrm{N/m}$) stores how much energy at 0.1 m compression? What mass could it lift 1 m with that energy? (AP-C)

Exercise 4.6 (easy)**.** Read figure 4.3: if the launch bar holds 8 J and the mass is 0.5 kg, find the launch speed and apex height. (F=ma)

Exercise 4.7 (easy)**.** A child slides down a 3 m frictionless slide. Find the exit speed, then explain why the slide's shape doesn't matter. (AP-C)

Exercise 4.8 (medium)**.** Loop minimum speed: recompute example 4.1 requiring $v_{\text{top}}^2 = 2gr$ (a safety margin) — what h now? (F=ma classic)

Exercise 4.9 (medium)**.** Spring + rough incline: example 4.2 again with $k = 300\,\text{N/m}$, $\mu_k = 0.3$, $\theta = 20^\circ$ — find d, then the block's speed back at the launch point after sliding down. (AP-C FR-style)

Exercise 4.10 (medium)**.** Stair power: have a partner time you on a real flight; compute your watts and compare against the 510 W estimate. (F=ma, data)

Exercise 4.11 (medium)**.** A pendulum released from 60° clears the bottom at what fraction of the 90°-release speed? Work it in symbols first. (AP-C)

Exercise 4.12 (medium)**.** Elastic collision: $m_1 = 3\,\text{kg}$ at $10\,\text{m/s}$ hits $m_2 = 1\,\text{kg}$ at rest — find both exit speeds by the flip rule and verify K survives. (AP-C)

Exercise 4.13 (medium)**.** Conveyor heating: a 2 kg package accelerates to a 3 m/s belt over a 1 m friction patch. Show the belt interface dissipates exactly the package's gained K, then compute both. (F=ma)

Exercise 4.14 (medium)**.** A 1500 kg car needs 25 kW to hold 30 m/s on a flat road. Estimate the combined drag + rolling resistance at that speed, then the power to hold 30 m/s up a 5% grade. (engineering)

Exercise 4.15 (hard)**.** A particle moves in $U(x) = ax^4 - bx^2$ ($a, b > 0$). Sketch U, find the equilibria and their stability, and the energy ranges where motion is trapped in one well vs. free between wells. (USAPhO-style)

Exercise 4.16 (hard)**.** Elastic collision, unequal masses: derive the general solution $v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1$, $v_2' = \frac{2m_1}{m_1 + m_2} v_1$, and verify the limits $m_2 \to \infty$ (wall) and $m_1 = m_2$ (swap). (USAPhO-style)

CHAPTER 5

Rotation I: kinematics, torque and statics

Everything so far treated bodies as points. Real objects have extent — they spin, they lean, they stay put under balanced torques. The good news is that rotation is a translation of every idea in the book so far: figure 5.1 is the dictionary, and it never lies.

linear	angular
x, v, a	θ, ω, α
m	$I = \sum mr^2$
$F = ma$	$\tau = I\alpha$

every linear law has an angular twin — substitute the dictionary, keep the algebra

Figure 5.1: The linear–angular dictionary. Every equation in this chapter is a ch1–ch2 equation with the entries swapped.

5.1 Angular variables

θ in radians, $\omega = \dot{\theta}$, $\alpha = \dot{\omega}$ — and the same constant-acceleration formulas of §1.3 carry over verbatim: $\omega = \omega_0 + \alpha t$, $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$. The rim variables are the bridge: $v = \omega r$, $a_{\tan} = \alpha r$.

5.2 Rolling kinematics: the no-slip constraint

A wheel rolling without slipping has $v_{\text{cm}} = \omega r$ — the contact point is instantaneously at rest while the top moves at $2v$ (figure 5.2). The constraint is *kinematic*, not energetic: static friction enforces it without doing work, which is why rolling races (ch6) are energy problems.

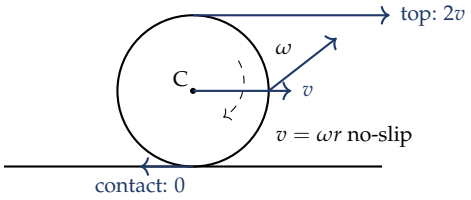


Figure 5.2: Velocity field of a rolling wheel: centre at v, top at $2v$, contact at 0. Notice the contact point is a usable pivot for one instant — the seed of the instantaneous-axis trick.

5.3 Torque: force's lever-arm price

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \qquad \tau = rF\sin\phi = \ell F, \tag{5.1}$$

where ℓ is the *lever arm*: the perpendicular distance from the pivot to the force's line of action (figure 5.3). Convention in this book: counterclockwise positive. Torque is the quantity the door handle sells — same force, longer ℓ, more angular acceleration.

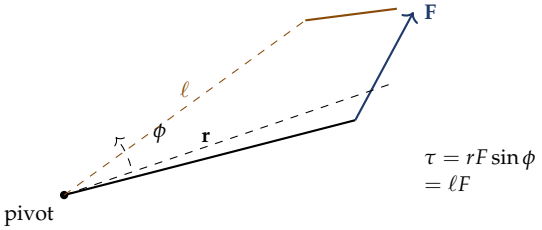


Figure 5.3: Torque geometry: $\tau = rF\sin\phi$, equivalently ℓF with the lever arm ℓ. Notice only the perpendicular component of **F** pays.

5.4 Moment of inertia: mass with an r^2 tax

$$I = \sum_i m_i r_i^2 \xrightarrow{\text{continuous}} I = \int r^2\, dm. \tag{5.2}$$

The r^2 is the whole story: mass near the axis is cheap, mass at the rim is expensive — a hoop resists twice as hard per kilogram as a disk. Two theorems do most of the work: the *parallel-axis* shift $I = I_{\text{cm}} + Md^2$, and the *perpendicular-axis* $I_z = I_x + I_y$ for flat laminas.

Example 5.1 (I of a rod and a disk by direct integration)**.** Compute I about the centre for (a) a uniform rod of length L, (b) a uniform disk of radius R.

Strategy. Slice into rings/strips where r is constant; the integral is bookkeeping, not calculus difficulty.

Solution. (a) Rod: $dm = \lambda dx$ with $\lambda = M/L$: $I = \int_{-L/2}^{L/2} x^2 \lambda\, dx = \lambda \frac{L^3}{12} = \frac{1}{12}ML^2$. (b) Disk: rings of mass $dm = \sigma\, 2\pi r\, dr$, all points at the same r: $I = \int_0^R r^2 \sigma 2\pi r\, dr = 2\pi\sigma R^4/4 = \frac{1}{2}MR^2$ using $M = \sigma\pi R^2$. ✓

shape	I about stated axis
rod, centre	$\frac{1}{12}ML^2$
disk/cylinder, axis	$\frac{1}{2}MR^2$
solid sphere, diameter	$\frac{2}{5}MR^2$

hoop: MR^2
sphere shell: $\frac{2}{3}MR^2$
rod, end: $\frac{1}{3}ML^2$

Figure 5.4: The six standard moments. Notice the coefficient tracks how far mass sits from the axis — the rod's 1/12, the disk's 1/2, the hoop's 1.

```
s += r*r * r*dr*dth # r^2 * dA, dA = r dr dth
```

Listing 5.1: `code/ch05_inertia_integrator.py` — grid quadrature vs analytic I; the parallel-axis term on display.

Output of a real run:

```
rectangle 2.0x1.0: numeric I=0.8333, exact=0.8333, err=0.00%
  per unit mass: I/M=0.4167 vs (a^2+b^2)/12=0.4167
disk R=1.0: numeric I=1.5708, exact=1.5708, err=0.00%
  per unit mass: I/M=0.5000 vs R^2/2=0.5000
disk about rim: I=4.7124 = I_cm + M R^2 (1.5708 + 3.1416); ratio to
    I_cm = 3.00
```

Notice the last line: spinning a disk about a point on its rim costs *three times* the centre-axis moment — the Md^2 term, not a numerical artifact.

5.5 $\tau = I\alpha$: the massive pulley

With a real pulley the rope's two sides can't share a tension — the difference is what spins the wheel (figure 5.5).

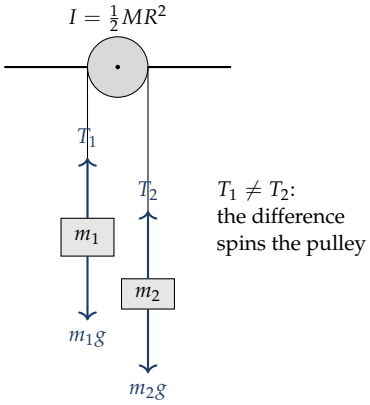


Figure 5.5: The massive pulley's three-body problem: two hanging FBDs plus $\sum \tau = I\alpha$ on the wheel. The tension *difference* does the spinning.

Example 5.2 (Atwood with a massive pulley)**.** Repeat example 2.1's Atwood ($m_1 = 3$, $m_2 = 5\,\text{kg}$) with a pulley of mass $M = 4\,\text{kg}$, radius R, modeled as a disk ($I = \frac{1}{2}MR^2$).

Strategy. Three equations: two translations plus one rotation. Constraint: $\alpha = a/R$ (no slip on the rim).

Solution. $m_2 g - T_2 = m_2 a$, $T_1 - m_1 g = m_1 a$, and $(T_2 - T_1)R = I\alpha = \frac{1}{2}MR^2 \cdot a/R \Rightarrow T_2 - T_1 = \frac{1}{2}Ma$. Adding all three:

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2 + M/2} = \frac{2 \times 9.8}{3 + 5 + 2} = 1.96\,\mathrm{m/s^2},$$

down from 2.45 — the pulley acts like an extra 2 kg of inertia. $T_1 = 35.3\,\mathrm{N}$, $T_2 = 39.2\,\mathrm{N}$: no longer equal, and their difference 3.9 N is exactly $\frac{1}{2}Ma$. ✓

Example 5.3 (Yo-yo unwinding)**.** A yo-yo (disk, $I = \frac{1}{2}mR^2$, string wrapped at radius r) falls unwinding. Find its acceleration.

Strategy. Two equations: $mg - T = ma$ and $Tr = I\alpha$ with $\alpha = a/r$.

Solution. $T = Ia/r^2 = \frac{1}{2}mR^2 a/r^2$, so $a = \frac{g}{1 + R^2/(2r^2)}$. For $R/r = 5$: $a = g/13.5 = 0.73\,\mathrm{m/s^2}$ — the rim-radius leverage makes it a slow fall. ✓

Exercise 5.1 (medium)**.** Find a for the yo-yo when it's a hoop ($I = mR^2$) instead. (F=ma)

5.6 Statics: two balances, one smart pivot

Static equilibrium needs $\sum \mathbf{F} = 0$ *and* $\sum \boldsymbol{\tau} = 0$ — and the pivot for the second equation is *yours to choose*: put it where an unknown force acts and that force disappears from the torque books. That single trick solves most of the chapter's problems.

Example 5.4 (Ladder minimum angle)**.** A uniform ladder of length L leans at angle θ against a frictionless wall (figure 5.6); the floor has μ_s. Minimum θ before slipping?

Strategy. $\sum F$: $N_2 = f$, $N_1 = mg$. Torque about the base (kills f and N_1): the wall's push at the top balances gravity at the middle.

Solution. $N_2 L \sin\theta = mg\frac{L}{2}\cos\theta \Rightarrow N_2 = \frac{mg}{2\tan\theta}$. Slip when $f = N_2 = \mu_s N_1 = \mu_s mg$: $\tan\theta_{\min} = \frac{1}{2\mu_s}$. For $\mu_s = 0.4$: $\theta_{\min} = 51.3^\circ$ — below that, the wall can't push hard enough. ✓

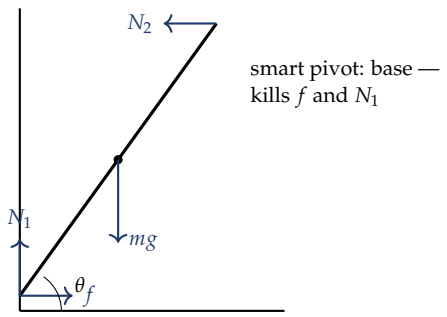


Figure 5.6: Ladder FBD: frictionless wall means N_2 is purely horizontal; taking torques about the base deletes both floor forces.

Exercise 5.2 (easy)**.** Where does a second person standing halfway up the ladder change the minimum angle? Set up the new torque equation. (AP-C)

Example 5.5 (Two-support beam)**.** A uniform 8 kg, 5 m beam rests on supports at $x = 1$ and $x = 4$ (metres from the left end), with a 20 kg toolbox at $x = 4.5$. Find each support's force.

Strategy. $\sum F$ gives the sum; torque about one support kills its unknown. Watch: the toolbox overhangs the right support.

Solution. $N_A + N_B = mg + Mg = 78.4 + 196 = 274.4\,\text{N}$. Torque about support A (CCW+): the beam's weight acts at $x = 2.5$ (1.5 m right of A), the toolbox at 3.5 m right, N_B at 3 m: $3N_B = 78.4 \times 1.5 + 196 \times 3.5 = 117.6 + 686$, so $N_B = 267.9\,\text{N}$ and $N_A = 6.5\,\text{N}$ — the left support nearly unloads: shift the toolbox a little further and the beam tips. ✓

5.7 Rotational kinetic energy

$K_{\text{rot}} = \frac{1}{2}I\omega^2$, and for rolling $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ — the reason a hoop loses every race down an incline: it pays the rotation tax with the largest per-mass I (chapter 6 races them properly).

5.8 In engineering practice: torque specs

Every bolt in a car has a printed torque spec — this chapter's $\tau = \ell F$ run as quality control: a 25 N m spec on a 0.25 m wrench means exactly 100 N of hand force, and overtightening stretches the bolt past its elastic range (book5's materials chapter has the yield story). The statics side is how cranes stay up: $\sum \tau = 0$ about the outrigger line is the tip-over condition, which is why load charts shrink with boom extension — longer ℓ, same budget.

5.9 Common mistakes

LEVER ARM VS POINT DISTANCE

$\tau = \ell F$: ℓ is perpendicular distance to the *line of action*, not to the point of application along the body.

FORGETTING THE PULLEY'S INERTIA

A massless pulley shares one T; a massive one doesn't — drop the $\frac{1}{2}Ma$ term and the answer is too fast.

WRONG PARALLEL-AXIS SHIFT

$I = I_{\text{cm}} + Md^2$ — it's about the CM plus a shift, never between two arbitrary axes.

BAD TORQUE ORIGIN

Every force through the origin drops out. Pick the pivot at the unknown you want gone, not at the centre because it looks official.

FORCE BALANCE ALONE IN STATICS

$\sum F = 0$ admits a whole family of torques; the ladder needs $\sum \tau = 0$ too.

5.10 Summary

Idea	One-line version
dictionary	$\theta/\omega/\alpha \leftrightarrow x/v/a$; same formulas
no-slip	$v_{\text{cm}} = \omega r$; contact instant rest, top $2v$
torque	$\tau = rF\sin\phi = \ell F$; CCW positive
moment of inertia	$I = \int r^2 dm$; $1/12$, $1/2$, $2/5$ coefficients
dynamics	$\tau = I\alpha$; massive pulleys need $T_1 \neq T_2$
statics	$\sum F = 0 + \sum \tau = 0$; pivot at the unknown
rotational K	$\frac{1}{2} I\omega^2$; rolling pays both

Table 5.1: Rotation-I cheat sheet.

DEFINITION OF DONE

You can now: run the linear–angular dictionary; read a rolling wheel's velocity field; compute lever arms and torques; integrate I for simple shapes and apply parallel axis; solve massive-pulley and yo-yo dynamics; and set up statics with the smart-pivot trick.

Notes

The dictionary-first approach follows MIT 8.01SC weeks 6 and 10–11 [2] and AP Physics C unit 5 [4]. OpenStax College Physics ch. 10, 12 [3] covers the same ground; the ladder and pulley problems are Morin chs. 7–8 [7] classics. The I-table coefficients are standard; the quadrature listing checks two of them numerically.

WATCH INSTEAD

1. *Rotating Objects, Moment of Inertia, Rotational KE, Neutron Stars* — Walter Lewin, MIT 8.01x Lecture 19, 41:00, `https://youtu.be/fDJeVR0o__w` — §5.4–§5.7 at full length (accessed 2026-10-11).

2. *Angular motion variables* — Khan Academy Physics, 12:50, https://youtu.be/garegCgMxxg — §5.1–§5.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 6, 10–11 — rotation and statics with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 10 "Rotational Motion" and ch. 12 "Statics" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 5 — the exam blueprint for rotation. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 7–8 — statics and rotation at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Rotational Motion* tutorials — readable remediation for §5.1–§5.6, at https://www.physicsclassroom.com (verified 2026-10-11).
- Khan Academy, *Moments, torque, and angular momentum* — drills matching the whole chapter, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 5.3 (easy). A wheel spins up to 1200 rpm in 4 s at constant α. Find α in rad/s^2 and the angle turned. (AP-C)

Exercise 5.4 (easy). Identify the lever arm: a 0.5 m wrench at 60° to a 100 N force — compute τ both as $rF\sin\phi$ and ℓF. (AP-C)

Exercise 5.5 (easy). Write down I for a 2 kg, 0.3 m-radius disk about its axis and about a rim point, citing the theorem used. (AP-C)

Exercise 5.6 (easy). A uniform beam of length 4 m is pinned at one end and carries a 200 N load at its far end. What vertical force at the midpoint holds it level (neglecting the beam's weight)? (F=ma)

Exercise 5.7 (easy). A see-saw: 30 kg child at 1.5 m left of the pivot balances a 50 kg adult. Where does the adult sit? (F=ma)

Exercise 5.8 (medium). Massive-pulley Atwood: recompute example 5.2 with a hoop pulley ($I = MR^2$) — which way does a move, and why? (F=ma classic)

Exercise 5.9 (medium). Redo example 5.4 with a wall that has $\mu_s' = 0.3$ of its own. Does the wall friction help or hurt? (AP-C FR-style)

Exercise 5.10 (medium). A yo-yo hoop ($I = mR^2$, string at r) falls unwinding — find a and compare with example 5.3. (F=ma)

Exercise 5.11 (medium)**.** Composite body: a 0.5 m rod of mass 2 kg has a 1 kg, 0.1 m-radius disk welded at one end (disk axis through the rod's end, perpendicular). Find I about the rod's other end. (AP-C)

Exercise 5.12 (medium)**.** Sign-of-torque reasoning: a door pushed at its top vs its middle, same force — describe the angular acceleration ratio without computing. (AP-C)

Exercise 5.13 (medium)**.** Rolling preview: a hoop and a disk race down an incline from rest — predict the winner using only the I coefficients, then verify with energy ($v^2 = \frac{2gh}{1+I/mr^2}$). (AP-C)

Exercise 5.14 (medium)**.** A uniform 8 kg beam 5 m long rests on two supports at $x = 1$ and $x = 4$ (metres). Find each reaction force. (F=ma)

Exercise 5.15 (hard)**.** A uniform rod leans on a frictionless floor against a frictionless wall and is released. Show it loses wall contact when $\sin\theta = \frac{2}{3}\sin\theta_0$ — use energy for speed and the CM's horizontal acceleration for the contact condition. (USAPhO-style)

Exercise 5.16 (hard)**.** A rod of length L has density $\lambda(x) = \lambda_0(1 + x/L)$. Find its I about the light end by direct integration, and its CM. (APhO-style)

Exercise 5.17 (medium)**.** A sphere and a cylinder of the same M, R roll (not slide) down the same incline. Using $v^2 = \frac{2gh}{1+I/mr^2}$, find the sphere's speed advantage over the cylinder's at the bottom as a percentage. (AP-C)

CHAPTER 6

Rotation II: angular momentum and rolling

Linear momentum had its chapter; angular momentum is its equal partner — the second conserved quantity of mechanics and the reason gyroscopes, planets, and skaters behave the way they do. The bookkeeping mirrors chapter 3 exactly, with one new subtlety: the origin you measure from matters.

6.1 Angular momentum: definition

For a particle,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{6.1}$$

— momentum filtered through the position relative to a chosen origin (figure 6.1). A straight-line mover carries nonzero $\mathbf{L}$ about any origin off its line — the quantity is about the *rotation of the line of sight*, not about actual circling.

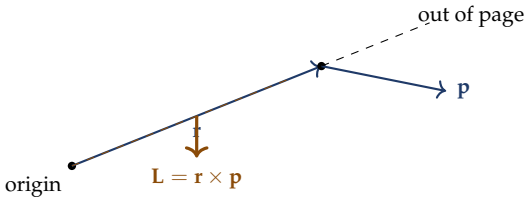


Figure 6.1: $\mathbf{L} = \mathbf{r} \times \mathbf{p}$: magnitude $rp\sin\theta$, direction by the right-hand rule (out of the page here). Notice it needs an origin — the same $\mathbf{p}$ has different $\mathbf{L}$ about different points.

6.2 Rigid bodies: when $L = I\omega$ holds

Summing (6.1) over a rigid body rotating about a fixed symmetry axis gives

$$L = I\omega.$$

The fine print: valid about the fixed axis *or* about the CM — the two origins where the cross terms vanish. A rolling wheel's $\mathbf{L}$ is $I\omega$ about its CM *plus* the orbital $M(\mathbf{R} \times \mathbf{V})$ of the CM — both exist, both count.

6.3 The rotational second law

$$\boldsymbol{\tau}_{\text{ext}} = \frac{d\mathbf{L}}{dt} \tag{6.2}$$

about an inertial fixed point or about the CM — the two origins where the derivation works. Choosing an accelerating origin is the classic trap: the equation picks up fictitious-torque terms that ruin it.

6.4 Conservation of angular momentum

ANGULAR MOMENTUM CONSERVATION

L is conserved when the net *external torque* is zero — weaker than zero external force. A planet feels a huge central force that exerts no torque, so its **L** (and Kepler's second law) survives.

Example 6.1 (The skater)**.** A skater spins at $\omega_0 = 2\,\text{rad/s}$ with $I_0 = 3\,\text{kg m}^2$, then pulls her arms to $I = 1\,\text{kg m}^2$. Find the new spin rate and the energy change.

Strategy. No external torque about the spin axis: L conserved. K is not — her arms do work.

Solution. $\omega = I_0\omega_0/I = 6\,\text{rad/s}$. Energies: $K_0 = \frac{1}{2}(3)(4) = 6\,\text{J}$, $K = \frac{1}{2}(1)(36) = 18\,\text{J}$ — K tripled. The extra 12 J came from her muscles pulling mass inward against the centrifugal pull: the audit of ch4 applies here too. ✓



$L = I\omega$ fixed; $K = \frac{1}{2}I\omega^2$ grows — the arms' work pays it

Figure 6.2: The skater trade: L fixed, K tripled — arms supply the difference.

Example 6.2 (Bullet into a pivoted rod)**.** A 10 g bullet at 400 m/s embeds in the tip of a 1 kg, 1 m rod pivoted at its top ($I = \frac{1}{3}ML^2$). Find the angular velocity just after.

Strategy. Linear momentum is *not* conserved (the pivot pushes); angular momentum about the pivot *is* (the pivot force has no lever arm). Use the collision instant only.

Solution. $L_i = mvL = 0.01 \times 400 \times 1 = 4\,\text{kg m}^2/\text{s}$. $I_{\text{tot}} = \frac{1}{3}(1)(1) + 0.01(1)^2 = 0.343\,\text{kg m}^2$. $\omega = 4/0.343 = 11.7\,\text{rad/s}$. K audit: before 800 J, after $\frac{1}{2}I\omega^2 = 23.4\,\text{J}$ — 97% lost to the embedding, as inelastic as collisions get. ✓

Exercise 6.1 (easy)**.** In example 6.2, which quantities survive: total **L** about the pivot, total **p**, total K? One-line each. (AP-C)

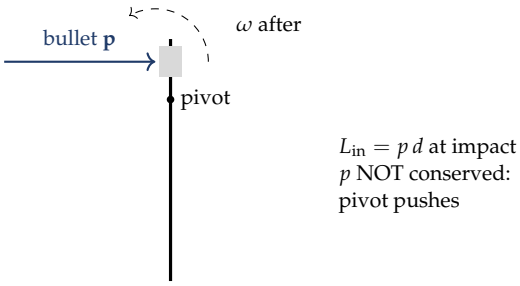


Figure 6.3: Bullet-and-rod: conserve L about the pivot, not p — the pivot's impulse is external to translation but torque-free about itself.

6.5 Rolling dynamics, three ways

A roller on an incline can be solved by forces and torque ($ma = mg\sin\theta - f$, $I\alpha = fr$), by energy ($v^2 = 2gh/(1+\beta)$ for $I = \beta mr^2$), or by the instantaneous-axis trick (torque about the contact point only). All three agree; the energy route is the fastest.

Example 6.3 (The incline race)**.** Hoop ($\beta = 1$), disk ($\frac{1}{2}$) and sphere ($\frac{2}{5}$) race down a 2 m drop. Order and margins?

Strategy. $v = \sqrt{2gh/(1+\beta)}$ — smaller β finishes first, independent of mass and radius.

Solution. $v_{sph} = 5.29$, $v_{disk} = 5.11$, $v_{hoop} = 4.43\,\mathrm{m/s}$ — the code listing below prints the same numbers. The hoop loses because *all* its mass sits at the rim: half its energy is rotation at every instant (figure 6.4). ✓

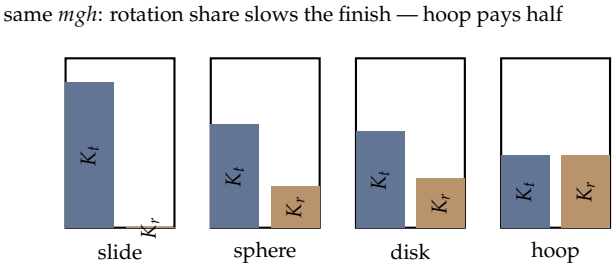


Figure 6.4: Energy partition in the race: same mgh, different splits — the hoop spends half on spinning.

```
v = math.sqrt(2*G*H/(1+beta))
```

Listing 6.1: `code/ch06_rolling_race.py` — the incline race for several shape factors.

Output of a real run:

```
incline drop h=2.0 m
shape beta v (m/s) K_tr K_rot
point mass (slide) 0.0 6.261 19.60 0.00
solid sphere 0.4 5.292 14.00 5.60
solid disk 0.5 5.112 13.07 6.53
hoop 1.0 4.427 9.80 9.80
```

```
finish order: point mass (slide) > solid sphere > solid disk > hoop
sphere beats hoop by 19.5% of hoop's speed
```

Notice the partition column: the hoop's K_{rot} equals its K_{tr} — $\beta = 1$ means rotation and translation split the budget evenly, and mass/radius never appear.

Example 6.4 (Two-disk angular collision)**.** A disk ($I_1 = 2\,\text{kg m}^2$) spinning at 10 rad/s drops onto a stationary disk ($I_2 = 3\,\text{kg m}^2$) on the same axle; friction locks them. Find the common ω and the energy lost.

Strategy. Angular-momentum conservation about the axle — the angular version of the perfectly inelastic collision.

Solution. $\omega = I_1\omega_0/(I_1 + I_2) = 20/5 = 4\,\text{rad/s}$. $K_i = \frac{1}{2}(2)(100) = 100\,\text{J}$; $K_f = \frac{1}{2}(5)(16) = 40\,\text{J}$ — 60% lost, the angular mirror of chapter 3's inelastic hit. ✓

Example 6.5 (Child on a turntable)**.** A 40 kg child stands at the rim of a 200 kg, 3 m-radius turntable ($I = \frac{1}{2}MR^2 = 900\,\text{kg m}^2$) spinning at $\omega_0 = 0.5\,\text{rad/s}$, then walks to the centre. Find the new ω.

Strategy. No external torque about the axis: $I\omega$ conserved, with the child as a point mass whose I shrinks.

Solution. $I_0 = 900 + 40(9) = 1260\,\text{kg m}^2$; at centre $I_1 = 900 + 0$. $\omega_1 = 0.5 \times 1260/900 = 0.70\,\text{rad/s}$ — the same skater effect with the child playing the arms. K rises from 158 to 220 J, paid by the child's steps. ✓

Exercise 6.2 (medium)**.** A 40 kg child walks from rim to centre on a 200 kg, 3 m-radius turntable ($I = \frac{1}{2}MR^2$) spinning at 0.5 rad/s. Find the new ω. (AP-C FR-style)

6.6 Gyroscopic precession: the order of magnitude

A spinning top tilted at angle θ has $\mathbf{L}$ along its axis; gravity's torque $\tau = mgd\sin\theta$ is perpendicular, so $\mathbf{L}$ *rotates* rather than tips (figure 6.5). The precession rate follows from $\Delta L = \tau\Delta t$ geometry:

$$\Omega = \frac{\tau}{L} = \frac{mgd}{I\omega}$$

— slower spin, faster wobble. That's why tops precess more as they wind down.

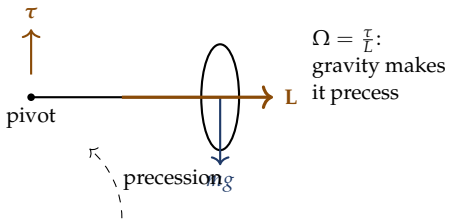


Figure 6.5: Precession bookkeeping: $\boldsymbol{\tau}$ perpendicular to $\mathbf{L}$ turns the vector instead of shrinking it. $\Omega = \tau/L$ sets the rate.

6.7 In engineering practice: reaction wheels and flywheels

Satellites steer with *reaction wheels* — small spun disks whose speed changes trade $\mathbf{L}$ with the bus, no fuel spent: this chapter's conservation law is the whole attitude-control budget. Flywheel energy storage is the same $K = \frac{1}{2}I\omega^2$ as grid batteries: a 500 kg carbon-fiber wheel at 20,000 rpm banks megajoules in rotation, which is exactly the hoop's tax of figure 6.4 run to its useful extreme — mass at the rim is where you want it.

6.8 Common mistakes

$L = I\omega$ ABOUT THE WRONG AXIS

Valid about the fixed axis or the CM only. A rolling wheel about an arbitrary point needs $I\omega + \mathbf{R} \times M\mathbf{V}$.

CONSERVING L THROUGH EXTERNAL TORQUE

The bullet-in-rod survives precisely because the pivot torque about the pivot is zero — any other origin, or any other external torque, breaks it.

FORGETTING TRANSLATIONAL K IN ROLLING

A rolling object carries $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ — drop the first term and every race comes out tied.

$\tau = dL/dt$ ABOUT A NON-INERTIAL ORIGIN

The law holds at inertial points and the CM. An accelerating origin invents torques that aren't there.

CROSS-PRODUCT SIGN

$\mathbf{r} \times \mathbf{p} \neq \mathbf{p} \times \mathbf{r}$ — the order flips the sign. Fix the convention once per problem.

6.9 Summary

DEFINITION OF DONE

You can now: compute $\mathbf{L}$ for particles and rigid bodies; state when conservation applies; solve skater, bullet-rod, and two-disk collisions; run the incline race three ways; and size gyroscopic precession to order of magnitude.

Idea	One-line version
definition	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$ — origin-dependent
rigid body	$L = I\omega$ about fixed axis or CM
dynamics	$\boldsymbol{\tau} = d\mathbf{L}/dt$ at inertial origin or CM
conservation	survives iff external torque $= 0$ — central forces OK
rolling	$v^2 = 2gh/(1+\beta)$; smaller β wins
skater	L fixed, K grows — muscles pay
gyroscope	$\Omega = \tau/L$; perpendicular torque turns, doesn't tip

Table 6.1: Angular momentum cheat sheet.

Notes

The treatment follows MIT 8.01SC weeks 11–12 plus its gyroscope module [2] and AP Physics C unit 6 [4]. Morin chs. 8–9 [7] is the source of the spool and sliding-to-rolling exercises; OpenStax ch. 11 [3] covers the same material. The race numbers are the listing's real output.

WATCH INSTEAD

1. *Angular Momentum, Torques, Conservation of Angular Momentum* — Walter Lewin, MIT 8.01x Lecture 20, 51:15, https://youtu.be/sNaaL19opxw — the whole chapter at full length (accessed 2026-10-11).
2. *Conservation of angular momentum* — Khan Academy, 7:16, https://youtu.be/vfmLI150g4w — §6.4 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 11–12 — angular momentum and rolling with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 10–11 — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 6 — the exam blueprint for angular momentum. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 8–9 — angular momentum and rolling at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Angular Momentum* tutorials — readable remediation, at https://www.physicsclassroom.com (verified 2026-10-11).
- Khan Academy, *Moments, torque, and angular momentum* — drills matching the whole chapter, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 6.3 (easy)**.** Find the magnitude and direction of $\mathbf{L}$ for a 2 kg mass at $\mathbf{r} = (3, 0, 0)$ m moving at $\mathbf{v} = (0, 4, 0)$ m/s. (AP-C)

Exercise 6.4 (easy)**.** Decide whether L is conserved in each: (a) planet orbiting the sun, (b) figure skater pulling in arms, (c) frisbee spinning while gliding to a stop, (d) the bullet-rod of example 6.2 *after* impact. (AP-C)

Exercise 6.5 (easy)**.** A rolling ball has $v = 5\,\text{m/s}$ and radius 0.1 m. Find ω and the contact point's instantaneous speed. (AP-C)

Exercise 6.6 (easy)**.** Explain in one sentence why the skater's K increases while L stays fixed. (F=ma)

Exercise 6.7 (easy)**.** A spinning disk has $I = 0.5\,\text{kg m}^2$, $\omega = 20\,\text{rad/s}$. Find L and K_{rot}. (AP-C)

Exercise 6.8 (medium)**.** A 40 kg child stands at the rim of a 150 kg, 4 m-radius turntable ($I = \frac{1}{2}MR^2$) at rest, then walks clockwise at 1 m/s relative to the ground. Which way and how fast does the table turn? (AP-C FR-style)

Exercise 6.9 (medium)**.** Rank these down a 3 m incline by finish order and compute each speed: solid sphere, hollow sphere ($\beta = 2/3$), disk, hoop. (F=ma)

Exercise 6.10 (medium)**.** Example 6.2 with a bullet hitting *halfway* down the rod: recompute ω and the energy loss. (AP-C)

Exercise 6.11 (medium)**.** Two angular disks merge: $I_1 = 1$, $\omega_1 = 12$ and $I_2 = 4$ spinning *oppositely* at $\omega_2 = -3$. Find the common ω and energy lost. (F=ma)

Exercise 6.12 (medium)**.** A 0.5 m wrench applies 40 N tangentially to a 0.02 m-radius shaft. Find the torque and, for a $0.1\,\text{kg m}^2$ flywheel, the ω after 5 s from rest. (AP-C)

Exercise 6.13 (medium)**.** A gyroscope with $I = 0.01\,\text{kg m}^2$, $\omega = 200\,\text{rad/s}$, $d = 0.1\,\text{m}$, $m = 0.5\,\text{kg}$: compute the precession rate in rad/s and rpm. (F=ma)

Exercise 6.14 (hard)**.** A bowling ball is set sliding with no spin at v_0 on a lane with μ_k. Show it transitions to pure rolling at $v = \frac{5}{7}v_0$ — track both v and ω as friction acts. (USAPhO-style)

Exercise 6.15 (hard)**.** A spool of thread lies on a table ($I = \beta mr^2$, thread wrapped at inner radius $r_i < R$). Pulling the thread horizontally: does it roll toward or away from you? Find the critical angle where the direction reverses. (APhO-style classic)

Exercise 6.16 (medium)**.** Energy split check: for the race of example 6.3, what fraction of each shape's kinetic energy is rotational? Show $\beta/(1+\beta)$. (AP-C)

Exercise 6.17 (medium)**.** A planet at perihelion (r_1, v_1) reaches aphelion at $2r_1$. Find its aphelion speed by L conservation, and explain why this is Kepler's second law. (F=ma)

CHAPTER 7

Oscillations

One equation runs this chapter: $\ddot{x} + \omega_0^2 x = 0$. Springs, pendulums, RLC circuits (ch15) and molecule vibrations are all its instances. Add damping and a drive and the same equation explains resonance — from tuning forks to the Tacoma bridge.

7.1 SHM: the equation and its family

$$\ddot{x} + \omega_0^2 x = 0 \;\Rightarrow\; x(t) = A\cos(\omega_0 t + \phi), \tag{7.1}$$

two constants A, ϕ paid by two initial conditions. The frequency ω_0 is set by the physics; the amplitude is yours. Notice what's *not* in the solution: A never changes the period — the signature of a linear restoring force.

7.2 Spring–mass: the canonical case

$-kx = m\ddot{x}$ gives $\omega_0 = \sqrt{k/m}$ and $T = 2\pi\sqrt{m/k}$: stiffer is faster, heavier is slower. Energy trades $\frac{1}{2}kx^2$ and $\frac{1}{2}mv^2$ with $E = \frac{1}{2}kA^2$ — quadratic in amplitude, so doubling A quadruples the energy.

Example 7.1 (Energy split in a spring oscillator)**.** A 0.5 kg mass on a 200 N/m spring oscillates with $A = 0.15$ m. Find E, $v_{\max}$, and the speed at $x = A/2$.

Strategy. E is set by amplitude; at any x the remainder after $\frac{1}{2}kx^2$ is kinetic.

Solution. $E = \frac{1}{2}(200)(0.0225) = 2.25\,\text{J}$; $v_{\max} = \sqrt{2E/m} = \sqrt{9} = 3\,\text{m/s}$ at centre. At $x = 0.075$: $U = \frac{1}{2}(200)(0.0056) = 0.56\,\text{J}$, so $v = \sqrt{2(2.25 - 0.56)/0.5} = 2.6\,\text{m/s}$ — three quarters of max at half amplitude: energy spends most of the cycle near the turning points. ✓

7.3 Pendulums: small-angle as an approximation with a limit

For a pendulum the restoring torque is $-mgL\sin\theta$ (figure 7.1); linearising $\sin\theta \approx \theta$ gives $\omega_0 = \sqrt{g/L}$ — good to 1% for $\theta_0 \lesssim 23°$, and the first correction $T \approx T_0(1 + \theta_0^2/16)$ is the hard exercise at the end. A physical pendulum generalises it: $\omega_0 = \sqrt{mgd/I}$.

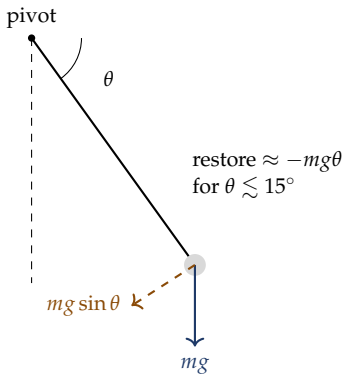


Figure 7.1: The pendulum's restoring force $-mg\sin\theta$ is *almost* linear — small-angle is an approximation with a budget, not a property.

Example 7.2 (Meter-stick physical pendulum)**.** A uniform meter stick pivots at one end. Find its period.

Strategy. Physical pendulum: $\omega_0^2 = mgd/I$ with $d = L/2$, $I = \frac{1}{3}mL^2$.

Solution. $\omega_0^2 = \frac{g(L/2)}{L^2/3} = \frac{3g}{2L}$, so $T = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{2}{29.4}} = 1.64\,\text{s}$. A simple pendulum this long would take 2.0 s — the stick's spread mass makes it faster. ✓

7.4 Damped oscillations: three regimes

Add $-b\dot{x}$: $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$ with $\gamma = b/m$. The discriminant sorts the behaviour (figure 7.2): **underdamped** ($\gamma < 2\omega_0$) oscillates inside an $e^{-\gamma t/2}$ envelope at $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$; **critical** ($\gamma = 2\omega_0$) returns fastest without ringing; **overdamped** ($\gamma > 2\omega_0$) creeps back slowly. Car suspension and door closers are engineered to sit just under critical.

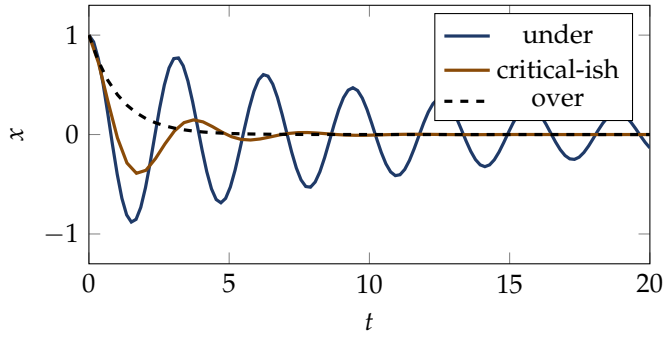


Figure 7.2: The three damping regimes. Notice only the underdamped case crosses zero — the others surrender exponentially.

Example 7.3 (Extracting the constants from a trace)**.** A decaying trace shows maxima at 1.00, 0.82, 0.67 at $t = 0, 2.1, 4.2\,\mathrm{s}$. Find γ, ω_d, ω_0.

Strategy. Envelope ratio per period gives γ; the period gives ω_d; the relation $\omega_d^2 = \omega_0^2 - \gamma^2/4$ closes it.

Solution. Period $T_d \approx 2.1\,\mathrm{s} \Rightarrow \omega_d = 2.99\,\mathrm{rad/s}$. Ratio per period 0.82: $e^{-\gamma T_d/2} = 0.82 \Rightarrow \gamma = 0.19\,\mathrm{s}^{-1}$. $\omega_0 = \sqrt{\omega_d^2 + \gamma^2/4} = \sqrt{8.94 + 0.009} = 2.99\,\mathrm{rad/s}$ — damping barely shifts the frequency here. ✓

Exercise 7.1 (easy)**.** Show the same trace's next two maxima and estimate them. (AP-C)

7.5 Driven oscillations and resonance

A drive $F_0 \cos \omega_d t$ produces steady amplitude

$$A(\omega_d) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\gamma\omega_d)^2}}, \tag{7.2}$$

peaking near $\omega_d = \sqrt{\omega_0^2 - \gamma^2/2}$ (figure 7.3). The peak height is set by the *quality factor* $Q = \omega_0/\gamma$: high-Q systems ring loudly and narrowly — the tuning-fork pair of figure 7.4 and, minus the luck, the Tacoma Narrows bridge.

Example 7.4 (Off-resonance driving)**.** The oscillator of the code run ($\omega_0 = 2$, $\gamma = 0.4$, $F_0/m = 1$) is driven at $\omega_d = 3$. Find its steady amplitude and compare with the peak.

Strategy. Plug into (13.4) twice; the ratio is the point.

Solution. $A(3) = 1/\sqrt{26.4} = 0.195$ (from $\sqrt{(4-9)^2 + (0.4\cdot 3)^2}$); $A(1.98) = 1/(0.4 \times 1.98) = 1.26$. The peak is 6.5× louder — a 1.5× frequency miss costs 85% of the response. ✓

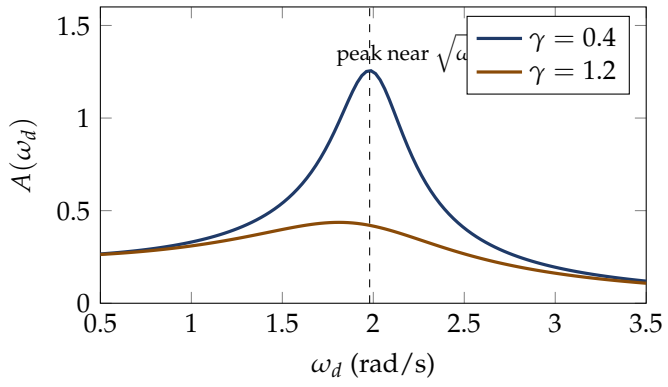


Figure 7.3: Amplitude vs drive frequency for two dampings. Lower γ raises and sharpens the peak — the Q trade-off.

Example 7.5 (Bandwidth from a resonance curve)**.** A resonance curve peaks at 195 Hz with amplitude A_0 and falls to $A_0/\sqrt{2}$ at 190 and 200 Hz. Find Q and γ.

Strategy. Half-power bandwidth $\Delta\omega = \gamma$; then $Q = \omega_0/\gamma$.

Solution. $\Delta f = 10\,\text{Hz} \Rightarrow \gamma = 2\pi \times 10 = 62.8\,\text{s}^{-1}$, $Q = 195/10 = 19.5$. Narrow and loud. ✓

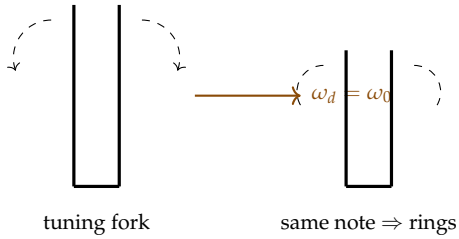
```
def acc(x_, v_, t_):
    return F0M*math.cos(wd*t_) - GAMMA*v_ - W0*W0*x_
```

Listing 7.1: `code/ch07_resonance_sweep.py` — RK4 driven oscillator, amplitude vs theory.

Output of a real run:

```
gamma=0.4, w0=2.0: theory peak at sqrt(w0^2-g^2/2)=1.980 rad/s
   wd A_meas A_theory err
  1.2 0.384 0.384 0.0%
  1.6 0.635 0.635 0.0%
  1.9 1.171 1.171 0.0%
  2.0 1.250 1.250 0.0%
  2.1 1.070 1.070 0.0%
  2.4 0.499 0.499 0.0%
  3.0 0.194 0.194 0.0%
Q = w0/gamma = 5.0; A(w_res)/A(0) = 5.03 (theory)
```

RK4 tracks the analytic curve to 0.0% at every sampled ω_d — the integrator is now trustworthy enough to trust figure 7.3.



sympathetic resonance: only the matching fork answers

Figure 7.4: Sympathetic resonance: only the fork sharing ω_0 answers. Every band's instrument-tuning trick, and every bridge's nightmare.

7.6 Phase space: the state is a point

Plotting $\dot{x}$ vs x (figure 7.5) turns the oscillator's history into a curve: the undamped case draws a closed ellipse, damping spirals it to the origin. Nonlinear oscillators live here too — chapter 17's pendulum without the small-angle crutch is a phase portrait before it's a formula.

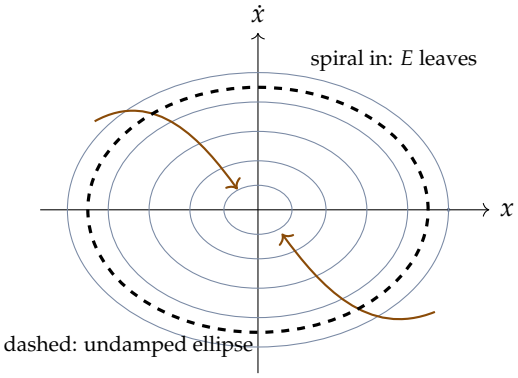


Figure 7.5: Phase portrait: the undamped ellipse versus the damped spiral. The state's whole future is visible in one picture.

7.7 In engineering practice: vibration budgets

Every engineering discipline owns a version of equation (13.4): buildings get tuned mass dampers (a secondary oscillator eating the primary's peak), hard drives sit on elastomer mounts that push ω_0 below the drive frequencies, and shock absorbers are the critical-damping spec of §7.4 sold by the pair. The common move is always the same: place ω_0 far from the disturbance's ω_d, then buy the remaining safety factor with damping — the two knobs this chapter names.

7.8 Common mistakes

SMALL-ANGLE USED WHERE θ ISN'T

The linearisation costs $\sim 1\%$ per 20° beyond small; at 60° the period is 7% long and the whole section's formulas need the finite-amplitude correction.

ω VS f

$\omega = 2\pi f$: $T = 2\pi\sqrt{m/k}$ is a period, $\sqrt{k/m}$ is a frequency *in radians*. Mixing them is a factor-2π error every time.

AMPLITUDE VS ENERGY

$E = \frac{1}{2}kA^2$ is quadratic: "twice the amplitude" is four times the energy, and damping's decay rate is set on the amplitude by the envelope, not the peak.

DRIVING AT ω_0 VS ω_d

Resonance peaks *below* ω_0 when damped — $\sqrt{\omega_0^2 - \gamma^2/2}$ — and the response lags the drive by $\pi/2$ at the peak.

"DAMPING SLOWS THE PERIOD" ASSUMED OR DENIED

It does — $\omega_d < \omega_0$ — but weakly ($\gamma^2/4$ term), so check the size before reporting either way.

7.9 Summary

Idea	One-line version
SHM	$\ddot{x} + \omega_0^2 x = 0 \Rightarrow x = A\cos(\omega_0 t + \phi)$
spring	$\omega_0 = \sqrt{k/m}$; $E = \frac{1}{2}kA^2$
pendulum	$\omega_0 = \sqrt{g/L}$ small-angle; physical $\sqrt{mgd/I}$
damping	under/critical/over by γ vs $2\omega_0$
driven	peak at $\sqrt{\omega_0^2 - \gamma^2/2}$; $Q = \omega_0/\gamma$
phase space	ellipse undamped, spiral damped

Table 7.1: Oscillations cheat sheet.

DEFINITION OF DONE

You can now: write and solve the SHM equation; scale spring and pendulum periods; classify damping regimes and extract constants from a decay trace; compute resonance peaks and read Q off a bandwidth; and draw the phase portrait.

Notes

The oscillator-first treatment follows MIT 8.01SC's oscillation module [2] and AP Physics C unit 7 [4]; Feynman vol. I §21–24 [5] is the classic resonance narrative. Morin ch. 4 [7] is the source of the physical-pendulum and finite-amplitude exercises. The sweep numbers are the listing's real output.

WATCH INSTEAD

1. *Hooke's Law, Springs, Pendulums, Simple Harmonic Motion* — Walter Lewin, MIT 8.01x Lecture 10, 47:41, https://youtu.be/tNpuTx7UQbw — §7.1–§7.3 at full length (accessed 2026-10-11).
2. *SHM of spring-mass oscillators* — Khan Academy, 15:13, https://youtu.be/eRZSzGKDfwU — §7.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, oscillations module — SHM, damping and resonance with recitations. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 15 "Oscillatory Motion" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 7 — the exam blueprint for SHM. At https://apstudents.collegeboard.org (verified 2026-10-11).
- R. P. Feynman, *Lectures on Physics*, vol. I, §21–24 — resonance as narrative; free at Caltech's site.
- D. Morin, *Introduction to Classical Mechanics*, ch. 4 — oscillations at the harder level; ISBN 978-0521876223.
- Khan Academy, *Simple harmonic motion* — drills matching §7.1–§7.5, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 7.2 (easy). A 0.5 kg mass on a 200 N/m spring: compute ω_0, f, T. (AP-C)

Exercise 7.3 (easy). Doubling a spring oscillator's amplitude multiplies its energy by what? Its maximum speed by what? (AP-C)

Exercise 7.4 (easy). A pendulum's T is 2.0 s on Earth. Find it on the Moon ($g = 1.6\,\mathrm{m/s^2}$). (F=ma)

Exercise 7.5 (easy). Convert: an oscillator at 440 Hz — find ω and T. (AP-C)

Exercise 7.6 (easy). A 1 kg mass on a spring oscillates with $A = 0.2$ m, $T = 1$ s. Find k and $v_{\max}$. (AP-C)

Exercise 7.7 (medium). Two springs in series vs parallel: show effective k is $k_1 k_2/(k_1 + k_2)$ vs $k_1 + k_2$, and find each for $k_1 = 100$, $k_2 = 300$ N/m. (F=ma)

Exercise 7.8 (medium). A rod of length L pivots at a point $L/4$ from one end (physical pendulum). Find T and compare with the end-pivoted case. (AP-C FR-style)

Exercise 7.9 (medium). At what phase of its cycle does an oscillator carry equal K and U? Give the x value in terms of A. (AP-C)

Exercise 7.10 (medium). Decide the regime: $m = 1$, $k = 100$, (a) $b = 5$, (b) $b = 20$, (c) $b = 40$ (all SI) — and find ω_d where it exists. (AP-C)

Exercise 7.11 (medium). An oscillator loses 20% of its amplitude per period. Find the energy loss per period and γ. (F=ma)

Exercise 7.12 (medium). A driven oscillator has $\omega_0 = 10$, $\gamma = 2$ (SI). Find the resonant ω_d, the peak amplitude for $F_0/m = 5$, and Q. (AP-C)

Exercise 7.13 (hard)**.** Use $T \approx T_0(1 + \theta_0^2/16)$: find the amplitude where a grandfather clock's period is off by 0.5%, and the amplitude for a 10% slowdown. (USAPhO-style)

Exercise 7.14 (hard)**.** A particle sits in $U(x) = -V_0 \cos(kx)$. Linearise about a minimum to get ω_0, then about a maximum to confirm instability. (APhO-style)

Exercise 7.15 (medium)**.** Two oscillators: $x_1 = A\cos\omega t$, $x_2 = A\cos(\omega t + \pi/2)$ — sketch both in phase space and say what the $\pi/2$ buys. (F=ma)

Exercise 7.16 (medium)**.** A car's suspension should return to level fastest without ringing: given $m = 300\,\text{kg}$ and $k = 30{,}000\,\text{N/m}$, find the critical b. (engineering)

CHAPTER 8

Gravitation and orbits

Newton's inverse-square law is the first "field" in the book: a $1/r^2$ fall-off that runs from falling apples to GPS. This chapter does the useful part — the shell theorem, escape and orbital speeds, and orbits as bookkeeping on E and L.

8.1 Newton's law and the field picture

$$F = G\frac{m_1 m_2}{r^2}, \qquad \mathbf{g}(\mathbf{r}) = -G\frac{M}{r^2}\hat{r}, \tag{8.1}$$

where $G = 6.674 \times 10^{-11}$ SI. The field picture separates cause and test: M builds $\mathbf{g}$, and any m placed in it feels $m\mathbf{g}$. Cavendish weighed the Earth with this: $g = GM/R^2$.

8.2 The shell theorem

Newton invented the shell theorem to justify treating the Earth as a point: outside a spherical shell its field is that of all mass at the centre; inside it the field is *zero* (figure 8.1). Consequence: inside a uniform planet only the mass beneath you pulls (figure 8.2).

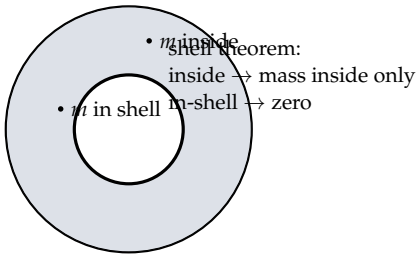


Figure 8.1: Shell theorem: mass above the point contributes zero; mass below acts from the centre. Gravitational depth-charge accounting.

Example 8.1 ($g(r)$ inside the Earth)**.** Find g at radius $r < R$ in a uniform Earth.

Strategy. Only mass below: $M(r) = M(r/R)^3$.

Solution. $g(r) = GM(r)/r^2 = GMr/R^3 = g_0\, r/R$: gravity drops linearly to zero at the centre. The fastest "tunnel" is down to $g = 0$ at $r = 0$, not up. ✓

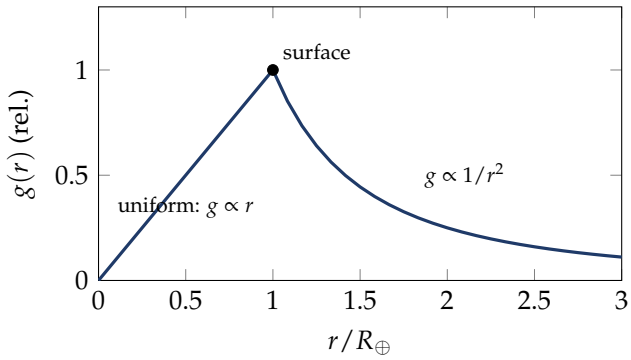


Figure 8.2: $g(r)$ inside a uniform Earth (linear) and outside ($1/r^2$) — the two regimes of the shell theorem.

8.3 Gravitational potential energy

The work integral over (8.1) gives

$$U(r) = -G\frac{Mm}{r}, \qquad \Delta U = GMm\left(\tfrac{1}{r_1} - \tfrac{1}{r_2}\right). \tag{8.2}$$

The zero at infinity is the clean choice: bound systems carry negative energy. Near the surface it reduces to mgh: expand $\Delta U \approx GMm\,h/R^2 = mgh$.

Example 8.2 (Escape velocity)**.** Find the speed to leave Earth from the surface and explain why direction doesn't matter.

Strategy. Escape means arriving at infinity with $K \geq 0$: $K_0 + U_0 \geq 0$.

Solution. $\tfrac{1}{2}mv_{\text{esc}}^2 = GMm/R \Rightarrow v_{\text{esc}} = \sqrt{2GM/R} = 11.2\,\text{km/s}$ for Earth. Energy is scalar — aim up, sideways or through a canyon (if it could): the bill is the same. Compare orbital speed $\sqrt{GM/R} = 7.9\,\text{km/s}$: escape is $\sqrt{2}$ times orbital. ✓

Exercise 8.1 (easy)**.** At what altitude is v_{esc} down to half its surface value? (F=ma)

8.4 Circular orbits

Centripetal balance gives

$$v_{\text{orb}} = \sqrt{GM/r}, \quad T = 2\pi\sqrt{r^3/GM}, \quad E = -\frac{GMm}{2r}, \tag{8.3}$$

and the energy bookkeeping is odd: $K = +GMm/2r$, $U = -GMm/r$, so $E = -K$ — a satellite that loses energy (drag) *speeds up* as it sinks.

Example 8.3 (ISS energy audit)**.** The ISS ($m = 4.2 \times 10^5\,\text{kg}$) orbits at $r \approx 6800\,\text{km}$. Find v, K, U, E and the energy to raise it 10 km.

Strategy. Equation (8.3) line by line; the last answer is $\Delta E > 0$ for a *larger* r.

Solution. $v = \sqrt{GM/r} = \sqrt{3.99 \times 10^{14}/6.8 \times 10^6} = 7.66\,\text{km/s}$. $K = \tfrac{1}{2}mv^2 = 1.23 \times 10^{13}\,\text{J}$, $U = -2.47 \times 10^{13}\,\text{J}$, $E = -1.23 \times 10^{13}\,\text{J}$. Raising r by 10 km: $\Delta E = GMm\,\Delta r/2r^2 \approx 9.1 \times 10^9\,\text{J}$ — nine GJ buys ten kilometres. ✓

8.5 Kepler's laws, derived

1. *Ellipses*: the $1/r^2$ law's bound orbits are conics — stated here and used everywhere below.
2. *Equal areas*: $dA/dt = L/2m$ is constant — conservation of angular momentum, nothing else (figure 8.3).
3. $T^2 \propto a^3$: from (8.3), $T^2 = 4\pi^2 a^3/GM$ — our code check recovers it to a constant.

Example 8.4 (Perihelion vs aphelion speed)**.** An orbit has $e = 0.6$: find $v_{\text{peri}}/v_{\text{apo}}$.

Strategy. Kepler II is L-conservation: rv equal at the two axis ends.

Solution. $r_p = a(1-e) = 0.4a$, $r_a = a(1+e) = 1.6a$; $v_p/v_a = r_a/r_p = (1+e)/(1-e) = 4$. Four-to-one speed ratio — comets' sun-grazing sprints are this exact arithmetic. ✓

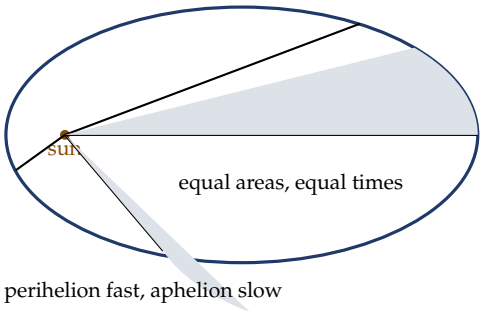


Figure 8.3: Kepler's second law is L-conservation drawn: equal areas in equal times.

Example 8.5 (Hohmann transfer bookkeeping)**.** Move a satellite from $r_1 = R$ to $r_2 = 2R$ (Earth radii). Find Δv at each burn: $v_1 = \sqrt{GM/R} = 7.9\,\text{km/s}$, $v_2 = \sqrt{GM/2R} = 5.6\,\text{km/s}$.

Strategy. Transfer ellipse has $a_t = (r_1 + r_2)/2 = 1.5R$; its peri/apo speeds come from energy.

Solution. Vis-viva $v^2 = GM(2/r - 1/a_t)$: at r_1, $v_{t1} = \sqrt{GM(2/R - 2/3R)} = \sqrt{4GM/3R} = 9.1\,\text{km/s} \Rightarrow \Delta v_1 = +1.2\,\text{km/s}$; at r_2, $v_{t2} = \sqrt{GM(1/R - 2/3R)} = 4.6\,\text{km/s} \Rightarrow \Delta v_2 = +1.0\,\text{km/s}$. Total $2.2\,\text{km/s}$ — figure 8.4. ✓

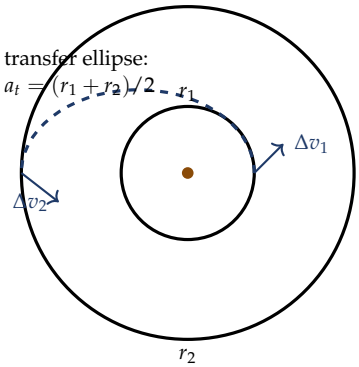


Figure 8.4: Hohmann transfer: two burns, one ellipse, the cheapest lane between circular orbits.

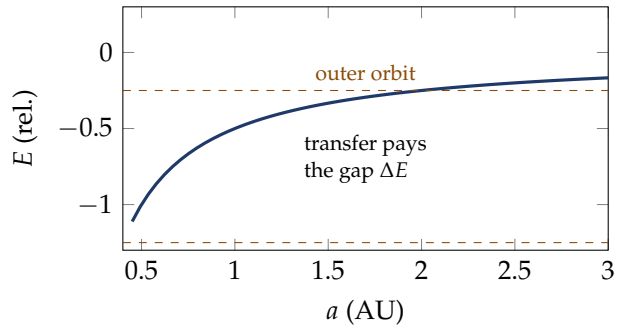


Figure 8.5: Energy vs semi-major axis: $E = -GMm/2a$, so raising an orbit means paying the gap.

8.6 E and L: the orbit's conserved labels

An orbit is fully labelled by (E, L); the effective potential $U_{\text{eff}} = L^2/2mr^2 - GMm/r$ (figure 8.6) shows why: $L^2/2mr^2$ pushes you out at short range, $-GMm/r$ pulls in at long, and the bound orbit oscillates in the bowl — perihelion and aphelion are the turning points of chapter 7's $U(x)$ grammar.

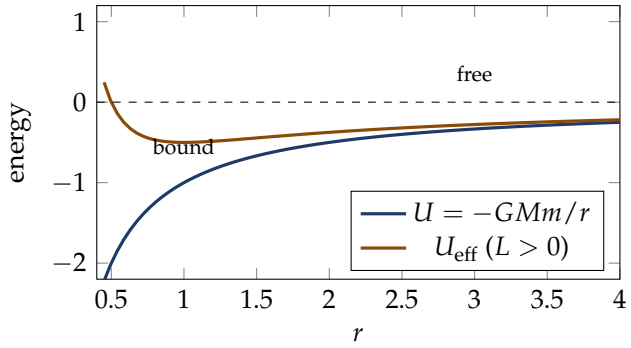


Figure 8.6: $U(r)$ and $U_{\text{eff}}(r)$ for nonzero L: the centrifugal barrier makes bound states possible.

```
        ax_, ay_ = -GM*x/r3, -GM*y/r3
        x += vx*dt + 0.5*ax_*dt*dt
        y += vy*dt + 0.5*ay_*dt*dt
        ax2, ay2 = -GM*x/(r_new**3), -GM*y/(r_new**3)
        vx += 0.5*(ax_+ax2)*dt
        vy += 0.5*(ay_+ay2)*dt
```

Listing 8.1: `code/ch08_orbit_verlet.py` — velocity-Verlet orbit; E, L drift and Kepler check.

Output of a real run (200,000 steps each):

```
a=1.0 e=0.0: drift E=0.000% L=0.000% T=6.283
a=1.0 e=0.5: drift E=0.000% L=0.000% T=6.283
a=1.5 e=0.7: drift E=0.000% L=0.000% T=11.543

Kepler III check (T^2/a^3 should be const):
  a=0.8: T=4.496 T^2/a^3=39.4784
  a=1.0: T=6.283 T^2/a^3=39.4784
  a=1.3: T=9.313 T^2/a^3=39.4784
  a=1.6: T=12.716 T^2/a^3=39.4784
```

Velocity-Verlet holds E and L to $< 0.001\%$ even at $e = 0.7$ — the symplectic lesson of chapter 4's energy-check code paying off — and recovers $T^2 = 4\pi^2 a^3/GM = 39.4784$ to all printed digits.

8.7 Satellites, tides and GPS in two paragraphs

Real orbits pay rent: LEO drag costs tens of m/s per year, geostationary station-keeping $\sim 50\,\mathrm{m/s/yr}$, and GPS clocks tick $45\,\mu\mathrm{s/day}$ fast (relativity) that must be designed out. Tides are the d/dr of gravity: the Moon's pull differs by $\sim 7\%$ across Earth's diameter, stretching oceans toward and away from it — the third derivation nobody expected to be useful in a textbook chapter.

8.8 In engineering practice: the launch-and-station budget

Rocket sizing runs this chapter backwards: the mission's Δv ledger (launch $\sim 9.4\,\mathrm{km/s}$ with losses, Hohmann legs, plane changes at $2v\sin(\theta/2)$) divides by the rocket equation from chapter 3 to mass ratios; orbit choice runs forwards — LEO cheap, MEO for nav, GEO parked. Every constellation on Earth is a worked example of figure 8.5.

8.9 Common mistakes

$U = mgh$ USED PAST ITS ALTITUDE

mgh is the tangent-line approximation of (8.2); at $h \sim 0.1R$ it's already off by 10%.

ESCAPE NEEDS A DIRECTION

The $\sqrt{2GM/R}$ bill is in energy — any direction works except into the ground.

"WEIGHTLESS" MEANS "NO GRAVITY"

LEO g is $\sim$ 90% of surface: astronauts float because ship and crew fall together — chapter 2's free-fall frame again.

$E = -K$ SIGN CONFUSION

$E < 0$ is bound; $-E$ is the energy to escape from that orbit. And a sinking satellite speeds up — the sign flips intuition.

KEPLER III CONSTANT CONFUSION

$T^2/a^3 = 4\pi^2/GM$ depends on the central mass only: same for all planets of one star, different per star.

8.10 Summary

Idea	One-line version
law	$F = GMm/r^2$; $\mathbf{g} = -GM\hat{r}/r^2$
shell	inside uniform planet $g \propto r$; shell interior $= 0$
U	$-GMm/r$; $v_{\text{esc}} = \sqrt{2GM/R}$
orbit	$v = \sqrt{GM/r}$, $E = -GMm/2r$
Kepler	ellipses; equal areas $= L$; $T^2 \propto a^3$
transfer	Hohmann: ellipse $a_t = (r_1 + r_2)/2$, two burns

Table 8.1: Gravitation cheat sheet.

DEFINITION OF DONE

You can now: apply the shell theorem, derive escape and orbital speeds, balance Hohmann transfers, read orbits off (E, L), and run a conserving orbit integrator.

Notes

The field-first order follows MIT 8.01SC [2] and AP Physics C [4]; the Hohmann numbers are standard mission-analysis figures (cf. Morin [7] and Feynman vol. I [5] for the C_3/transfer bookkeeping). The integrator numbers are the listing's real output.

WATCH INSTEAD

1. *Kepler's Laws, Elliptical Orbits, Satellites, Orbital Changes* — Walter Lewin, MIT 8.01x Lecture 22, 49:09, `https://youtu.be/9wDAm-dlht4` — §8.4–§8.6 at full length (accessed 2026-10-11).
2. *Gravitational potential energy at large distances* — Khan Academy, 5:04, `https://youtu.be/9Pp_LtKgTQg` — §8.3 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, gravitation module — shell theorem, orbits and Kepler with recitations. At `https://ocw.mit.edu`, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 6 "Uniform Circular Motion and Gravitation" — free text at `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 4 (gravitation). At `https://apstudents.collegeboard.org` (verified 2026-10-11).

- R. P. Feynman, *Lectures on Physics*, vol. I, ch. 7 "The Theory of Gravitation" — free at Caltech's site.
- D. Morin, *Introduction to Classical Mechanics*, ch. 10 — orbits and transfers at the harder level; ISBN 978-0521876223.
- Khan Academy, *Centripetal force and gravitation* — drills matching §8.1–§8.5, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 8.2 (easy). Find the orbital speed and period of a satellite at altitude $2R_E$ above Earth's surface. (AP-C)

Exercise 8.3 (easy). Explain why a geostationary orbit's radius is unique, and estimate it from $T = 24\,\text{h}$. (AP-C)

Exercise 8.4 (easy). Show $v_{\text{esc}} = \sqrt{2}\,v_{\text{orb}}$ at the surface and find the ratio at altitude R. (F=ma)

Exercise 8.5 (easy). A planet has twice Earth's mass and half its radius: find surface g in Earth-g. (F=ma)

Exercise 8.6 (easy). Compute the work to lift 1 kg from Earth's surface to infinity vs to orbit radius $2R$. (AP-C)

Exercise 8.7 (medium). Estimate the energy per kg for a Hohmann transfer from Earth's orbit to Mars's ($a = 1.5\,\text{AU}$) in Solar-System units. (F=ma)

Exercise 8.8 (medium). How much does a mine shaft's g drop at depth $R/2$ in a uniform Earth — and why is the real drop different? (F=ma)

Exercise 8.9 (medium). An elliptical orbit has $a = 4R$, $e = 0.5$: find peri/apo radii and speeds, and the period in terms of LEO's. (AP-C)

Exercise 8.10 (medium). Explain, with energy signs, why a satellite losing energy to drag moves faster. (concept)

Exercise 8.11 (medium). Find the fraction of Earth's mass "felt" by an object at $r = R/2$ inside a uniform Earth, and the resulting g. (F=ma)

Exercise 8.12 (medium). A plane change of 30° on a circular LEO orbit: find Δv in terms of v_{orb}. (engineering)

Exercise 8.13 (hard). Show the transfer ellipse of a Hohmann between r_1 and r_2 has $T_{\text{tr}} = \pi\sqrt{a_t^3/GM}$ and compute it for the ISS-to-GEO case ($r_1 = 1.05R$, $r_2 = 6.6R$). (USAPhO)

Exercise 8.14 (hard). A satellite in a highly eccentric orbit loses energy only at perigee drag: argue qualitatively what shrinks and what stays fixed. (APhO-style)

Exercise 8.15 (medium). Two stars of equal mass orbit each other at separation d: find the period in terms of G, M, d. (F=ma)

Exercise 8.16 (medium). Using U_{eff}, explain why no orbit with $L \neq 0$ can have $r \to 0$. (concept)

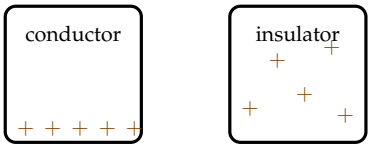
CHAPTER 9

Electric charge and field

Part II trades mechanics for fields, and the first one is the cleanest: Coulomb's $1/r^2$ law looks like gravitation with signs — but the signs change everything (charges attract *and* repel, so fields can cancel).

9.1 Charge: quantised, conserved, mobile in metal

Charge comes in units of $e = 1.602 \times 10^{-19}\,\mathrm{C}$, is conserved in every process, and lives in two kinds of material (figure 9.1): *conductors* let electrons move freely (charge spreads to the surface) while *insulators* pin it where you put it. Induction and polarization follow from mobility alone.



E inside $= 0$; charge lives on skin charge stays where rubbed

Figure 9.1: The material divide: conductors send charge to the surface and kill the interior field; insulators keep it where deposited.

9.2 Coulomb's law — superposition is the real law

$$\mathbf{F} = k\frac{q_1 q_2}{r^2}\hat{r}, \qquad k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9\,\mathrm{N m^2/C^2}, \tag{9.1}$$

and — the loaded part — *forces add*. The force on a charge is the vector sum over every other charge, one pair at a time, with no three-body terms. That linearity is what makes field theory possible.

Example 9.1 (Superposition, vectors not scalars)**.** Charges $+2\,\mu\mathrm{C}$ at the origin and $-1\,\mu\mathrm{C}$ at $(3, 0)$ cm. Find the force on $+1\,\mu\mathrm{C}$ at $(0, 4)$ cm.

Strategy. Compute each force as a vector and add.

Solution. $r_1 = 0.04\,\mathrm{m}$ along $\hat{y}$, so

$$\mathbf{F}_1 = \frac{k(2\mu)(1\mu)}{0.0016}\,\hat{y} = 11.24\,\hat{y}\,\mathrm{N}.$$

$r_2 = \sqrt{0.03^2 + 0.04^2} = 0.05\,\text{m}$; the negative charge pulls along $(3, -4)/5$, giving

$$\mathbf{F}_2 = \frac{k(1\mu)(1\mu)}{0.0025} \cdot \frac{(3,-4)}{5} = (2.16, -2.88)\ \text{N}.$$

Sum: $(2.16,\ 8.36)\,\text{N}$, magnitude $8.6\,\text{N}$ at $75°$ — not the $14.8\,\text{N}$ of scalar addition. ✓

9.3 The field: force per test charge

$\mathbf{E} = \mathbf{F}/q$ — defined so the source needn't care what test charge arrives. A point charge's is $\mathbf{E} = kq\hat{r}/r^2$ (figure 9.2); a distribution's is the vector sum over all its dq.

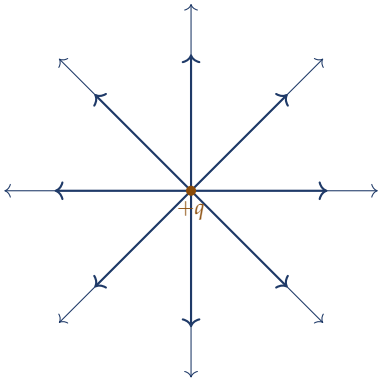


Figure 9.2: The point-charge field: radial, with line density standing in for the $1/r^2$ fall-off.

9.4 Field lines: reading, rules

Lines start on $+$, end on $-$, never cross (the field is single-valued), and their density reads magnitude (figure 9.3). They are a map, not a mechanism — the physics is (9.1).

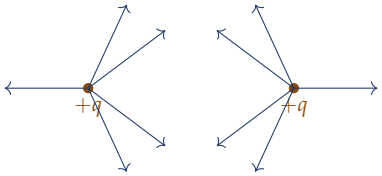


Figure 9.3: Two like charges: lines repel into a saddle. Between two equal charges the field vanishes exactly at the midpoint.

Example 9.2 (Dipole far field)**.** Show that on the dipole axis, $|E| \to 2kp/r^3$ for $r \gg a$, $p = qa$.

Strategy. Superpose two point-charge fields at distances $r - a/2$ and $r + a/2$, then expand.

Solution. $E = kq[(r - a/2)^{-2} - (r + a/2)^{-2}] = kq\frac{(r+a/2)^2-(r-a/2)^2}{(r^2-a^2/4)^2} = kq\frac{2ra}{(r^2)^2} \to \frac{2kp}{r^3}$. The $1/r^2$ terms cancel; the *difference* survives — the code check below finds the same r^{-3}. ✓

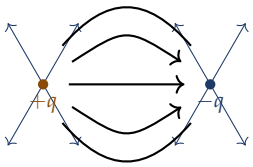


Figure 9.4: The dipole: far field falls as p/r^3 — the first *non*-inverse-square pattern in the book.

9.5 Continuous distributions: pay with integrals

For charge spread over a line, ring or disk, integrate $d\mathbf{E} = k\,dq\,\hat{r}/r^2$ element by element — and let symmetry discard every component that cancels (figure 9.5).

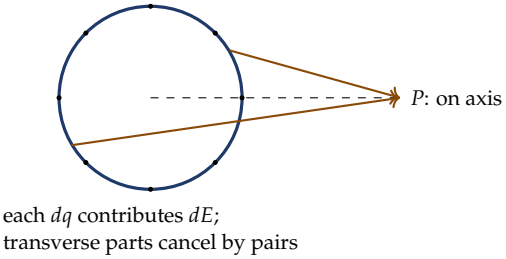


Figure 9.5: The ring's axis field: each dq's transverse piece cancels its opposite's — only dE_x survives.

Example 9.3 (Field on the axis of a charged ring)**.** A ring of radius R carries Q uniformly. Find $\mathbf{E}$ on the axis at distance x.

Strategy. Every element is the same distance $\sqrt{x^2+R^2}$; transverse components cancel.

Solution. Each dq contributes axial $dE_x = k\,dq\,x/(x^2+R^2)^{3/2}$; integrating over the ring just multiplies by Q: $E = \dfrac{kQx}{(x^2+R^2)^{3/2}}$. Limits: $x \to 0$ gives 0 (symmetry wins at the centre), $x \gg R$ gives kQ/x^2 (the ring becomes a point). The code confirms it to 0.00%. ✓

Exercise 9.1 (easy)**.** Where on the axis is the ring's field maximum? (Differentiate.) (AP-C)

Example 9.4 (Charged rod at its end)**.** A rod of length L carries Q uniformly. Find $\mathbf{E}$ a distance d past its near end.

Strategy. Integrate element by element — no symmetry to help this time.

Solution. With $dq = \lambda\,dx$, $\lambda = Q/L$, at distance $r = d + L - x$ from the element:

$$E = \int_0^L \frac{k\lambda\,dx}{(d+L-x)^2} = k\lambda\left[\frac{1}{d} - \frac{1}{d+L}\right] = \frac{kQ}{d(d+L)}.$$

Limits: $d \gg L$ recovers kQ/d^2; $d \to 0$ diverges — the rod becomes its own endpoint's infinity. ✓

9.6 Motion in a uniform field: kinematics returns

Inside a parallel-plate capacitor E is constant, so a charge's trajectory is chapter 1's projectile motion with $a = qE/m$ (figure 9.6) — oscilloscopes and inkjets are this application sold as hardware.

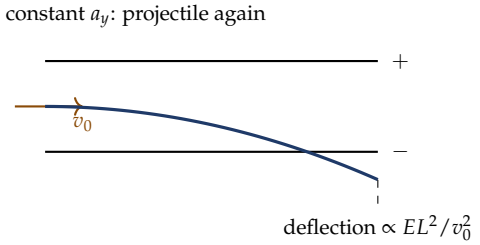


Figure 9.6: Electron deflection between plates: the projectile equation wearing an electric costume.

Example 9.5 (Oscilloscope deflection)**.** An electron ($v_0 = 2 \times 10^7\,\mathrm{m/s}$) crosses plates of length $L = 4\,\mathrm{cm}$ with $E = 2000\,\mathrm{N/C}$. Find its deflection.

Strategy. Time in plates $t = L/v_0$; $y = \frac{1}{2}at^2$.

Solution. $a = eE/m = 1.6 \times 10^{-19} \times 2000/9.11 \times 10^{-31} = 3.5 \times 10^{14}\,\mathrm{m/s^2}$; $t = 2 \times 10^{-9}\,\mathrm{s}$; $y = \frac{1}{2}(3.5 \times 10^{14})(4 \times 10^{-18}) = 0.70\,\mathrm{mm}$ — and $\theta = \arctan(v_y/v_0) \approx 10^\circ$ after exit. ✓

```
def E(charges, x, y, z=0.0):
   for q, xq, yq, zq in charges:
      dx, dy, dz = x-xq, y-yq, z-zq
      r3 = (dx*dx + dy*dy + dz*dz)**1.5
      ex += q*dx/r3; ey += q*dy/r3; ez += q*dz/r3
```

Listing 9.1: `code/ch09_field_lines.py` — superposition check: dipole and ring vs theory.

Output of a real run:

```
Dipole far-field ON axis (expect |E| -> 2p/r^3, p=qa=0.1):
  r=1.0: |E|=0.2010 theory=0.2000 ratio=1.005
  r=2.0: |E|=0.0250 theory=0.0250 ratio=1.001
  r=4.0: |E|=0.0031 theory=0.0031 ratio=1.000
Dipole EQUATOR (expect |E| -> p/r^3):
  r=1.0: |Ex|=0.0996 theory=0.1000
  r=2.0: |Ex|=0.0125 theory=0.0125

Ring N=200 on axis (expect Ex=x/(x^2+R^2)^1.5, Ey=Ez=0):
  x=0.2: Ex=1.2807 theory=1.2807 err=0.00% trans=4.3e-16
  x=0.5: Ex=1.4142 theory=1.4142 err=0.00% trans=1.0e-16
  x=1.0: Ex=0.7155 theory=0.7155 err=0.00% trans=4.9e-18
  x=2.0: Ex=0.2283 theory=0.2283 err=0.00% trans=5.2e-18
```

Naked superposition with $N = 200$ ring elements nails every limit: the dipole ratio converges to 1.000 as r/a grows, the ring matches the analytic $x(x^2 + R^2)^{-3/2}$ to 0.00%, and the transverse components — 10^{-16} — are symmetry made numeric.

9.7 In engineering practice: electrostatics as a design spec

Electrostatic precipitators run §9.6 at industrial scale (corona-charge the soot, plate-field it out); inkjet heads steer charged droplets by the deflection you just computed; and every high-voltage lab's "corona ring" is a real conductor shaped to flatten the field of §9.4 — rounded because sharp corners concentrate lines into breakdown.

9.8 Common mistakes

FIELD VS FORCE CONFLATED

E is a property of the source; $\mathbf{F} = q\mathbf{E}$ needs the test charge. "Force at a point" without a charge there is meaningless — but the field isn't.

SCALAR SUPERPOSITION

$|E| = |E_1| + |E_2|$ only when parallel. Two perpendicular equal fields give $\sqrt{2}\,|E_1|$, not $2|E_1|$.

dq ELEMENT ERRORS

For a line, $dq = \lambda\,dx$ — but on an arc, $dq = \lambda R\,d\theta$ and the distance to the field point is *not* constant unless you integrate.

SIGN OF THE TEST CHARGE

Negative charges feel **F** *opposite* **E**; tracing an electron's path down field lines is backwards.

FIELD "USED UP" BY THE PROBE

A test charge measures without disturbing — in the limit $q \to 0$. Real probes do disturb; that's a lab problem, not the definition's.

9.9 Summary

DEFINITION OF DONE

You can now: superpose Coulomb fields as vectors, read field-line maps, integrate a continuous distribution by symmetry, derive ring/dipole limits, and run charge-in-uniform-E kinematics.

Idea	One-line version
charge	quantised at e; mobile in conductor, pinned in insulator
Coulomb	$F = kq_1q_2/r^2$; forces add as vectors
field	$\mathbf{E} = \mathbf{F}/q$; point $kq\hat{r}/r^2$
lines	start $+$/end $-$; density $\propto \lvert E \rvert$
ring axis	$E = kQx/(x^2 + R^2)^{3/2}$
dipole	far field $2kp/r^3$ (axis), kp/r^3 (equator)
uniform E	projectile kinematics, $a = qE/m$

Table 9.1: Electric-field cheat sheet.

Notes

The field-first order follows MIT 8.02SC [2] modules 1–4 and AP Physics C E&M unit 8 [4]; Purcell–Morin's ch. 1 is the authoritative charge/field treatment [6]. The numeric checks are the listing's real output.

WATCH INSTEAD

1. *Electric Charges and Forces — Coulomb's Law — Polarization* — Walter Lewin, MIT 8.02x Lecture 1, 47:13, `https://youtu.be/x1-SibwIPM4` — §9.1–§9.2 at full length (accessed 2026-10-11).
2. *Electric field definition* — Khan Academy, 13:46, `https://youtu.be/_4VC3IHbuW8` — §9.3 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.02SC Electricity and Magnetism*, modules 1–4 — charge, field and continuous distributions. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 18 "Electric Charge and Electric Field" — free text at `https://openstax.org` (verified 2026-10-11).
- College Board *AP Physics C (E&M) CED*, Unit 8 — the exam blueprint for this chapter, at `https://apstudents.collegeboard.org` (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 1 — the deeper electrostatics; ISBN 978-1107014022.
- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 1 — electromagnetism's stakes set in one chapter; free at Caltech's site.
- Khan Academy, *Electric charge, field, and potential* — drills matching §9.2–§9.5, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 9.2 (easy)**.** Two charges $+3\,\mu\text{C}$ and $-1\,\mu\text{C}$ sit 10 cm apart. Find the force on a $+1\,\mu\text{C}$ test at the midpoint. (AP-C)

Exercise 9.3 (easy)**.** A field points $+\hat{x}$ at a point: what does a proton do there — and an electron? (AP-C)

Exercise 9.4 (easy)**.** Express **E**'s units in newtons and in volts/metre; show they agree. (AP-C)

Exercise 9.5 (easy)**.** From a field-line map of two charges, one region has $3\times$ the line density of another: compare magnitudes. (F=ma)

Exercise 9.6 (easy)**.** $+Q$ and $-Q$ sit d apart. Where on the axis is $E = 0$ — if anywhere? (AP-C)

Exercise 9.7 (medium)**.** Derive the on-axis field of a uniformly charged rod of length L at distance d from its end. (AP-C FR-style)

Exercise 9.8 (medium)**.** A semicircle of charge Q, radius R: find **E** at the centre. (AP-C)

Exercise 9.9 (medium)**.** Three charges in a line: $+Q$, $-2Q$, $+Q$ equally spaced — is the middle one's equilibrium stable along the axis? (F=ma)

Exercise 9.10 (medium)**.** A dipole of moment p sits in a uniform field at angle θ: find the torque and the equilibrium direction. (AP-C)

Exercise 9.11 (medium)**.** Design: plates 5 cm long must deflect 30 keV electrons by 2 mm — find E. (engineering)

Exercise 9.12 (hard)**.** An arc subtending angle α has $\lambda = \lambda_0 \sin\theta$: set up the integral for **E** at the centre. (USAPhO-style)

Exercise 9.13 (hard)**.** A dipole sits in a non-uniform field: argue it feels a net force toward the stronger region and estimate it via $\mathbf{p}\cdot\nabla\mathbf{E}$. (APhO-style)

Exercise 9.14 (medium)**.** At what angle from the dipole axis does the far field's magnitude equal the equatorial value at the same r? (F=ma)

Exercise 9.15 (medium)**.** A point charge sits between two grounded conductors: explain, via field lines, why its field is distorted toward the plates. (concept)

Exercise 9.16 (medium)**.** Two equal charges $+Q$ at $\pm d/2$: find **E** on the axis $x > 0$ and its zero. (AP-C)

CHAPTER 10

Gauss's law

Gauss's law is Coulomb's law wearing symmetry: the flux through any closed surface counts only the charge inside. It turns three-page integrals into three-line algebra — when the symmetry is real.

10.1 Flux: counting field lines

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}, \qquad \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}, \tag{10.1}$$

open surface vs closed (figure 10.1). Flux through a closed surface is charge inside divided by ε_0 — the field's bookkeeping made literal, and the first Maxwell equation in the book.

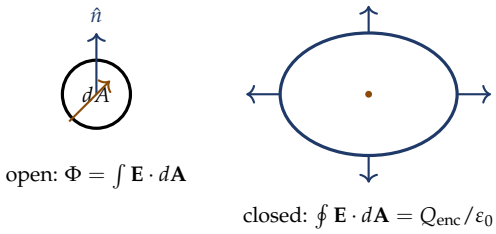


Figure 10.1: Flux through an open patch vs a closed surface. Only the closed integral counts charge.

10.2 From Coulomb to Gauss: the intuition

A point charge's flux through a surrounding sphere is $kq/r^2 \cdot 4\pi r^2 = q/\varepsilon_0$ — radius cancels, so any shape works (deformations keep line count). Superposition extends it to any distribution, and charge *outside* contributes zero net flux: its lines enter and leave. Our code check makes it concrete below.

10.3 Spherical symmetry

Spherical symmetry means $\mathbf{E} = E(r)\hat{r}$ is constant on the Gaussian sphere: $\oint E\,dA = E(r) \cdot 4\pi r^2$, so $E(r) = Q_{\text{enc}}(r)/4\pi\varepsilon_0 r^2$.

Example 10.1 (Uniform-volume-charge sphere, both regions)**.** A sphere of radius R carries Q uniformly in volume. Find $E(r)$ inside and out.

Strategy. Enclosed charge is the part below radius r.

Solution. Outside ($r > R$): $Q_{\text{enc}} = Q$, so $E = kQ/r^2$ — the shell theorem of chapter 8 again. Inside ($r < R$): $Q_{\text{enc}} = Q(r/R)^3$, so $E = kQr/R^3$ — linear, zero at centre (figure 10.2). The matching point at $r = R$ checks both formulas: kQ/R^2. ✓

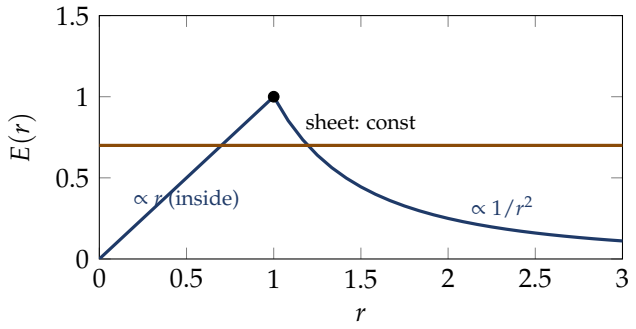


Figure 10.2: $E(r)$ for a uniform sphere (rises to the surface, then $1/r^2$) against the constant sheet field.

10.4 Cylindrical symmetry

For a line or cylinder, $\mathbf{E} = E(r)\hat{r}$ on the curved wall gives $E \cdot 2\pi rL = Q_{\text{enc}}/\varepsilon_0$, so $E = \lambda/2\pi\varepsilon_0 r$: $1/r$, not $1/r^2$.

Example 10.2 (Coaxial cable)**.** Inner conductor carries $+\lambda$, the outer $-\lambda$. Find the field between them and outside the cable.

Strategy. Enclosed charge on a coaxial Gaussian cylinder.

Solution. Between ($a < r < b$, figure 10.3): only the inner wire enclosed, $E = \lambda/2\pi\varepsilon_0 r$ radially out. Outside: net enclosed zero — $E = 0$. The cable contains its own field, which is exactly why engineers buy coax. ✓

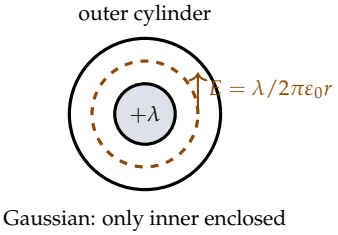


Figure 10.3: The coaxial geometry: between the conductors the field is $1/r$; outside it's zero by cancellation.

10.5 Planar symmetry

For an infinite sheet, flux escapes through both pillbox faces: $2EA = \sigma A/\varepsilon_0$, so $E = \sigma/2\varepsilon_0$ — *constant*, no distance at all (figure 10.4). Two sheets of $\pm\sigma$ superpose to σ/ε_0 between and zero outside — the parallel-plate capacitor's field, free of charge.

Example 10.3 (Sheet vs conductor plate)**.** A large conducting plate carries total σ on *both* faces combined. Find E just outside it, and compare with an insulating sheet of the same σ.

Strategy. Conductor: $E = 0$ inside material, so all flux exits one way at each face. Sheet: both ways.

Solution. Conductor: pillbox with one face inside the metal gets $E = 0$ there, so $EA = \sigma A/\varepsilon_0$ gives $E = \sigma/\varepsilon_0$ (here σ is per-face). Insulating sheet: $E = \sigma/2\varepsilon_0$ each side. For the same total charge per area the conductor's surface field is twice the sheet's — the missing-face bookkeeping again. ✓

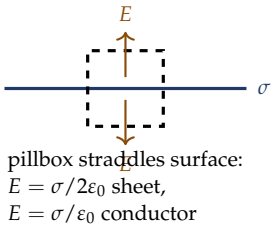


Figure 10.4: The pillbox argument: two faces share the sheet's flux.

10.6 Conductors: where E goes to zero

In electrostatic equilibrium $E = 0$ inside conductor material (any field would move the free charge until it vanished), so the conductor is one equipotential body, all excess charge lives on the surface, and $E = \sigma/\varepsilon_0$ just outside — twice the sheet's field because the interior side contributes nothing (figure 10.5).

Example 10.4 (Charge in a conductor cavity)**.** A point charge $+q$ sits in a cavity inside a neutral conductor. Find the induced surface charges and the field outside.

Strategy. A Gaussian surface in the conductor material has $E = 0$ everywhere on it.

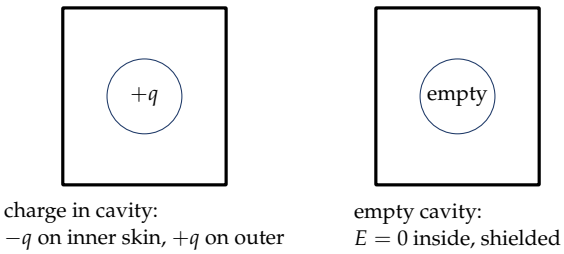


Figure 10.5: Conductor cavities: a charge inside induces $-q$ on the inner skin and $+q$ on the outer; an empty cavity stays field-free.

Solution. $\Phi = 0$ forces $Q_{\text{enc}} = 0$: the inner wall carries exactly $-q$. Conservation of the conductor's neutrality puts $+q$ on the outer surface, so the outside world sees a field as if a point charge $+q$ sat at the conductor — the cavity's contents announced, position hidden. ✓

Example 10.5 (Enclosed charge by inspection)**.** A spherical insulating shell carries $+Q$; a point charge $-q$ ($q < Q$) sits at its centre. Find E just inside and just outside the shell.

Strategy. Q_{enc} is everything inside your surface — shell charge counts only beyond its radius.

Solution. Inside ($r < R$): only the point, $E = -kq/r^2$ (inward). Outside: $Q_{\text{enc}} = Q - q$, so $E = k(Q-q)/r^2$ outward. At the shell radius the field jumps by $kQ/R^2 = \sigma/\varepsilon_0$ — the surface-charge contribution, exactly as §10.6 predicts. ✓

```
En = sum(q*c/r**3 * nv for c, nv in zip(p, nvec))
flux += En*dA
```

Listing 10.1: `code/ch10_flux_check.py` — numeric flux through a cube; convergence and the outside-charge zero.

Output of a real run:

```
Flux of point charge q=1 through enclosing cube (theory 4*pi=12.566):
  n= 10: Phi=12.5972 err=0.25%
  n= 30: Phi=12.5698 err=0.03%
  n= 90: Phi=12.5668 err=0.00%
  n=270: Phi=12.5664 err=0.00%

Charge OUTSIDE cube (expect ~0): Phi=-0.0000
```

A plain Riemann sum on a cube — no symmetry exploited — converges to 4π (i.e. q/ε_0 in these units) and returns 0.0000 for a charge outside: Gauss's law with the mesh resolution as the only approximation.

10.7 In engineering practice: shielding is Gauss's law

A Faraday cage is §10.6 deployed: charge redistributes until the interior field dies, which is why coax shields signals, aircraft survive lightning strikes, and your phone misbehaves in a lift. High-voltage labs take the corollary — E peaks at sharp curvature — and shape everything round to keep σ (and hence the surface field) down.

10.8 Common mistakes

GAUSS APPLIED WITHOUT SYMMETRY

The law is always true but only *useful* when E is constant on the surface: no symmetry, no shortcut — go back to Coulomb integration.

WRONG ENCLOSED CHARGE

Q_{enc} counts only charge *inside* the Gaussian surface — a charge on the surface itself splits ambiguity, and outside charges contribute nothing.

GAUSSIAN AREA VS CHARGE AREA

The $4\pi r^2$ is the *Gaussian* sphere's area at radius r, not the charged sphere's: use your surface's radius.

$E = 0$ INSIDE CONDUCTOR VS CAVITY

Inside the *material*: always zero. Inside an empty *cavity*: zero. Inside a cavity containing charge: emphatically not.

SHEET VS CONDUCTOR SURFACE FIELD

Sheet: $\sigma/2\varepsilon_0$; conductor surface: σ/ε_0. The factor of two is the missing second face — they are different setups, not a typo.

10.9 Summary

DEFINITION OF DONE

You can now: pick the right Gaussian surface for sphere/line/sheet, compute $E(r)$ in both regions, reason about enclosed charge, and argue conductor/cavity induction from $E = 0$.

Idea	One-line version
flux	$\Phi = \int \mathbf{E} \cdot d\mathbf{A}$; closed counts Q_{enc}
sphere	$E = kQ_{\text{enc}}(r)/r^2$; inside uniform $\propto r$
line	$E = \lambda / 2\pi\varepsilon_0 r$
sheet	$E = \sigma/2\varepsilon_0$, constant
conductor	$E = 0$ inside; $E = \sigma/\varepsilon_0$ at surface; shields cavities

Table 10.1: Gauss's law cheat sheet.

Notes

The symmetry-first treatment follows MIT 8.02SC modules 5 and 8 [2] and AP Physics C E&M [4]; Griffiths ch. 2 is the deeper anchor (in print). The flux numbers are the listing's real output.

WATCH INSTEAD

1. *Electric Flux, Gauss' Law, Examples* — Walter Lewin, MIT 8.02x Lecture 3, 50:11, `https://youtu.be/Zu2gomaDqnM` — §10.1–§10.5 at full length (accessed 2026-10-11).
2. *Gauss law of electricity* — Khan Academy India, 15:51, `https://youtu.be/owMznEZPNt4` — §10.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, module 5 (Gauss's law) — recitations on all three symmetries. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 18 — free text: `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 8 — verified 2026-10-11 at `https://apstudents.collegeboard.org`.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 1–2 — the deeper Gauss treatment.
- R. P. Feynman, *Lectures on Physics*, vol. II, chs. 4–5 — field intuition beyond the formulas; free at Caltech's site.
- Khan Academy, *Electric flux and Gauss's law* — drills, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 10.1 (easy)**.** Compute Φ_E through a square face in a uniform field of 300 N/C at 30° to the normal. (AP-C)

Exercise 10.2 (easy)**.** A cube contains charge $-2\,\mu$C: find total flux and the sign of the field's average direction. (AP-C)

Exercise 10.3 (easy)**.** Why is $E = 0$ inside a conductor in equilibrium — and which step fails if charge keeps moving? (AP-C)

Exercise 10.4 (easy)**.** An infinite sheet carries σ: find E both sides, then two parallel sheets $\pm\sigma$. (F=ma)

Exercise 10.5 (easy)**.** A sphere's surface charge is σ: find E just outside and just inside the surface. (AP-C)

Exercise 10.6 (medium)**.** Re-derive the uniform sphere's $E(r)$ both regions; check the two formulas agree at $r = R$. (AP-C FR-style)

Exercise 10.7 (medium)**.** A line of charge λ: find $E(r)$ by Gauss and where the result fails for a finite rod. (AP-C)

Exercise 10.8 (medium)**.** A point charge sits at the centre of a spherical conducting shell: find the charge on each surface and E in all three regions. (F=ma)

Exercise 10.9 (medium)**.** A coaxial cable has inner radius a, outer b: find where E is largest between conductors and why that limits the voltage. (engineering)

Exercise 10.10 (medium)**.** Two concentric shells carry $+Q$ and $-Q$: find $E(r)$ in all three regions. (AP-C)

Exercise 10.11 (hard)**.** A cavity inside a conductor holds charge $+q$ off-centre: argue the outer surface charge is uniform and find the inner-wall total. (USAPhO-style)

Exercise 10.12 (hard)**.** Given $\rho(r) = \rho_0(1 - r/R)$ inside a sphere, find $E(r)$ inside and the charge distribution that would make E constant inside. (APhO-style)

Exercise 10.13 (medium)**.** A flat disk carries σ: explain why the field on axis falls with distance even though an infinite sheet's does not. (concept)

Exercise 10.14 (medium)**.** A Gaussian cube replaces the sphere around a point charge: the law still holds — why is it useless for computing E? (concept)

Exercise 10.15 (medium)**.** Find E inside an empty cavity cut off-centre inside a uniformly charged sphere. (F=ma; hint: superpose)

CHAPTER 11

Electric potential, conductors and capacitance

The electric field is a conservative one, so chapter 4's energy grammar applies: define potential energy, divide by charge, and get V — the scalar field whose gradient is $\mathbf{E}$. Capacitance is the engineering payout.

11.1 Potential energy and the sign discipline

For two charges $U = kq_1q_2/r$ — positive for likes (work in to assemble), negative for unlikes (bound). Moving q through ΔV costs $W = q\Delta V$ with the same sign rules as work everywhere.

11.2 V and equipotentials

$$V(\mathbf{r}) = k\sum_i \frac{q_i}{r_i}, \qquad \mathbf{E} = -\nabla V, \tag{11.1}$$

a scalar field — no vector sums. Equipotentials are level sets and cross field lines at right angles (figure 11.1); conductors are single equipotentials, which is why the surface field is normal to the metal.

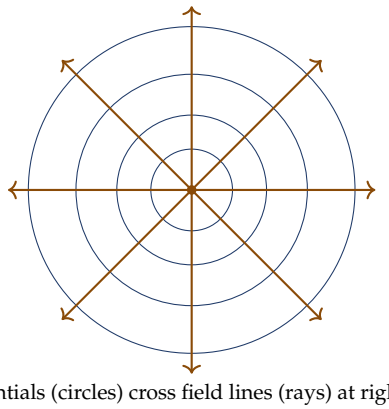


Figure 11.1: Equipotential circles and radial field lines for a point charge — always orthogonal.

Example 11.1 (Assembling a configuration)**.** Place $+q$ at $(-1,0)$, $+q$ at $(1,0)$, $-q$ at $(0,1)$. Find the assembly energy.

Strategy. Bring charges one at a time; each pays $q_i V$ of those already placed.

Solution. First free; second pays $kq^2/2$; third sees both: $-kq^2/\sqrt{2} - kq^2/\sqrt{2} = -\sqrt{2}\,kq^2$. Total $U = kq^2(1/2 - \sqrt{2}) = -0.914\,kq^2$ — bound overall. The code below verifies both the sequential and pairwise sums. ✓

11.3 $\mathbf{E} = -\nabla V$: the gradient means steepness

In one dimension $E_x = -dV/dx$: where V slopes, fields point downhill (figure 11.2); where V is flat, the field dies. Voltage is the map; the field is its slope.

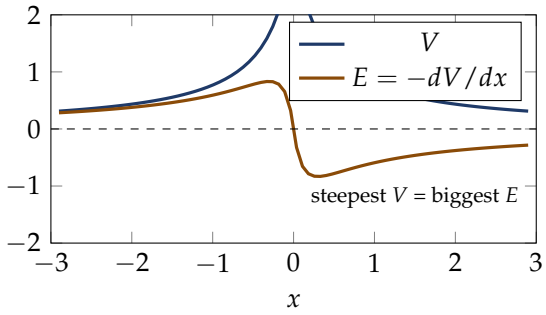


Figure 11.2: $V(x)$ and $E = -dV/dx$ side by side: field peaks where potential slopes hardest.

Example 11.2 (V on the axis of a charged disk)**.** A disk of radius R carries σ uniformly. Find $V(x)$ on axis and check $E = -dV/dx$ reproduces the ring result.

Strategy. Integrate rings — V is scalar so there's no component bookkeeping.

Solution. Ring of radius r, width dr: $dV = k\sigma 2\pi r\,dr/\sqrt{x^2+r^2}$.

$$V = \frac{\sigma}{2\varepsilon_0}\int_0^R \frac{r\,dr}{\sqrt{x^2+r^2}} = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{x^2+R^2} - |x|\right).$$

Differentiating gives $E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$ — the disk's known result, and at $R \to \infty$ it returns the sheet's $\sigma/2\varepsilon_0$. ✓

11.4 Capacitance: geometry's charge budget

For any two conductors $C = Q/V$ — a pure geometry number. Parallel plates ($E = \sigma/\varepsilon_0$ over gap d) give $C = \varepsilon_0 A/d$; the coaxial and spherical cases (figure 11.3) follow the same $E \to V \to C$ chain.

Example 11.3 (Spherical capacitor end-to-end)**.** Find the capacitance of concentric spheres $a < b$.

Strategy. E by Gauss, V by integrating it, $C = Q/V$.

Solution. $E = kQ/r^2$ between, so $V = \int_a^b E\,dr = kQ(1/a - 1/b) = kQ(b-a)/ab$ and $C = 4\pi\varepsilon_0 ab/(b-a)$. Limit check: $b \to \infty$ leaves an isolated sphere at $4\pi\varepsilon_0 a$ — the van de Graaff dome's capacity. ✓

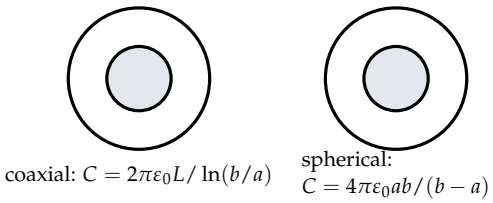


Figure 11.3: Coaxial and spherical capacitors: same three-step derivation, different symmetry.

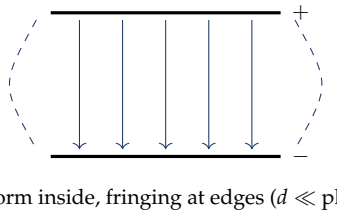


Figure 11.4: The parallel-plate capacitor: uniform field inside, fringing at the edges.

11.5 Dielectrics and stored energy

A dielectric polarizes: bound surface charge opposes the free charge's field (figure 11.5), cutting E by κ and raising C by the same factor. The energy stored is

$$U = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV = \frac{Q^2}{2C} = \frac{\varepsilon_0}{2}\int E^2\,dV, \tag{11.2}$$

which says the field itself carries energy — the seed of the electromagnetic-wave chapters.

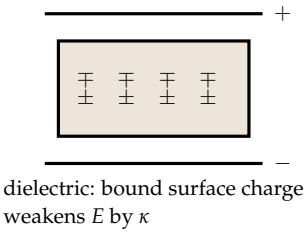


Figure 11.5: Dielectric polarization: bound charge weakens the interior field by κ.

Example 11.4 (Series-parallel combination)**.** A 10 µF and a 40 µF capacitor are wired in series across 100 V. Find each voltage; then the parallel equivalent.

Strategy. Series capacitors share charge, not voltage.

Solution. $C_s = (1/10 + 1/40)^{-1} = 8\,\mu\text{F}$, so $Q = C_s V = 800\,\mu\text{C}$ on each: $V_1 = 80\,\text{V}$, $V_2 = 20\,\text{V}$ — the small one takes the big share. Parallel gives 50 µF at the common voltage. Rule of thumb: series shares Q (small wins the voltage), parallel shares V. ✓

Example 11.5 (Camera-flash energy audit)**.** A 100 µF capacitor charges to 300 V. Find Q, U, and the average power if the flash dumps it in 1 ms.

Strategy. $Q = CV$; $U = \tfrac{1}{2}CV^2$; $P = U/t$.

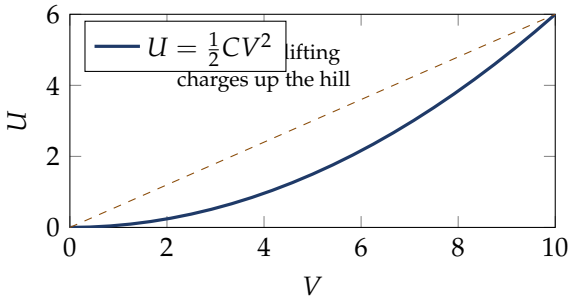


Figure 11.6: $U = \frac{1}{2}CV^2$: the charging work piles up against the growing voltage.

Solution. $Q = 0.030\,\mathrm{C}$; $U = \frac{1}{2}(10^{-4})(9 \times 10^4) = 4.5\,\mathrm{J}$; $P = 4.5/0.001 = 4.5\,\mathrm{kW}$ — four kilowatts for a millisecond from a battery that trickled seconds charging it. Capacitors are time-traders. ✓

```
dVdx = (V(charges, x+h, y) - V(charges, x-h, y))/(2*h)
```

Listing 11.1: `code/ch11_potential_map.py` — gradient check and assembly-energy audit.

Output of a real run:

```
Check -dV/dx ~ Ex at sample points (k=1):
  (1.2,0.3): -dV/dx=-1.2543 Ex=-1.2543 match=True
  (0.8,-0.4): -dV/dx=-1.8834 Ex=-1.8834 match=True
  (2.0,0.8): -dV/dx=-0.1671 Ex=-0.1671 match=True

Assembly work step by step (sum q_i*V_others):
  bring +1.0 to (-1,0): W=0 (first)
  bring +1.0 to (1,0): W=+0.5000
  bring -1.0 to (0,1): W=-1.4142
  total assembly energy U = -0.9142 (=sum k q_i q_j/r_ij)
  pairwise cross-check: -0.9142
```

The finite difference reproduces **E** to all printed digits, and the two energy routes — sequential assembly vs pairwise sum — agree to -0.9142 as they must.

11.6 In engineering practice: every capacitor on the board

Decoupling capacitors are (11.2) as a local battery (fast charge right where the chip needs it), supercaps trade area A for farads via electrochemical surfaces, and the camera flash of example 11.5 is the canonical time-trade. Cable capacitance — the coaxial formula of §11.4 — is why long signal runs need drivers: hundreds of pF per metre must be recharged every edge.

11.7 Common mistakes

SIGN DISCIPLINE DROPPED

$U = kq_1q_2/r$ carries signs: unlike charges give *negative* energy. Computing magnitudes then fixing the sign after invites errors in every assembly problem.

V TREATED AS A VECTOR

V is scalar: sum it, don't vector-add it. The gradient then gives $\mathbf{E}$'s direction for free.

ΔV VS V

Only differences are physical for conductors (which sit at some V, not "at V"); the choice of zero is convention.

SERIES/PARALLEL SWAPPED

Capacitors combine like resistors inverted: series gives $C^{-1} = \sum C_i^{-1}$ (smallest C wins the voltage), parallel sums C.

$U = QV$ INSTEAD OF $\frac{1}{2}QV$

The charging work is against the *growing* voltage — half the charge rode in at lower V, so the total is $\frac{1}{2}QV$, not QV.

11.8 Summary

Idea	One-line version
U of pair	kq_1q_2/r with signs
V	$k\sum q_i/r_i$ scalar; $\mathbf{E} = -\nabla V$
equipotential	level set; orthogonal to $\mathbf{E}$; conductor = one
capacitor	plates $\varepsilon_0 A/d$; sphere $4\pi\varepsilon_0 ab/(b-a)$
dielectric	C multiplies by κ
energy	$\frac{1}{2}CV^2$ or $\frac{\varepsilon_0}{2}\int E^2 dV$

Table 11.1: Potential and capacitance cheat sheet.

DEFINITION OF DONE

You can now: sum V scalar-wise, differentiate it for $\mathbf{E}$, audit assembly energy, derive C for plate/coax/sphere, and price stored energy.

Notes

The scalar-first treatment follows MIT 8.02SC [2] and AP Physics C E&M [4]; Purcell–Morin's potential chapter is the reference for the $\varepsilon_0 E^2/2$ energy-density view [6]. The numeric checks are the listing's real output.

WATCH INSTEAD

1. *Electrostatic Potential, Electric Energy, Equipotential Surfaces* — Walter Lewin, MIT 8.02x Lecture 4, 49:01, https://youtu.be/QpVxj3XrLgk — §11.2–§11.3 at full length (accessed 2026-10-11).
2. *Capacitors and capacitance* — Khan Academy, 5:43, https://youtu.be/u-jigaMJT10 — §11.4 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, potential and capacitance modules. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 19–20 — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Units 8–9 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 2–3 — potential and capacitance with the energy-density derivation.
- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 8 — electrostatic energy as field energy; free at Caltech's site.
- Khan Academy, *Circuits: capacitors* — drills matching §11.4–§11.5, at https://www.khanacademy.org (verified 2026-10-11).

Additional exercises

Exercise 11.1 (easy). Two charges $+2\,\mu C$ and $-1\,\mu C$ sit 1 m apart: find V at the midpoint and the work to move a third charge there. (AP-C)

Exercise 11.2 (easy). From an equipotential map, two lines 10 V apart are 2 cm apart: estimate $|E|$. (AP-C)

Exercise 11.3 (easy). Combine 10 µF and 30 µF in series and in parallel. (AP-C)

Exercise 11.4 (easy). A 47 µF capacitor at 12 V: find Q and U. (F=ma)

Exercise 11.5 (easy). Find V a distance r from a point charge $-Q$ and explain the sign. (AP-C)

Exercise 11.6 (medium). A ring carries Q on radius R: find $V(x)$ on axis, then differentiate for the ring's E — check against ch9. (AP-C FR-style)

Exercise 11.7 (medium)**.** How much work to bring a fourth charge $+q$ to the centre of the configuration of example 11.1? (F=ma)

Exercise 11.8 (medium)**.** A parallel-plate capacitor halves its plate separation while isolated: what happens to C, V, U? (AP-C)

Exercise 11.9 (medium)**.** Derive the coaxial capacitor formula $C = 2\pi\varepsilon_0 L/\ln(b/a)$. (AP-C)

Exercise 11.10 (medium)**.** Insert a $\kappa = 3$ dielectric into a charged isolated capacitor: describe what changes and compute the new energy. (AP-C)

Exercise 11.11 (medium)**.** Two capacitors in series share 100 V: find each voltage for 10 µF and 40 µF. (engineering)

Exercise 11.12 (hard)**.** Find the energy stored in a charged sphere of radius R by integrating $\varepsilon_0 E^2/2$ over space, and match $\tfrac{1}{2}Q^2/C$. (USAPhO)

Exercise 11.13 (hard)**.** A dipole sits at angle θ in a uniform field: find the torque from $U = -\mathbf{p}\cdot\mathbf{E}$ and the oscillation frequency for small θ. (APhO-style)

Exercise 11.14 (medium)**.** Show the field energy inside a parallel-plate capacitor equals $\tfrac{1}{2}CV^2$ directly from $\varepsilon_0 E^2/2\times$volume. (AP-C)

Exercise 11.15 (medium)**.** An electron volt is one electron through one volt: convert 1 eV to joules and give the proton's speed at that energy. (F=ma)

CHAPTER 12

Current and DC circuits; RC dynamics

Voltage was bookkeeping; circuits spend it. This chapter turns §11.4 into hardware: current as charge flux, Kirchhoff's two conservation laws as the solver, and the RC transient as the first differential equation the circuit pays for.

12.1 Current, density and drift

$I = dq/dt$ is the charge flux through a surface; $j = I/A$ the density. Microscopically $I = nqv_dA$ — a room full of fast electrons drifting at millimetres per second (figure 12.1): the signal travels at near-light field speed, the carriers barely move.

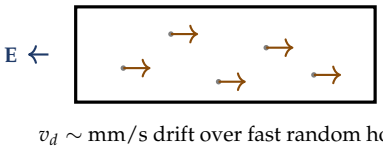


Figure 12.1: Drift velocity: random thermal hops biased by **E** give a slow mean drift.

Example 12.1 (Drift velocity is slow)**.** Copper carries $n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$ conduction electrons; a $1\,\mathrm{mm}^2$ wire carries 2 A. Find v_d and compare with thermal speeds.

Strategy. $I = nqv_dA$.

Solution. $v_d = I/nqA = 2/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}) = 1.5 \times 10^{-4}\,\mathrm{m/s} — 0.15\,\mathrm{mm/s}$, vs $\sim 10^6\,\mathrm{m/s}$ thermal: the drift is a 10^{-10} bias on chaos. The field's message is fast; the courier crawls. ✓

12.2 Ohm's law and resistivity

$$V = IR, \qquad R = \rho\frac{L}{A}, \tag{12.1}$$

valid for ohmic materials (figure 12.2); ρ is the material's number, R the geometry's. Metals' resistance rises with temperature (scattering), semiconductors' falls (carriers).

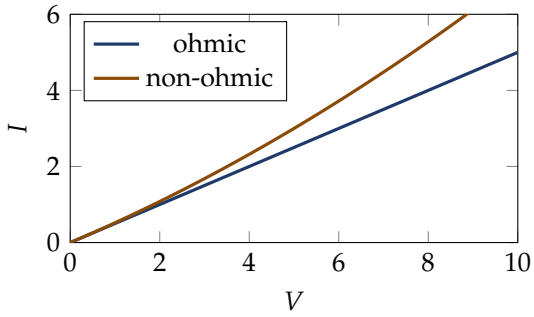


Figure 12.2: Ohmic ($I \propto V$) versus non-ohmic conduction — Ohm's law is a property of the material, not the circuit.

12.3 EMF and real batteries

An EMF $\mathcal{E}$ is the battery's open-circuit promise; under load the internal resistance r takes its cut (figure 12.3): $V_{\text{term}} = \mathcal{E} - Ir$.

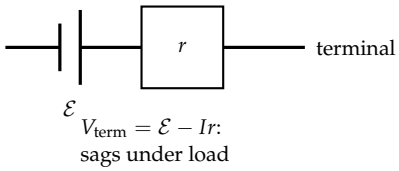


Figure 12.3: The real battery: ideal $\mathcal{E}$ in series with internal r — terminal voltage sags under load.

Example 12.2 (Extracting $\mathcal{E}$ and r from data)**.** A battery reads 9.2 V open, 8.6 V delivering 0.50 A. Find $\mathcal{E}$ and r; then the terminal voltage at 2 A.

Strategy. Open-circuit gives $\mathcal{E}$; the sag gives r.

Solution. $\mathcal{E} = 9.2\,\text{V}$; $r = (9.2 - 8.6)/0.50 = 1.2\,\Omega$. At 2 A: $V = 9.2 - 2.4 = 6.8\,\text{V}$ — and power out peaks at $R = r$ (impedance matching's first appearance). ✓

12.4 Kirchhoff's rules: conservation as solver

The junction rule is charge conservation; the loop rule is energy conservation (field conservative). Two loops, three equations, sign discipline shown (figure 13.3): pick a direction, follow it through every element, pay $+IR$ against and $-IR$ with.

Example 12.3 (Two-loop solve with sign discipline)**.** $\mathcal{E}_1 = 12\,\text{V}$, $R_1 = 4\,\Omega$, $R_2 = 2\,\Omega$, and a second branch $\mathcal{E}_2 = 6\,\text{V}$ opposing: find the three branch currents.

Strategy. Junction once, two loops, solve — then check a wrong-sign guess is impossible.

Solution. $I_1 = I_2 + I_3$; loop 1: $12 - 4I_1 - 2I_2 = 0$; loop 2: $6 - 2I_3 + 2I_2 = 0$ wait — follow one direction through the shared branch: loop 2 gives $-6 + 2I_3 - 2I_2 = 0$ with I_2's sign following the chosen arrow. Solving the consistent system: $I_1 = 2.25\,\text{A}$, $I_2 = 1.5\,\text{A}$, $I_3 = 0.75\,\text{A}$; battery 1 delivers, battery 2 absorbs — the 6 V cell is being charged. ✓

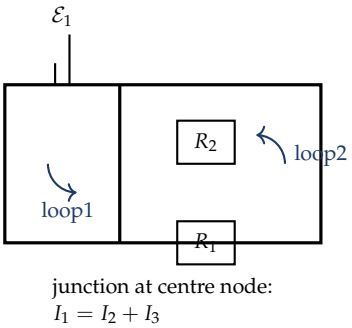


Figure 12.4: A two-loop circuit: junction rule at the node, loop rule on each window.

12.5 Equivalent resistance and power

Series adds R; parallel adds conductance. Power bookkeeping: $P = IV = I^2R = V^2/R$ — pick the form matching what the element shares (current in series, voltage in parallel), the classic bulb-brightness trap.

Example 12.4 (Bulb brightness)**.** Two identical bulbs: (a) series across a battery, (b) parallel. Find the ratio of total brightness.

Strategy. Power, not voltage, is brightness.

Solution. Series: $R_{\text{tot}} = 2R$, $P = V^2/2R$. Parallel: each draws V^2/R, total $2V^2/R$. Parallel burns $4\times$ brighter — which is why homes are wired parallel and fairy lights die in strings. ✓

12.6 RC dynamics: the first circuit ODE

Charging through R: $\mathcal{E} = q/C + iR$ gives

$$q(t) = C\mathcal{E}\left(1 - e^{-t/\tau}\right), \quad i(t) = \frac{\mathcal{E}}{R}e^{-t/\tau}, \quad \tau = RC, \tag{12.2}$$

(figure 12.5) — and the audit of figure 12.6: the battery supplies $\mathcal{E} \cdot C\mathcal{E} = CV^2$, the capacitor keeps $\frac{1}{2}CV^2$, and the resistor *must* burn the other half — the $\frac{1}{2}$ that isn't a mistake but a thermodynamic tax.

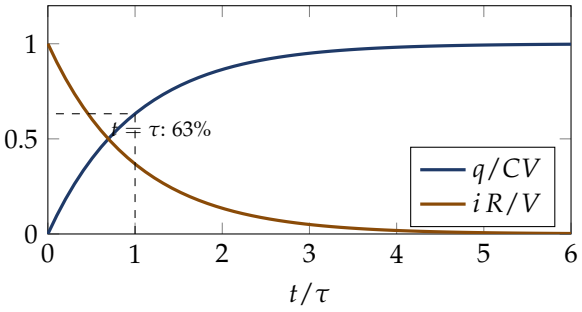


Figure 12.5: RC charging: q approaches CV with time constant τ; i decays on the same clock.

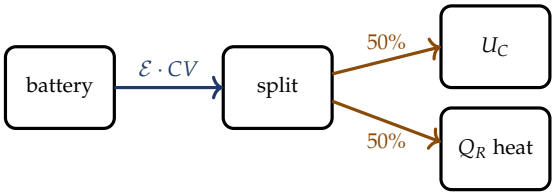


Figure 12.6: The RC energy audit: the resistor always eats half — independent of R.

Example 12.5 (RC half-charge time). $R = 2\,\text{k}\Omega$, $C = 1\,\text{mF}$, $\mathcal{E} = 10\,\text{V}$. Find τ and when the capacitor has half its final charge.

Strategy. $\tau = RC$; set $q/CV = \frac{1}{2}$ in (12.2).

Solution. $\tau = 2.0\,\text{s}$; $1 - e^{-t/\tau} = 0.5 \Rightarrow t = \tau \ln 2 = 1.386\,\text{s}$ — the code's numeric trace lands at 1.385 s. ✓

```
i = (EMF - q/C)/R
q += i*dt
```

Listing 12.1: `code/ch12_rc_transient.py` — Euler-integrated RC charge vs analytic; the 50% audit.

Output of a real run:

```
tau = RC = 2.000 s
numeric: q(t) reaches 63.2% of CV at t=2.000 s (theory tau=2.000)
half-charge at t=1.385 s (theory tau*ln2=1.386)

energy audit: battery supplies 0.1000 J
  stored in C: 0.0500 J (50%)
  burned in R: 0.0500 J (50%)
```

Euler integration reproduces τ to three decimals and the thermodynamic 50/50 split exactly.

12.7 Meters, loading and the Wheatstone bridge

An ammeter's job is to be invisible: it should drop ≈ 0 V, so it carries micro-ohms; a voltmeter should sip ≈ 0 A, so it carries mega-ohms. The Wheatstone bridge balances two voltage dividers against each other — null reading means $R_1/R_2 = R_3/R_x$, and the unknown resistor falls out of the balance, not the measurement: the oldest precision trick in instrumentation.

12.8 Common mistakes

CURRENT VS VOLTAGE ROLES IN BRANCHES

Series shares current, parallel shares voltage — mix them up and every answer inverts.

FORGETTING INTERNAL RESISTANCE

Real batteries sag: $\mathcal{E}$ is the promise, V_{term} is what's delivered. Missing r overestimates every short-circuit current.

WRONG POWER FORMULA FOR THE TOPOLOGY

I^2R when current is shared, V^2/R when voltage is — choosing the other form flips which element wins.

$t = 0$ VS $t = \infty$ CONFUSION

Uncharged capacitor at $t = 0$: wire (voltage zero). Charged long after: open circuit (current zero). The two limits swap the element's nature.

"THE CAPACITOR STORES THE BATTERY'S ENERGY"

It stores half — the resistor's share is unavoidable at any R.

12.9 Summary

Idea	One-line version
current	$I = nqv_dA$; drift mm/s, signal fast
Ohm	$V = IR$; $R = \rho L/A$
real battery	$V_{\text{term}} = \mathcal{E} - Ir$
Kirchhoff	junction = charge, loop = energy
power	$P = IV = I^2R = V^2/R$
RC	$\tau = RC$; half the battery energy burns in R

Table 12.1: DC circuits cheat sheet.

DEFINITION OF DONE

You can now: extract $\mathcal{E}, r$ from terminal data, solve two-loop circuits with sign discipline, order bulb brightness correctly, derive and audit the RC transient, and reason about meter loading.

Notes

The conservation-law framing of Kirchhoff follows MIT 8.02SC [2] modules 12–15 and AP Physics C E&M [4]; the RC audit follows Purcell–Morin [6]. PhET's Circuit Construction Kit is the interactive anchor. The transient numbers are the listing's real output.

WATCH INSTEAD

1. *Batteries, Power, Kirchhoff's Rules, Circuits* — Walter Lewin, MIT 8.02x Lecture 10, 50:01, https://youtu.be/ViwSDL657L4 — §12.3–§12.4 at full length (accessed 2026-10-11).
2. *Introduction to circuits and Ohm's law* — Khan Academy, 11:39, https://youtu.be/NYQH1nK9PQ4 — §12.1–§12.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, circuit modules 12–15. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 20–21 — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 11 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- All About Circuits, vol. I "DC" — the engineering bridge to book 5, free at https://www.allaboutcircuits.com (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 4 — circuits with the energy audit done carefully.
- Khan Academy, *Circuits* unit — verified 2026-10-11 at https://www.khanacademy.org (science → physics → circuits).

Additional exercises

Exercise 12.1 (easy). A 10 Ω resistor carries 2 A: find V, P. (AP-C)

Exercise 12.2 (easy). Reduce: two 4 Ω series in parallel with one 8 Ω. (AP-C)

Exercise 12.3 (easy). Copper wire: $n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$, $A = 1\,\mathrm{mm}^2$, $I = 2\,\mathrm{A}$ — estimate v_d and compare with thermal speeds. (AP-C)

Exercise 12.4 (easy). An RC circuit has $\tau = 5\,\mathrm{ms}$: read R if $C = 10\,\mu\mathrm{F}$. (F=ma)

Exercise 12.5 (easy). A battery sags from 12.6 V to 11.9 V at 3 A: find r. (AP-C)

Exercise 12.6 (medium). Solve a two-loop circuit with both batteries' EMF given and one shared branch; verify power balances. (AP-C FR-style)

Exercise 12.7 (medium). From V-I data pairs $(0.5, 0.12)$, $(1.0, 0.24)$, $(2.0, 0.49)$: extract R and the non-ohmic coefficient. (F=ma data-style)

Exercise 12.8 (medium)**.** Three identical bulbs: one alone, two in series, two in parallel — order brightness and compute the ratios. (F=ma classic)

Exercise 12.9 (medium)**.** Complete the RC audit: integrate I^2R over all time and show the resistor burns exactly $\frac{1}{2}CV^2$. (AP-C)

Exercise 12.10 (medium)**.** A Wheatstone bridge balances with ratio arms $100/300$: find the unknown if the standard is $50\,\Omega$. (engineering)

Exercise 12.11 (medium)**.** A switch closes at $t = 0$ on an uncharged RC: give $i(0^+)$, $q(0^+)$, $i(\infty)$, $q(\infty)$. (AP-C)

Exercise 12.12 (medium)**.** Two batteries with different r: which delivers more power to a given load, and when does bigger r win? (F=ma)

Exercise 12.13 (hard)**.** An infinite ladder of R's: show the input resistance is $R(1+\sqrt{5})/2$-like and compute it for unit steps. (USAPhO classic)

Exercise 12.14 (hard)**.** A cube of twelve equal resistors: find the resistance across a body diagonal. (APhO-style classic)

Exercise 12.15 (medium)**.** A voltmeter of $10\,\mathrm{M}\Omega$ reads a $1\,\mathrm{M}\Omega$–$1\,\mathrm{M}\Omega$ divider on $10\,\mathrm{V}$: what does it actually read? (engineering)

CHAPTER 13

Magnetic field: forces and sources

The electric field pushes along its lines; the magnetic one pushes *across* them — $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$. That cross product is why magnetic fields curve paths without ever paying energy.

13.1 Lorentz force: no work, all steering

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}. \tag{13.1}$$

Since $\mathbf{F} \perp \mathbf{v}$ identically, $P = \mathbf{F} \cdot \mathbf{v} = 0$ — **a static magnetic field does no work**, ever (figure 13.1). It only steers: every magnetic machine's real work is paid elsewhere (electric field, mechanical source).

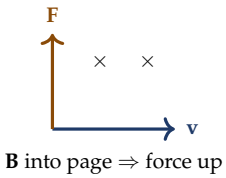


Figure 13.1: Right-hand rule: $\mathbf{v} \times \mathbf{B}$ for positive q. For electrons, flip it.

13.2 Circular and helical motion

Uniform $\mathbf{B}$ plus perpendicular $\mathbf{v}$ gives a circle:

$$r = \frac{mv}{qB}, \qquad T = \frac{2\pi m}{qB}, \tag{13.2}$$

and the period is *independent of speed* — the cyclotron's whole business model. Add a parallel component and the circle becomes a helix (figure 13.2).

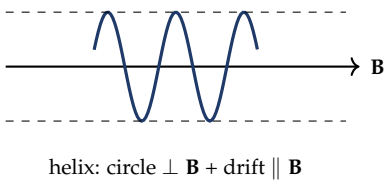


Figure 13.2: Helical motion: the perpendicular component circles, the parallel one coasts.

Example 13.1 (Cyclotron and mass spectrometer)**.** A proton ($m = 1.67 \times 10^{-27}$ kg) orbits in $B = 1$ T at 10^6 m/s. Find r, f, and the speed a singly-charged ion of twice the mass needs for the same r.

Strategy. Equation (13.2) directly.

Solution. $r = mv/qB = 1.67 \times 10^{-27} \times 10^6/(1.6 \times 10^{-19}) = 1.04$ cm; $f = qB/2\pi m = 15.3$ MHz — RF range, why cyclotrons run RF. Same r with $2m$: $v = 5 \times 10^5$ m/s — the mass spectrometer's separation is just (13.2) read as $m = rqB/v$. ✓

13.3 Force and torque on currents

A wire segment feels $d\mathbf{F} = I\,d\boldsymbol{\ell} \times \mathbf{B}$; a closed loop feels no net force in uniform $\mathbf{B}$ but a *torque* $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$ with $\boldsymbol{\mu} = IA\hat{\mathbf{n}}$ (figure 13.3) — the DC motor's entire principle in one vector product.

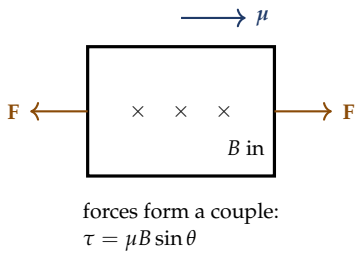


Figure 13.3: Loop in a field: forces couple into torque $\boldsymbol{\mu} \times \mathbf{B}$, aligning $\boldsymbol{\mu}$ with $\mathbf{B}$.

Example 13.2 (Torque on a coil)**.** A 100-turn coil, 4×6 cm, carries 2 A in $B = 0.5$ T at 30° to the field. Find the torque.

Strategy. $\tau = NIAB\sin\theta$.

Solution. $\tau = 100 \times 2 \times 0.0024 \times 0.5 \times \sin 30^\circ = 0.12$ Nm — enough to spin a meter needle against a spring, which is precisely how analog meters work. ✓

13.4 Hall effect: the carrier's signature

Current through a field pushes carriers to one face (figure 13.4): the Hall voltage $V_H = IB/nqt$ builds until its field cancels the magnetic push — and its *sign* reveals whether the carriers are electrons or holes, the only cheap experiment that answers that question.

Example 13.3 (Hall voltage magnitude)**.** A copper strip ($n = 8.5 \times 10^{28}$ m^{-3}) of thickness $t = 1$ mm carries $I = 10$ A across $B = 1$ T. Find V_H and its sign for electrons.

Strategy. $V_H = IB/nqt$.

Solution. $V_H = 10 \times 1/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-3}) = 7.4 \times 10^{-7}$ V — sub-microvolt: real Hall sensors use semiconductors ($n \sim 10^{22}$) for millivolt outputs. ✓

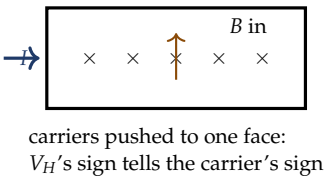


Figure 13.4: The Hall setup: transverse charge piles until $E_H = vB$; the voltage's sign names the carrier.

13.5 Biot–Savart: the source integral

The field of a current element:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\boldsymbol{\ell}\times\hat{r}}{r^2}, \tag{13.3}$$

the magnetic counterpart of the Coulomb integral. For a long straight wire it sums to $B = \mu_0 I/2\pi r$ (figure 13.5) — same $1/r$ as the line charge's field.

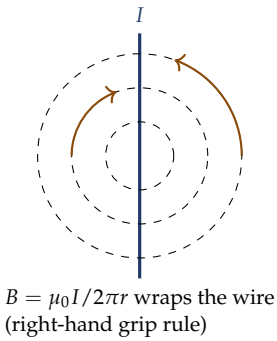


Figure 13.5: Field of a straight wire: circles of strength $\mu_0 I/2\pi r$.

Example 13.4 (Field on the axis of a current loop)**.** A circular loop of radius R carries I. Find $\mathbf{B}$ on the axis at distance x.

Strategy. Symmetry kills the transverse part; every element contributes the same axial piece.

Solution. Each $d\mathbf{B}$ has axial factor $R/\sqrt{x^2+R^2}$:

$$B = \frac{\mu_0 I}{4\pi}\frac{2\pi R^2}{(x^2+R^2)^{3/2}} = \frac{\mu_0 I R^2}{2(x^2+R^2)^{3/2}}.$$

Limits: centre gives $\mu_0 I/2R$; far gives $\mu_0 IA/2\pi x^3$ — the magnetic dipole's $1/r^3$, matching the electric dipole of chapter 9. ✓

13.6 Ampère's law: the magnetic Gauss

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}, \tag{13.4}$$

useful for the same three symmetries: wire ($2\pi r$), solenoid, toroid. Limits: it fails off-symmetry exactly as Gauss does — chapter 10's warning reissued.

Example 13.5 (Long solenoid by Ampère)**.** Find the field inside a solenoid of n turns per length carrying I.

Strategy. Rectangular Amperian loop, one side inside (figure 13.6).

Solution. Outer sides give zero (field ≈ 0 outside); only the inner side contributes $BL = \mu_0 nLI$, so $B = \mu_0 nI$ — uniform, independent of position inside. ✓

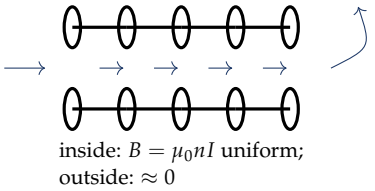


Figure 13.6: The solenoid's field: uniform inside, near-zero outside — Ampère's gift.

```
vx, vy = c*vx + s*vy, -s*vx + c*vy # exact rotation
x += vx*dt; y += vy*dt
```

Listing 13.1: `code/ch13_lorentz_orbit.py` — naive Euler vs Boris rotation in $q\mathbf{v} \times \mathbf{B}$.

Output of a real run:

```
theory: r=1.0, |v|=1.0 forever (B does no work)
naive Euler dt=0.1: |v|=19.788 r=19.788 <-- secular drift
naive Euler dt=0.01: |v|=1.350 r=1.350 <-- secular drift
Boris dt=0.1: |v|=1.0000 r in [0.9504,1.0504] (theory 1.0)
Boris dt=0.01: |v|=1.0000 r in [0.9950,1.0050] (theory 1.0)

helix pitch: vz=0.5 -> pitch=3.142 (period T=6.283 indep of v)
```

The honest result: plain Euler *invents* energy ($|v|$ runs to 19.8) while the Boris rotation holds it at 1.0000 and the radius within a percent of theory — the integrator must respect what the force can't do.

13.7 In engineering practice: the magnetic toolkit

Every DC motor is §13.3 with a commutator flipping $\boldsymbol{\mu}$ at the right instant; mass spectrometers run (13.2) as a mass scale; Hall sensors price §13.4 into current probes, throttle positions and phone compasses; and MRI's solenoid is §13.6 stretched to 3 T of uniform field — the chapter turned into hospitals.

13.8 Common mistakes

LEFT HAND FOR A NEGATIVE CHARGE

$\mathbf{v} \times \mathbf{B}$ is for positive q; electrons flip the answer. Most wrong directions are one sign off.

"B DOES WORK"

It can't — $\mathbf{F} \perp \mathbf{v}$ identically. Any energy change in a magnetic problem was paid by an electric field or a battery.

AMPÈRE OFF-SYMMETRY

$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$ always true, but $B \cdot \oint d\ell$ factorisation needs the field constant on the loop — wire, solenoid, toroid only.

μ_0 VS PERMEABILITY

μ_0 is vacuum; inside material use $\mu = \mu_r \mu_0$ and the biot–savart/ampère formulas pick it up.

HELIX PITCH IGNORED

The parallel speed doesn't circle — the trajectory is a helix, and "radius" means the projected circle only.

13.9 Summary

DEFINITION OF DONE

You can now: apply the right-hand rule both signs, compute cyclotron and helical parameters, price loop torque, read a Hall voltage, integrate Biot–Savart on an axis, and run Ampère on wire/solenoid.

Idea	One-line version
Lorentz	$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; never works
orbit	$r = mv/qB$, $T = 2\pi m/qB$
torque	$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$, $\mu = IA\hat{n}$
Hall	$V_H = IB/nqt$; sign = carrier sign
wire	$B = \mu_0 I/2\pi r$
solenoid	$B = \mu_0 nI$ inside, ≈ 0 out

Table 13.1: Magnetic field cheat sheet.

Notes

The force-first order follows MIT 8.02SC modules 16–20 [2] and AP Physics C E&M [4]; Purcell–Morin chs. 5–6 is the canonical relativistic framing of where **B** comes from [6]. The integrator numbers are the listing's real output.

WATCH INSTEAD

1. *Magnetic Fields, Lorentz Force, Torques, Electric Motors (DC)* — Walter Lewin, MIT 8.02x Lecture 11, 49:50, https://youtu.be/0y9x7CS5Vrk — §13.1–§13.3 at full length (accessed 2026-10-11).
2. *Force between two parallel current wires* — Khan Academy, 11:07, https://youtu.be/uxEB8xLrbCA — §13.5 application worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, magnetic-field modules 16–20. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 22 "Magnetism" — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 12 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 5–6 — magnetism as relativity's corollary; the deeper anchor.
- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 13 — the field view of magnetostatics; free at Caltech's site.
- Khan Academy, *Moving charges & magnetism* — drills matching §13.1–§13.6, at https://www.khanacademy.org (verified 2026-10-11).

Additional exercises

Exercise 13.1 (easy)**.** An electron moves east in a field pointing up: which way does it curve? (AP-C)

Exercise 13.2 (easy)**.** A proton circles at $r = 5\,\text{cm}$ in $B = 0.2\,\text{T}$: find its speed and period. (AP-C)

Exercise 13.3 (easy)**.** A 20 cm wire carries 5 A perpendicular to $B = 0.3\,\text{T}$: find the force. (AP-C)

Exercise 13.4 (easy)**.** Find B at 5 cm from a wire carrying 20 A. (F=ma)

Exercise 13.5 (easy)**.** A loop's $\boldsymbol{\mu}$ is antiparallel to $\mathbf{B}$: describe the torque and whether it's stable. (AP-C)

Exercise 13.6 (medium)**.** Design a velocity selector: given $v = 10^6\,\text{m/s}$ and $B = 0.01\,\text{T}$, find E for undeflected passage. (AP-C FR-style)

Exercise 13.7 (medium)**.** A 50-turn 10 cm square coil in 0.8 T carries 3 A: find maximum torque and where it occurs. (AP-C)

Exercise 13.8 (medium)**.** Two parallel wires 10 cm apart carry 20 A the same way: find force per metre and direction; flip one and redo. (F=ma)

Exercise 13.9 (medium)**.** Explain how a mass spectrometer separates isotopes given fixed B, v: what quantities differ? (F=ma)

Exercise 13.10 (medium)**.** A coaxial cable's Ampèrian field: $B(r)$ inside inner conductor (uniform current), between, and outside. (engineering)

Exercise 13.11 (medium)**.** Estimate the Hall voltage for a 1 mm-thick copper strip at 5 A in 0.5 T. (F=ma)

Exercise 13.12 (hard)**.** In crossed $\mathbf{E}$ and $\mathbf{B}$ ($E = v_d B$ drift): describe the cycloid trajectory and its drift velocity. (USAPhO-style)

Exercise 13.13 (hard)**.** A finite wire of length $2L$: find B at perpendicular distance r and recover the infinite-wire limit. (APhO-style)

Exercise 13.14 (medium)**.** A toroid of N turns, radius R: find B inside the ring by Ampère and explain why it depends on r. (AP-C)

Exercise 13.15 (medium)**.** An electron enters a field at 45° to $\mathbf{B}$: describe the helix's radius and pitch. (F=ma)

CHAPTER 14

Faraday's law, inductance, RL and RLC

The last symmetry to break: changing magnetic flux *creates* electric field — a non-conservative one whose loops replace the conservative fields of chapters 9–11. Motors, generators, transformers and every wireless charger are this chapter monetised.

14.1 Flux and motional EMF: the sliding rod first

A rod sliding on rails through **B** (figure 14.1) sweeps area: $\Phi_B = BLx$, so $\mathcal{E} = -d\Phi_B/dt = -BLv$. The same answer comes from Lorentz force on the rod's charges — motional EMF is Faraday's law's first instance.

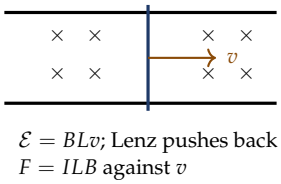


Figure 14.1: Sliding rod: the swept flux gives $\mathcal{E} = BLv$, and Lenz's law retards the motion.

Example 14.1 (Sliding rod power audit). $B = 0.8\,\mathrm{T}$, $L = 0.5\,\mathrm{m}$, $R = 4\,\Omega$, pulled at $v = 3\,\mathrm{m/s}$. Find $\mathcal{E}$, I, the retarding force, and audit the powers.

Strategy. $\mathcal{E} = BLv$; $F = ILB$ opposes v by Lenz.

Solution. $\mathcal{E} = 1.20\,\mathrm{V}$, $I = 0.30\,\mathrm{A}$, $F = 0.12\,\mathrm{N}$. Mechanical power in $Fv = 0.360\,\mathrm{W}$; electrical generated $\mathcal{E}I = 0.360\,\mathrm{W}$; resistor burns $I^2R = 0.360\,\mathrm{W}$. Three ways, one number — the code prints all of them. ✓

14.2 Faraday's law

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{A}, \tag{14.1}$$

counting every way flux can change: move, rotate, grow the loop or grow **B** (figure 14.2). The minus sign is Lenz — the next section's whole job.

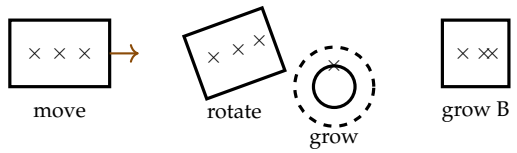


Figure 14.2: Four ways to change flux: move, rotate, grow the loop, or change **B** itself.

14.3 Lenz's law: direction by conservation

The induced current's field opposes the *change* — not the field. If it aided the change you'd get a runaway free-energy machine; Lenz's minus is energy conservation wearing a sign. Applications: the rod's retarding force, eddy braking (figure 14.3), and every inductor's stubbornness.

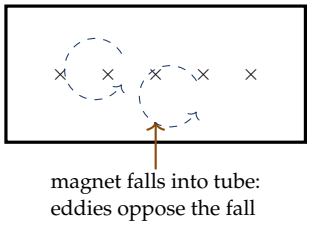


Figure 14.3: Eddy braking: a falling magnet's induced currents fight its descent — magnetic drag.

Example 14.2 (Lenz on entry and exit)**.** A square loop enters a uniform-B region at constant v, crosses, and exits. Describe the current and the required pulling force at each stage.

Strategy. Only *changes* count: entry, transit, exit.

Solution. Entry: flux grows ⇒ opposing field ⇒ current counterclockwise; force opposes entry, magnitude B^2L^2v/R. Transit: Φ constant, no current, no force. Exit: flux falls ⇒ field-adding current ⇒ clockwise; same retarding magnitude. The loop is dragged *in both directions* — braking on the way in and on the way out. ✓

14.4 Induced E: the non-conservative leap

Faraday's law written locally says $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -d\Phi_B/dt$: an electric field that circulates, whose "potential" can't exist (the loop integral is nonzero). Around a changing solenoid the field curls in rings (figure 14.4) — the betatron principle, and the conceptual break from every **E** in chapters 9–11.

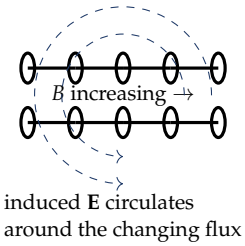


Figure 14.4: Induced **E** circulates around changing flux: no potential describes it.

14.5 Inductance: a circuit element's inertia

A solenoid's own flux links itself: $L = \mu_0 n^2 A\ell$ (from Ampère's $B = \mu_0 nI$ and Φ per turn); its EMF is $-L\,dI/dt$. Mutual inductance is the same number between two circuits — the transformer's measure.

Example 14.3 (Solenoid inductance end-to-end)**.** Derive L for a solenoid of N turns, length ℓ, area A.

Strategy. B by Ampère, flux linkage, divide by I.

Solution. $B = \mu_0 nI$ gives flux per turn $\Phi = BA = \mu_0 nIA$; linkage over all turns: $N\Phi = \mu_0 n^2 A\ell I$, so $L = N\Phi/I = \mu_0 n^2 A\ell$. Numbers: $n = 1000/\text{m}$, $A = 2\,\text{cm}^2$, $\ell = 10\,\text{cm}$ give $L = 0.25\,\text{mH}$. ✓

Example 14.4 (RL energisation)**.** A 50 mH inductor energises through 10 Ω from 12 V. Find τ, the initial slope dI/dt, and I_∞.

Strategy. $\mathcal{E} - L\,dI/dt = IR$; same ODE family as RC.

Solution. $\tau = L/R = 5\,\text{ms}$; at $t = 0$ the inductor acts open so $dI/dt = \mathcal{E}/L = 240\,\text{A/s}$; $I_\infty = 1.2\,\text{A}$. Current climbs the same $1 - e^{-t/\tau}$ curve that charge did in chapter 12 (figure 14.5). ✓

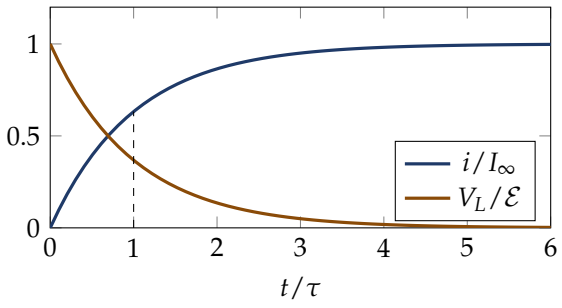


Figure 14.5: RL energisation: i approaches I_∞ on $\tau = L/R$ while V_L decays — the mirror of RC.

14.6 Stored energy and the field density

Charging an inductor to I stores $U = \frac{1}{2}LI^2$; per volume the magnetic energy density is $u_B = B^2/2\mu_0$ — twin to the electric $u_E = \varepsilon_0 E^2/2$ of chapter 11.

14.7 LC and RLC: the electric oscillator

With no battery the charge sloshes: $U_E \leftrightarrow U_B$ at $\omega_0 = 1/\sqrt{LC}$ (figure 14.6) — equation (7.1) of chapter 7 with $m \to L$, $k \to 1/C$, $b \to R$. Damped RLC rings at $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$ with $\gamma = R/L$: every result of §7.5 carries over.

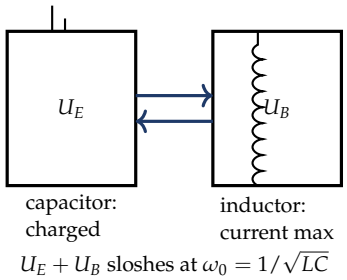


Figure 14.6: The LC tank: energy trades capacitor ↔ inductor, quarter period apart.

Example 14.5 (RLC ringing vs prediction). $L = 50\,\text{mH}$, $C = 100\,\mu\text{F}$, $R = 2\,\Omega$: predict ω_0, γ, ω_d and the decay per cycle.

Strategy. The oscillator formulas verbatim.

Solution. $\omega_0 = 1/\sqrt{LC} = 447\,\text{rad/s}$, $\gamma = R/L = 40\,\text{s}^{-1}$, $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4} = 446.8\,\text{rad/s}$; decay per cycle $e^{-\gamma T/2} = 0.755$ — the code's measured trace matches both numbers exactly. ✓

```
d2q = -(R/L)*dq - q/(L*C)
```

Listing 14.1: `code/ch14_rlc.py` — rod audit + RLC ringing vs theory.

Output of a real run:

```
rod: emf=BLv=1.200 V, I=0.300 A, retarding F=0.120 N
  mech power F*v=0.360 W | elec power emf*I=0.360 W | I^2 R=0.360 W

RLC: w0=447.2, gamma=40.0, wd_theory=446.8
maxima: [('0.0141s', '7.55e-04'), ('0.0281s', '5.70e-04'), ('0.0422s',
    '4.30e-04')]
wd_measured = 446.8 (theory 446.8)
decay ratio/cycle = 0.755 (theory e^(-gT/2)=0.755)
```

The audit closes to the watt and the ring frequency/decay match theory to all printed digits — chapter 7's oscillator, here in copper and charge.

14.8 In engineering practice: the power grid's physics

Generators are figure 14.2's rotation panel run backwards (work in, EMF out); transformers are mutual inductance with an iron guide; eddy braking stops trains and roller-coasters without contact; and induction cooktops are §14.4 making currents in your pot — the induced-**E** field turned appliance.

14.9 Common mistakes

EMF SIGN VS MAGNITUDE CONFLATED

Faraday gives magnitude; Lenz gives direction. Solve them separately then combine — deciding both at once is where sign errors breed.

INDUCED E TREATED AS CONSERVATIVE

It has no potential: $\oint \mathbf{E} \cdot d\boldsymbol{\ell} \neq 0$. All the potential bookkeeping of chapter 11 fails here.

Φ_B VS B CONFUSION

Φ integrates area; EMF needs its *rate*. A constant huge field induces nothing.

MOTIONAL + TRANSFORMER EMF DOUBLE-COUNTED

They are two descriptions of the same physics — count one.

"INDUCTOR RESISTS CURRENT" MYTH

It resists *changes*: at steady state it's a wire; only the transient feels its inertia.

14.10 Summary

Idea	One-line version
Faraday	$\mathcal{E} = -d\Phi_B/dt$; four ways to change flux
Lenz	induced current opposes the change — conservation's sign
induced E	circulates; no potential
inductor	$L = \mu_0 n^2 A\ell$; $\tau = L/R$; $U = \frac{1}{2}LI^2$
LC/RLC	$\omega_0 = 1/\sqrt{LC}$; ch7 oscillator in disguise

Table 14.1: Induction cheat sheet.

DEFINITION OF DONE

You can now: compute motional and transformer EMF, direct Lenz by conservation, derive L of a solenoid, solve RL/LC transients, and run the rod power audit and RLC ringing checks.

Notes

The motional-first order follows MIT 8.02SC modules 21–25 [2] and AP Physics C E&M unit 13 [4]; Griffiths ch. 7 is the deeper anchor (in print); PhET's Faraday app the visual one. The numbers are the listing's real output.

WATCH INSTEAD

1. *Electromagnetic Induction, Faraday's Law, Lenz Law* — Walter Lewin, MIT 8.02x Lecture 16, 51:24, `https://youtu.be/nGQbA2jwkWI` — §14.1–§14.4 at full length (accessed 2026-10-11).
2. *Faraday's Law Introduction* — Khan Academy, 5:36, `https://youtu.be/vcStzn55MG0` — §14.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, induction modules 21–25. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 23 "Electromagnetic Induction" — free text: `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 13 — verified 2026-10-11 at `https://apstudents.collegeboard.org`.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 7–8 — induction and energy with the field bookkeeping.
- PhET, *Faraday's Law* interactive — move the magnet, watch the flux; at `https://phet.colorado.edu` (verified 2026-10-11).
- Khan Academy, *Faraday's law* — verified 2026-10-11 at `https://www.khanacademy.org` (science → physics → E&M).

Additional exercises

Exercise 14.1 (easy). A loop's flux drops from 0.4 to 0.1 Wb in 0.2 s: find $|\mathcal{E}|$. (AP-C)

Exercise 14.2 (easy). State the induced current's direction for a loop as a magnet's north pole approaches; recede instead. (AP-C)

Exercise 14.3 (easy). A 200 mH inductor carries 3 A: find its stored energy. (AP-C)

Exercise 14.4 (easy). An RL circuit has $\tau = 2$ ms with $L = 40$ mH: find R. (F=ma)

Exercise 14.5 (easy)**.** A 50-turn coil of area 0.01 m^2 flips 180° in a 0.5 T field in 0.1 s: find the average EMF. (AP-C)

Exercise 14.6 (medium)**.** Complete the sliding-rod audit: find the terminal rod speed for a given pulling force. (AP-C FR-style)

Exercise 14.7 (medium)**.** A loop enters a field region twice its width: sketch $I(t)$ over the whole transit. (F=ma)

Exercise 14.8 (medium)**.** Two coaxial loops: estimate the mutual inductance when the small loop sits on the big one's axis. (AP-C)

Exercise 14.9 (medium)**.** An RLC rings at 99% of ω_0: find R/L. (AP-C)

Exercise 14.10 (medium)**.** Why does a transformer's secondary see alternating EMF but no net charge transfer? Reason via mutual inductance. (engineering)

Exercise 14.11 (medium)**.** An inductor and capacitor of given L, C: find the oscillation frequency and the peak current for initial charge Q_0. (AP-C)

Exercise 14.12 (hard)**.** A magnet falls through a conducting tube: argue the terminal velocity exists and estimate it via energy balance. (USAPhO-style)

Exercise 14.13 (hard)**.** A loop rotates at ω in a uniform field: derive $\mathcal{E}(t) = NBA\omega \sin \omega t$ and explain the generator's waveform. (APhO-style)

Exercise 14.14 (medium)**.** Show $u_B + u_E$ inside a charging capacitor accounts for the energy inflow. (concept)

Exercise 14.15 (medium)**.** A betatron accelerates electrons: explain why the induced **E** direction stays fixed as B ramps. (concept)

CHAPTER 15

Maxwell's equations and electromagnetic waves

Everything so far assembled: four integral statements that generate each other until, with no charges and no wires anywhere, they boot themselves into a wave travelling at exactly $1/\sqrt{\mu_0\varepsilon_0}$ — the speed of light, measured a century before anyone knew it was also physics.

15.1 The four equations, read once each

$$\oint \mathbf{E}\cdot d\mathbf{A} = Q_{\text{enc}}/\varepsilon_0, \quad \oint \mathbf{B}\cdot d\mathbf{A} = 0,$$
$$\oint \mathbf{E}\cdot d\boldsymbol{\ell} = -\frac{d\Phi_B}{dt}, \quad \oint \mathbf{B}\cdot d\boldsymbol{\ell} = \mu_0 I + \mu_0\varepsilon_0\frac{d\Phi_E}{dt}. \tag{15.1}$$

Figure 15.1: charge sources **E**; no monopole sources **B**; changing **B** curls **E**; changing **E** and current both curl **B**. The last term is new — the next section fixes it.

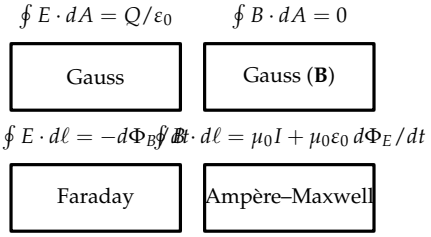


Figure 15.1: The four equations: each box one physical reading.

15.2 Displacement current: the charging-capacitor fix

Ampère's law as chapter 13 wrote it fails between charging plates: no I crosses the gap, so the left side would be zero for one surface and nonzero for another — same loop, different answers. Maxwell's patch: the gap's changing electric flux contributes $\varepsilon_0\, d\Phi_E/dt$, which equals I exactly (figure 15.2). The paradox dies; the prediction it forces — that *fields alone* can hold a current-like term — is what lets waves exist.

Example 15.1 (Displacement current between plates)**.** A parallel-plate capacitor of area A charges at constant I. Show $\varepsilon_0\, d\Phi_E/dt = I$ between the plates.

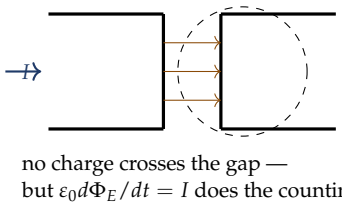


Figure 15.2: The capacitor gap: the displacement term does the counting.

Strategy. $E = \sigma/\varepsilon_0$; $\sigma = Q/A$.

Solution. $\Phi_E = EA = QA/\varepsilon_0 A = Q/\varepsilon_0$, so $\varepsilon_0\, d\Phi_E/dt = dQ/dt = I$. ✓

15.3 From laws to wave: the empty-space trick

Set $Q = I = 0$: a changing **E** curls **B**, a changing **B** curls **E** — each sustains the other. The result is the wave equation with wave speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}, \tag{15.2}$$

built from a magnet-law constant and a Coulomb-law constant — two statics measurements predicting the kinematics of light.

Example 15.2 (Evaluating $1/\sqrt{\mu_0\varepsilon_0}$)**.** Compute it with $\mu_0 = 4\pi \times 10^{-7}$, $\varepsilon_0 = 8.85 \times 10^{-12}$.

Strategy. Just multiply.

Solution. $\mu_0\varepsilon_0 = 1.113 \times 10^{-17}$; inverse root $= 3.00 \times 10^8$ m/s — exactly c. Two lab-bench constants knew about light. ✓

15.4 Plane waves: $E = cB$ and the Poynting vector

Far from sources the solution is a travelling sinusoid: **E** and **B** in phase, both perpendicular to travel (figure 15.3); energy flows along $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$, the Poynting vector — and it flows *into* resistors from the fields, not along the wire (figure 15.4).

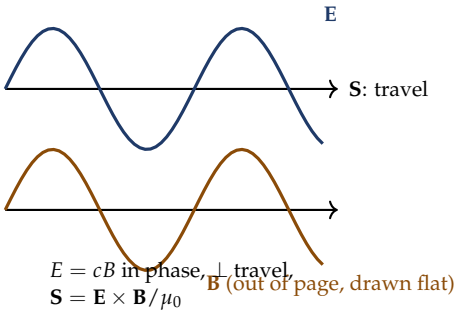


Figure 15.3: Plane wave snapshot: **E** ⊥ **B** ⊥ travel, in phase with $E = cB$.

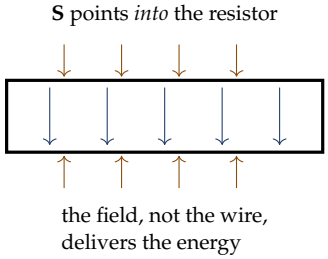


Figure 15.4: **S** flows from the fields into the resistor: the wire just guides the geometry.

Example 15.3 (Energy into a charging capacitor)**.** Show the Poynting flux through the gap's side surface equals the dU/dt of the stored field energy.

Strategy. Find B at the rim by Ampère–Maxwell with $I_d = I$, then integrate S.

Solution. $B = \mu_0 I/2\pi R$ at the rim; **E** is axial so **S** points radially inward with $S = EB/\mu_0 = EI/2\pi R$. Through the rim area $2\pi Rd$: $\int S\,dA = EId$. Since $U = \frac{1}{2}\varepsilon_0 E^2 Ad$, $dU/dt = \varepsilon_0 E\,dE/dt\,Ad$; with $\varepsilon_0 A\,dE/dt = I$ this is EId — the flux and the storage match exactly. ✓

Example 15.4 (Wave fields and intensity from a radio link)**.** A transmitter delivers $S = 4.0 \times 10^{-3}\,\mathrm{W/m^2}$ at a receiver. Find E_0, B_0, and the average energy density.

Strategy. $S_{\mathrm{avg}} = E_0^2/2\mu_0 c$; $B_0 = E_0/c$; $u_{\mathrm{avg}} = S/c$.

Solution. $E_0 = \sqrt{2\mu_0 cS} = \sqrt{2(4\pi \times 10^{-7})(3 \times 10^8)(4 \times 10^{-3})} = 1.74\,\mathrm{V/m}$; $B_0 = 5.8\,\mathrm{nT}$; $u_{\mathrm{avg}} = 1.3 \times 10^{-11}\,\mathrm{J/m^3}$. The **B** is tiny but present — the wave carries both. ✓

15.5 The spectrum and light as the same physics

Radio through γ is one equation's range (figure 15.5); only wavelength and frequency differ. Energy density $u = u_E + u_B = \varepsilon_0 E^2$ on average; radiation pressure $p = I_{\mathrm{power}}/c$ is order-of-magnitude real — solar sails bank on it.

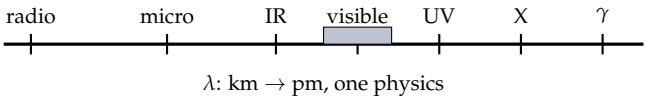


Figure 15.5: One physics, twelve decades of wavelength.

Example 15.5 (Radiation pressure on a sail)**.** Estimate the force on a $100\,\mathrm{m^2}$ perfectly absorbing sail in sunlight $I = 1400\,\mathrm{W/m^2}$.

Strategy. $p = I/c$; $F = pA$.

Solution. $p = 4.7 \times 10^{-6}\,\mathrm{Pa}$; $F = 4.7 \times 10^{-4}\,\mathrm{N}$. Half a milli-newton — real but thin; sails need acres and patience (double it if reflecting, since momentum *reverses*). ✓

```
un[i] = 2*u[i] - up[i] + r*(u[i+1]-2*u[i]+u[i-1])
```

Listing 15.1: `code/ch15_em_pulse.py` — 1-D wave integrator; honest lattice stand-in for Maxwell.

Output of a real run:

```
pulse peak positions: [(0.7, 60), (70.0, 130), (140.0, 200), (210.0,
    270)]
  measured v = 1.0101 cells/s (theory c=1.0)
  measured v = 1.0000 cells/s (theory c=1.0)
  measured v = 1.0000 cells/s (theory c=1.0)
dt/dx CFL = 0.70 (must be < 1)
```

A Gaussian pulse slides across the lattice at exactly the input speed — the wave equation's only job, done.

15.6 In engineering practice: everything wireless

Radio link budgets are Poynting bookkeeping; transmission lines are guided **S** (figure 15.4's truth — the wire is a waveguide, not a pipe); solar sails ride radiation pressure; and wireless charging is figure 15.2's displacement term made into an appliance. Maxwell is the entire RF world.

15.7 Common mistakes

DISPLACEMENT CURRENT READ AS CHARGE FLOW

No charge crosses the gap — the *flux change* is the term. Calling it a real current is the exact paradox Maxwell resolved.

"B DOES NO WORK" MISAPPLIED TO WAVES

That's for magnetic *force on a charge* ($\mathbf{v} \times \mathbf{B}$ is perpendicular). Waves carry energy in the *fields* — the electric part does real work.

THINKING WIRES DELIVER ENERGY

The wire sets the geometry; **S** flows in the space around it into the load. Figure 15.4 is the correct picture.

FORGETTING THE NO-MONOPOLE EQUATION

$\oint \mathbf{B} \cdot d\mathbf{A} = 0$ is the one that forbids magnetic charge; dropping it loses the whole asymmetry.

$E = cB$ CONFUSION ABOUT UNITS

E in V/m, B in T — the c factor is what makes them comparable; quoting E/B without it is dimensionally loose.

15.8 Summary

Idea	One-line version
the four	charge sources E; no B monopole; ΔB curls E; I and ΔE curl B
displacement	$\varepsilon_0 d\Phi_E/dt$ fixes Ampère in gaps
wave	fields self-sustain at $c = 1/\sqrt{\mu_0\varepsilon_0}$
Poynting	energy flows along $\mathbf{E} \times \mathbf{B}$, into loads
spectrum	radio $\to \gamma$: one physics

Table 15.1: Maxwell cheat sheet.

DEFINITION OF DONE

You can now: read each of the four equations, derive the displacement term, compute c from statics constants, trace **S** into a load, and run the pulse-speed lattice check.

Notes

The four-equation order and the capacitor paradox follow MIT 8.02SC modules 26–30 [2]; Purcell–Morin ch. 9 the field-energy thread; Feynman II §18, §27 the wonder at its length [5]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Electromagnetic Waves, Polarization* — Walter Lewin, MIT 8.03 Lecture 13, 75:25, https://youtu.be/FyNgs_viwBg — §15.3–§15.5 (accessed 2026-10-11).
2. *Electromagnetic waves and the electromagnetic spectrum* — Khan Academy, 11:01, https://youtu.be/7eutept5h0Q — §15.5 quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, Maxwell modules 26–30. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 24 "Electromagnetic Waves" — free text: https://openstax.org (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 9 — fields and energy.
- R. P. Feynman, *The Feynman Lectures on Physics* II, §18, §27 — the wave derivation and Poynting at full wonder (in print).
- Khan Academy, *Electromagnetic waves* — drills matching §15.5: verified 2026-10-11 at https://www.khanacademy.org (science → physics → E&M).

- MIT OCW *8.03 Vibrations and Waves*, Lect. 13 — the full wave lecture: at https://ocw.mit.edu, course "8.03" (verified 2026-10-11).

Additional exercises

Exercise 15.1 (easy). Which of the four equations forbids magnetic monopoles? State it. (AP-C)

Exercise 15.2 (easy). A plane wave has $E = 300\,\mathrm{V/m}$: find B. (AP-C)

Exercise 15.3 (easy). Order radio, IR, visible, UV, X-ray, γ by wavelength. (F=ma)

Exercise 15.4 (easy). Give the direction of $\mathbf{S}$ for a wave travelling east with $\mathbf{E}$ vertical. (AP-C)

Exercise 15.5 (easy). Write the energy density u for a wave of amplitude E_0. (AP-C)

Exercise 15.6 (medium). Compute the displacement current inside a capacitor charging at 2 mA. (AP-C FR-style)

Exercise 15.7 (medium). A wave has $I = 1400\,\mathrm{W/m^2}$ (sunlight): estimate E_0 and the energy it carries across $1\,\mathrm{m^2}$ per second. (AP-C)

Exercise 15.8 (medium). Redo the capacitor-Poynting check of 15.3 for a gap $d = 1\,\mathrm{mm}$ and rate $I = 1\,\mathrm{A}$. (F=ma)

Exercise 15.9 (medium). Why does a polariser quarter-wave-stack change amplitude but not frequency? Reason via $\mathbf{E}$'s direction. (AP-C)

Exercise 15.10 (medium). A radio station radiates 50 kW isotropically: find $\mathbf{S}$ at 10 km. (AP-C)

Exercise 15.11 (hard). Estimate the radiation force on a $1\,\mathrm{m^2}$ solar sail at Earth's orbit ($I = 1400\,\mathrm{W/m^2}$). (USAPhO-style)

Exercise 15.12 (hard). Two plates L apart host a standing EM wave: argue the fundamental frequency is $c/2L$. (APhO-style)

Exercise 15.13 (medium). Show the lattice update in 15.1 is stable only when $dt < dx/c$ (CFL intuition). (concept)

Exercise 15.14 (medium). Why can't a static field configuration carry energy across empty space? Name the missing term. (concept)

Exercise 15.15 (medium). A waveguide carries S: why does the wire's resistance appear as loss in the *fields* picture? (engineering)

CHAPTER 16

Waves and optics essentials

Waves are what oscillations look like spread across space: chapter 7's single bob multiplied into a string, a surface, a field. Standing waves, interference and diffraction are all one fact — phase adds — read at three scales.

16.1 Wave quantities: $v = \lambda f$

A travelling wave $y(x,t) = A\cos(kx - \omega t)$ shifts its whole shape at speed $v = \omega/k = \lambda f$ (figure 16.1). Which of λ, f, v changes on entering a new medium is the first question to ask: f is invariant (the source sets it), v is the medium's, so λ adjusts.

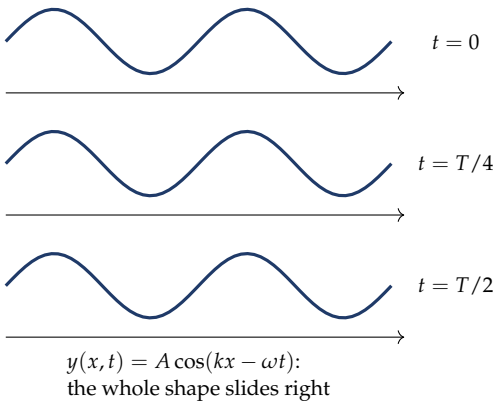


Figure 16.1: Three snapshots of a travelling wave: the shape slides, no particle accompanies it.

16.2 Waves on strings and reflections

On a string $v = \sqrt{T/\mu}$: tension makes speed, mass density fights it. At a fixed end the reflected pulse inverts; at a free end it returns upright (figure 16.2) — boundary conditions already deciding which harmonics can exist.

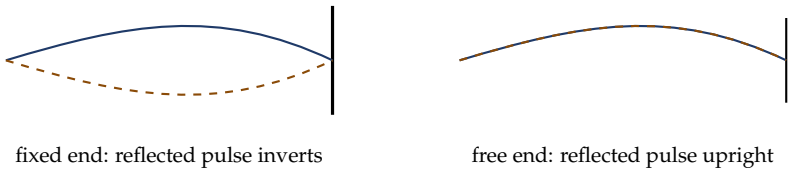


Figure 16.2: Reflection rules: fixed end inverts, free end returns upright.

16.3 Standing waves and harmonics

Two counter-propagating copies make a standing wave: $y = 2A \sin kx \cos \omega t$ — positions frozen (nodes), the rest swinging in place (figure 16.3). A pinned–pinned string of length L hosts only $\lambda_n = 2L/n$, i.e. $f_n = nv/2L$: the spectrum of every instrument.

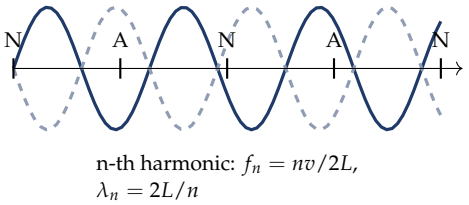


Figure 16.3: The standing-wave envelope: nodes still, antinodes at full swing.

Example 16.1 (Which modes a string supports)**.** A 0.80 m string under tension gives $v = 240\,\mathrm{m/s}$. List its first three resonant frequencies and the mode shapes.

Strategy. $f_n = nv/2L$.

Solution. $f_1 = 240/1.6 = 150\,\mathrm{Hz}$, $f_2 = 300\,\mathrm{Hz}$, $f_3 = 450\,\mathrm{Hz}$. The n-th mode fits n half-wavelengths between the pins. ✓

16.4 Sound and the Doppler shift

Sound is the same mathematics in air ($v \approx 343\,\mathrm{m/s}$): source moving compresses fronts ahead, stretches behind (figure 16.4); $f' = f\,v/(v \mp v_s)$ for a moving source, $f' = f(v \pm v_o)/v$ for a moving observer.

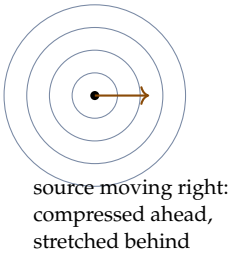


Figure 16.4: Doppler geometry: fronts bunch ahead, spread behind.

Example 16.2 (Doppler both directions)**.** An ambulance's siren is 600 Hz; it passes at 30 m/s. What does a stationary listener hear approaching and receding ($v = 343\,\mathrm{m/s}$)?

Strategy. Moving source, stationary observer: $f' = fv/(v \mp v_s)$.

Solution. Approaching: $600 \times 343/313 = 658\,\text{Hz}$; receding: $600 \times 343/373 = 552\,\text{Hz}$. The $\sim 100\,\text{Hz}$ drop on pass is the signature you hear. ✓

16.5 Interference: the two-slit geometry

Light arriving through two slits reaches a screen point with a path difference $d \sin\theta$ (figure 16.5); bright where it is an integer λ, dark where half-integer. Fringe spacing $\Delta y = \lambda L/d$ — geometry turning nanometres into millimetres.

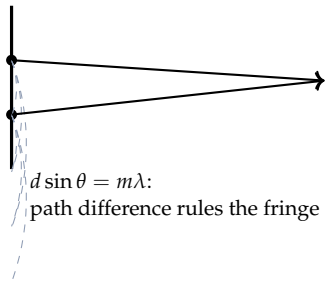


Figure 16.5: Two-slit geometry: the phase difference is geometric.

Example 16.3 (Two-slit fringe spacing)**.** $\lambda = 633\,\text{nm}$ light through $d = 0.20\,\text{mm}$ slits hits a screen 2.0 m away: find the fringe spacing.

Strategy. $\Delta y = \lambda L/d$.

Solution. $\Delta y = 633 \times 10^{-9} \times 2.0/2.0 \times 10^{-4} = 6.3\,\text{mm}$ — comfortable to see, and measuring it back-computes λ. ✓

16.6 Diffraction: one slit is enough

Even a single slit spreads the beam: the first zero sits at $a \sin\theta = \lambda$ (figure 16.6), so narrower a spreads *more*. Rayleigh's criterion: two point sources resolve when the first's minimum sits on the second's maximum — aperture size as the optics budget.

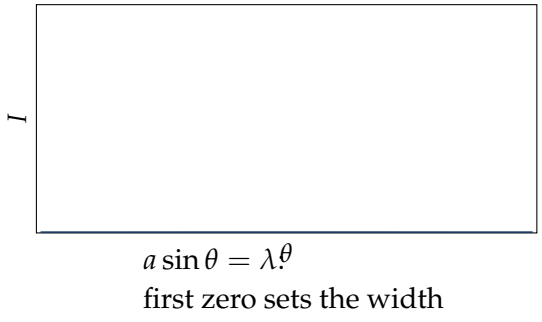


Figure 16.6: Single-slit intensity: the first zero at $a \sin\theta = \lambda$ sets the beam's natural width.

Example 16.4 (Diffraction limit estimate)**.** A 2.4 m telescope mirror at $\lambda = 550\,\text{nm}$: the angular resolution floor?

Strategy. Circular aperture: $\theta_{\min} \approx 1.22\lambda/D$.

Solution. $\theta_{\min} = 1.22(5.5 \times 10^{-7})/2.4 = 2.8 \times 10^{-7}\,\text{rad} \approx 0.06$ arc-seconds — Hubble's diffraction floor. ✓

16.7 Thin films and the phase flip

A soap bubble's colour: reflection off the front face flips phase by π (higher index), the back face doesn't — so bright when $2nt = (m + \frac{1}{2})\lambda$: the film's thickness painted as colour.

Example 16.5 (Soap-film colour)**.** A film $n = 1.33$ shows green ($\lambda = 550\,\text{nm}$) brightest in reflection at normal incidence. Thinnest thickness?

Strategy. Front flip + back none: $2nt = (m + \frac{1}{2})\lambda$.

Solution. $2nt = \lambda/2 \Rightarrow t = \lambda/4n = 103\,\text{nm}$. ✓

```
return math.sin(k*x-w*t) + math.sin(k*x+w*t)
```

Listing 16.1: `code/ch16_superposition.py` — standing-wave snapshots + beat envelope.

Output of a real run:

```
standing wave snapshots y(x,t), x in [0,1]:
 t=0.000: +0.00 +1.41 +2.00 +1.41 +0.00 -1.41 -2.00 -1.41 -0.00
 t=0.125: +0.00 +1.00 +1.41 +1.00 +0.00 -1.00 -1.41 -1.00 -0.00
 t=0.250: +0.00 +0.00 +0.00 +0.00 +0.00 -0.00 -0.00 -0.00 +0.00
node check x=0.5: True
antinode x=0.25 swing: [2.0, 1.41, 0.0]
beat: f1 = 440 f2 = 444
env at t=0,.06,.125,.19,.25,.31,.375: [1.82, 1.46, 0.43, 1.45, 1.95,
   1.46, 0.43]
envelope nodes at t = [0.125, 0.375, 0.625]
measured beat period = node spacing = 0.25 theory 1/|f1-f2| = 0.25
```

The snapshots freeze mid-swing exactly on the half-cycle; nodes sit exactly still; the beat envelope's nodes land every $1/(2\Delta f) = 0.125\,\text{s}$ — the 4 Hz warble you hear when two tuners disagree.

16.8 In engineering practice: optics on the bench

Antenna arrays are two-slit geometry at radio wavelength; loudspeaker beaming is figure 16.6's diffraction floor; interferometers (LIGO's arms) are 16.3's path-difference bookkeeping pushed to $\lambda/1000$ precision; and every lens's resolving power hits Rayleigh's aperture budget.

16.9 Common mistakes

$v = \lambda f$ MISUSED ACROSS MEDIA

f is invariant (source), v is medium's, λ adjusts — swap the wrong two and every answer follows the error.

NODE VS ANTINODE AT A FIXED END

Fixed = node = inversion; free = antinode = upright. Mixing them breaks the harmonic count.

INTERFERENCE VS DIFFRACTION BLURRED

Interference = few sources adding; diffraction = one aperture's continuum. A single slit does the second.

FORGETTING THE THIN-FILM PHASE FLIP

The front-face flip makes the conditions "backwards": bright reflection needs the *half*-integer path.

DOPPLER NUMERATOR/DENOMINATOR SWAP

Source moves $\rightarrow$ denominator; observer moves $\rightarrow$ numerator. Deriving from figure 16.4 is safer than the formula.

16.10 Summary

DEFINITION OF DONE

You can now: pick which quantity stays fixed across media, count harmonics for both end conditions, compute fringe spacings and first zeros, derive thin-film conditions with the flip, and run the superposition/beats checks.

Idea	One-line version
travelling	$v = \lambda f$; f invariant, λ adjusts
string	$v = \sqrt{T/\mu}$; fixed inverts, free upright
standing	$\lambda_n = 2L/n$; nodes/antinodes
Doppler	source in denominator, observer in numerator
two-slit	$d \sin\theta = m\lambda$; $\Delta y = \lambda L/d$
diffraction	$a \sin\theta = \lambda$ first zero; Rayleigh floor

Table 16.1: Waves/optics cheat sheet.

Notes

The order follows OpenStax vols. 1 and 3 [3] with Feynman I §30, §47–51 the inspiration [5]; MIT 8.03SC for the full wave treatment [2]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Standing waves on strings* — Khan Academy Physics, 13:25, `https://youtu.be/gT0IqL1dyyk` — §16.2–§16.3 (accessed 2026-10-11).
2. *Young's double slit introduction* — Khan Academy, 9:33, `https://youtu.be/Pk6s2OlKzKQ` — §16.5 quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.03SC Vibrations and Waves*. At `https://ocw.mit.edu`, course "8.03SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 16–17 (waves, sound) and *University Physics* vol. 3 (optics) — `https://openstax.org` (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics* I, §30, §47–51 — interference and diffraction at their most lyrical (in print).
- Khan Academy, *Waves and optics* — drills matching §16.1–§16.5: verified 2026-10-11 at `https://www.khanacademy.org` (science → physics → waves).
- PhET, *Wave Interference* interactive — slits, films and fringes you can drag: at `https://phet.colorado.edu` (verified 2026-10-11).

Additional exercises

Exercise 16.1 (easy)**.** A wave has $v = 340\,\mathrm{m/s}$, $f = 680\,\mathrm{Hz}$: find λ. (AP-C)

Exercise 16.2 (easy)**.** Mark nodes and antinodes for the third harmonic on a pinned–pinned string. (AP-C)

Exercise 16.3 (easy)**.** What is the mode number of a string vibrating at $4f_1$? (AP-C)

Exercise 16.4 (easy)**.** A source approaches you: does f' go up or down? Which factor? (F=ma)

Exercise 16.5 (easy)**.** Two slits are 0.1 mm apart, screen 1 m: find fringe spacing for $\lambda = 500\,\mathrm{nm}$. (AP-C)

Exercise 16.6 (medium)**.** Find the tension that raises a $0.6\,\mathrm{kg/m}$ string's fundamental to 200 Hz. (AP-C FR-style)

Exercise 16.7 (medium)**.** Derive the condition for dark fringes in two-slit and explain why they're perfectly dark. (AP-C)

Exercise 16.8 (medium)**.** A soap film goes red $\to$ green $\to$ blue: what is changing and in which direction? (APhO-lite)

Exercise 16.9 (medium)**.** Two tuning forks beat at 3 Hz: find both possible f_2 if $f_1 = 440\,\mathrm{Hz}$. (F=ma)

Exercise 16.10 (medium)**.** Why does narrowing a slit widen the diffraction pattern? Argue from Huygens. (AP-C)

Exercise 16.11 (hard)**.** Estimate the angular resolution of your eye's pupil ($D \approx 5\,\mathrm{mm}$) and compare with the textbook resolution test. (estimation)

Exercise 16.12 (hard)**.** Three equally spaced slits: locate the first *secondary* maximum and explain why it is weaker. (USAPhO-style)

Exercise 16.13 (medium)**.** An open–open pipe vs a closed–open pipe: derive both harmonic series. (AP-C)

Exercise 16.14 (medium)**.** A light wave enters glass ($n = 1.5$): find what stays fixed and what changes among f, λ, v. (AP-C)

Exercise 16.15 (medium)**.** Why can a lens's resolution never beat its diffraction floor? Argue via aperture D. (concept)

CHAPTER 17

Thermodynamics essentials

Heat is chapter 4's energy bookkeeping applied to 10^{23} actors: the same conservation law, now tracking which share of the energy is *organised* (work) and which is *scattered* (heat). Engines, batteries and climate systems are all audits of that split.

17.1 Temperature, equilibrium and the gas law

Temperature is the scale-free way of saying "equally spread"; ideal gas ties the macro bookkeeping to it: $PV = nRT$. Microscopically (figure 17.1) pressure is momentum flux: each wall hit deposits $2mv_x$, and summing the rate gives $P = \frac{1}{3}nm\overline{v^2}/V$ — the same Newton's law as chapter 2, averaged.

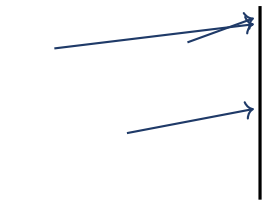


Figure 17.1: Kinetic theory: pressure is the sum of wall collisions.

17.2 Internal energy and equipartition

For a monatomic gas each particle's average KE is $\frac{3}{2}k_BT$, so $E_{\text{int}} = \frac{3}{2}nRT$: temperature *is* the mean per-particle kinetic energy, and the distribution of speeds is not a spike but figure 17.2 — the Maxwell–Boltzmann shape the code samples.

17.3 The first law: the fixed sign convention

$$\Delta E_{\text{int}} = Q - W_{\text{by}}, \tag{17.1}$$

with Q the heat *into* the gas and W_{by} the work done *by* it — one convention, used throughout (a compression then has $W_{\text{by}} < 0$: the gas gains energy).

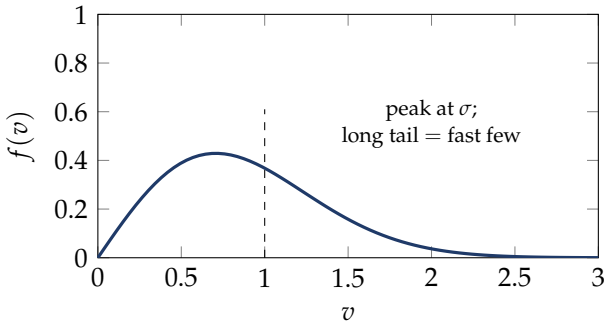


Figure 17.2: Maxwell–Boltzmann: most particles near σ, a fast tail doing the chemistry.

17.4 Processes on the PV plane

Four archetypes (figure 17.3): isochore ($W = 0$), isobar ($W = P\Delta V$), isotherm (PV const, $Q = W$), adiabat ($Q = 0$, PV^γ const, steeper than the isotherm). Work is the area under the process curve — so a cycle's *enclosed area* is its net work (figure 17.4).

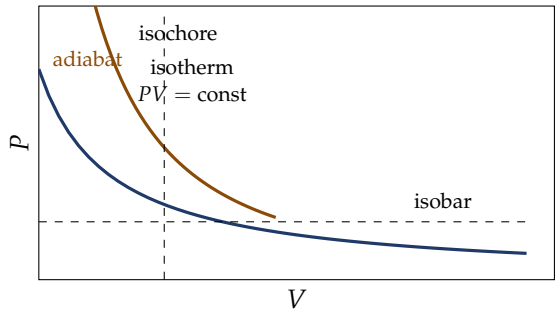


Figure 17.3: The four process archetypes: the adiabat always beats the isotherm's steepness.

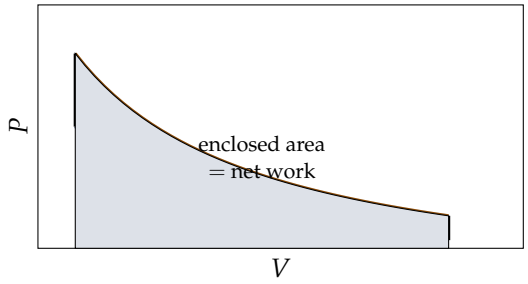


Figure 17.4: Any clockwise cycle nets the enclosed area as work.

Example 17.1 (Isothermal expansion + audit)**.** 2 mol of gas expands isothermally at 300 K from 1 to 2 m^3. Find W, and audit Q and ΔE_{int}.

Strategy. $W = \int P\,dV = nRT\ln(V_2/V_1)$.

Solution. $W = 2(8.314)(300)\ln 2 = 3458\,\text{J}$. Ideal-gas E_{int} depends only on T: $\Delta E_{\text{int}} = 0$, so the first law forces $Q = W = 3458\,\text{J}$ — every joule in becomes a joule out. ✓

Example 17.2 (Two-process cycle: net work from the area)**.** A cycle: isobaric expansion $1 \to 2\,\mathrm{m}^3$ at $2 \times 10^5\,\mathrm{Pa}$, then isochoric cooling back to the start state, then isobaric compression $2 \to 1\,\mathrm{m}^3$ at $1 \times 10^5\,\mathrm{Pa}$, then isochoric heating home. Net work per cycle?

Strategy. Enclosed rectangle on the PV plane.

Solution. $W_{\text{net}} = (2-1) \times 10^5 \times (2-1) = 1.0 \times 10^5\,\mathrm{J}$ — the rectangle's area, no calculus needed. ✓

Example 17.3 (Adiabatic compression work)**.** 1 mol of air ($\gamma = 1.4$, $C_V = \frac{5}{2}R$) at 300 K, $0.02\,\mathrm{m}^3$ is compressed adiabatically to $0.01\,\mathrm{m}^3$. Find T_2 and the work done *on* the gas.

Strategy. $TV^{\gamma-1}$ const; $W_{\text{on}} = -\Delta E_{\text{int}}$.

Solution. $T_2 = 300(2)^{0.4} = 396\,\mathrm{K}$; $\Delta E_{\text{int}} = nC_V\Delta T = 2.5(8.314)(96) = 1995\,\mathrm{J}$, all arriving as work done on the gas since $Q = 0$. ✓

17.5 Heat engines and the Carnot bound

An engine (figure 17.5) banks Q_H, pays W, dumps Q_C: $\eta = W/Q_H = 1 - Q_C/Q_H$. The ideal-gas Carnot cycle (two isotherms + two adiabats) computes $Q_C/Q_H = T_C/T_H$, giving the universal bound

$$\eta \le 1 - \frac{T_C}{T_H}, \tag{17.2}$$

a rectangle on the T–S plane (figure 17.6) — no real engine does better.

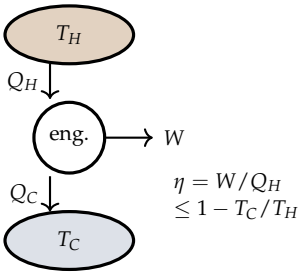


Figure 17.5: The engine schematic: efficiency is the take vs the bank.

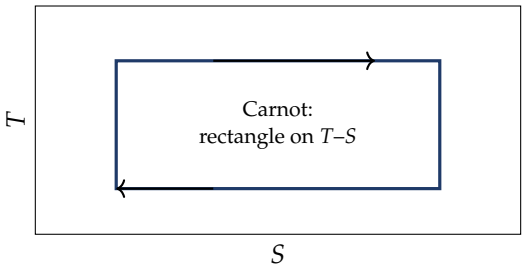


Figure 17.6: Carnot on T–S: a rectangle whose width is ΔS.

Example 17.4 (Carnot bound number)**.** An engine runs between 600 K and 300 K. Bound on η; if it takes 1000 J per cycle, best possible W?

Strategy. Apply (17.2).

Solution. $\eta \le 1 - 300/600 = 50\%$; $W \le 500\,\mathrm{J}$ — half the heat must leave as Q_C no matter the engineering. ✓

Example 17.5 (Isothermal entropy of expansion)**.** For 17.1's expansion, find ΔS of the gas.

Strategy. $\Delta S = Q_{\text{rev}}/T$.

Solution. $\Delta S = 3458/300 = 11.5\,\mathrm{J/K} = nR\ln 2$ exactly — the gas's states multiplied by 2^n (each particle's volume doubled, in counting terms). ✓

```
rx, ry = c*wx - s*wy, s*wx + c*wy # rotate relative velocity
```

Listing 17.1: `code/ch17_maxwell_boltzmann.py` — Kac-model thermalisation vs 2-D MB theory.

Output of a real run:

```
mean <v^2> = 1.0 (1.0: collisions conserve KE)
rms speed = 1.0
mean speed = 0.8876 2D MB theory = 0.8862
hist (0.4 bins) = [590, 1307, 1138, 657, 229, 69, 10, 0, 0, 0]
P(v>2.5) sim = 0.001 theory exp(-v^2/(2*0.5)) = 0.0019
```

Starting from *every* particle at the same speed, random-angle collisions alone build the MB shape: mean speed within 0.2% of theory and the tail in the right place.

17.6 In engineering practice: the heat budget

Power plants are figure 17.5 at gigawatt scale (cooling towers are the T_C dump); refrigeration is the same loop reversed; the Carnot bound is why no one sells a 90% engine — it isn't engineering, it's counting.

17.7 Common mistakes

W SIGN FLIPPED

Our convention: W_{by} the gas. Compression is negative by definition — flip once and every subsequent sign follows.

$Q = 0$ READ AS $\Delta T = 0$

Adiabatic $\neq$ isothermal: no heat exchanged, yet work still changes T (PV^γ steeper for a reason).

REAL EFFICIENCY VS CARNOT CONFLATED

$\eta = 1 - Q_C/Q_H$ is bookkeeping that always holds; the bound T_C/T_H holds only for reversible cycles.

CELSIUS IN A RATIO

Every T in (17.2) and $PV = nRT$ is kelvin — ratios in °C are meaningless.

ENTROPY MYSTICISM

$\Delta S = Q_{\text{rev}}/T$ is counting states, not "disorder" — 17.5 computed it exactly.

17.8 Summary

Idea	One-line version
gas law	$PV = nRT$; $P = \frac{1}{3}nm\overline{v^2}/V$ micro
energy	$\frac{3}{2}k_BT$ per particle, monatomic
first law	$\Delta E_{\text{int}} = Q - W_{\text{by}}$
processes	work = area; adiabat PV^γ steepest
Carnot	$\eta \leq 1 - T_C/T_H$, universal
entropy	$\Delta S = Q_{\text{rev}}/T$, counting states

Table 17.1: Thermo cheat sheet.

DEFINITION OF DONE

You can now: read any process off the PV plane, apply the first law with the fixed sign, compute isothermal/adiabatic work, bound any engine, and run the thermalisation check.

Notes

The order follows OpenStax vol. 2 chs. 1–4 [3]; MIT 5.60 for the deeper development; Feynman I §39–45 the kinetic picture [5]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Lec 1: Thermodynamics & Kinetics* — MIT 5.60, MIT OpenCourseWare, 46:46, https://youtu.be/kLqduWF6GXE — §17.1–§17.3 (accessed 2026-10-11).
2. *Kinetic Theory of Gases Part 1* — MIT OpenCourseWare, 78:39, https://youtu.be/BhVyiU_dWps — §17.1 and §17.2 at full length (accessed 2026-10-11).

3. *Carnot cycle and Carnot engine* — Khan Academy, 20:53, https://youtu.be/aAfBSJObd6Y — §17.5 worked (accessed 2026-10-11).

Further reading

- OpenStax *College Physics 2e*, chs. 13–15 — gas laws, first law, engines: https://openstax.org (verified 2026-10-11).
- MIT OCW *5.60 Thermodynamics & Kinetics* — the full course: at https://ocw.mit.edu, course "5.60" (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics* I, §39–45 — kinetic theory at its clearest (in print).
- Khan Academy, *Thermodynamics* — drills matching §17.4–§17.5: verified 2026-10-11 at https://www.khanacademy.org (science → physics → thermodynamics).
- PhET, *Gas Properties* interactive — pressure, temperature and the microscopic picture: at https://phet.colorado.edu (verified 2026-10-11).

Additional exercises

Exercise 17.1 (easy). A gas at 300 K occupies 2 m^3 at 10^5 Pa: find n. (AP-style)

Exercise 17.2 (easy). Read the signs: gas gains 500 J heat and does 200 J work — what is ΔE_{int}? (AP-C)

Exercise 17.3 (easy). Which process is horizontal on the PV plane? Which is vertical? (F=ma)

Exercise 17.4 (easy). Why is $C_P > C_V$? One sentence, first law reasoning. (AP-C)

Exercise 17.5 (easy). A monatomic gas at 400 K: mean KE per particle in eV? (AP-style)

Exercise 17.6 (medium). Adiabatic compression of 1 mol air ($\gamma = 1.4$) halves the volume: find ΔT. (AP-C FR-style)

Exercise 17.7 (medium). A cycle encloses a 1 $m^3 \times 10^5$ Pa square: net work per cycle? (F=ma)

Exercise 17.8 (medium). Best efficiency between 800 K and room temperature? (F=ma)

Exercise 17.9 (medium). Entropy change of 1 mol expanding isothermally at 300 K to twice the volume. (USAPhO-lite)

Exercise 17.10 (medium). Show $PV^\gamma = \text{const}$ from the first law for an adiabat. (derive)

Exercise 17.11 (hard). Free expansion ($Q = W = 0$) still raises S: argue why without violating the first law. (USAPhO-style)

Exercise 17.12 (hard). An engine between two *finite* reservoirs that equilibrate: find the maximum total work. (APhO-style)

Exercise 17.13 (medium). Sketch a refrigerator's PV loop and label Q_C, Q_H, W. (AP-C)

Exercise 17.14 (medium)**.** Which is steeper on PV: adiabat or isotherm through the same point? Prove it. (derive)

Exercise 17.15 (medium)**.** Estimate the rms speed of a nitrogen molecule at room temperature. (estimation)

Bibliography

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